

National Platform for Physical AI Oncology Trials

Draft 1.0

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Source Documents and Their Significance

This National Platform draws upon a comprehensive body of completed work spanning regulatory adaptations, clinical trial standards, patient-centered documentation, national infrastructure designs, and supporting research. Each source document listed below is referenced throughout this paper by its designated name. The following overview defines the role and significance of each document within the platform.

Regulatory Adaptation Documents

Adaption: ICH Harmonised Guideline [1]. A complete adaptation of the ICH E6(R3) Guideline for Good Clinical Practice to Physical AI oncology clinical trials. This document establishes principles of Physical AI clinical practice, investigator responsibilities, sponsor responsibilities, and data governance requirements across four major sections and three appendices. It is referenced throughout Sections 4, 8, and 15 as the foundational good clinical practice standard for Physical AI implementations.

Physical AI Adaption: 21 CFR Part 50 [2]. An adaptation of the federal regulation governing Protection of Human Subjects to Physical AI clinical trial contexts. This document addresses informed consent requirements specific to robotic and AI-assisted procedures, IRB review standards for Physical AI investigations, pediatric protections, and a comprehensive Physical AI system classification framework. It is referenced in Sections 5, 9, and 10 as the human subjects protection standard.

Physical AI Adaption: 21 CFR Part 312 [3]. An adaptation of the Investigational New Drug Application regulation to Physical AI trial contexts spanning ten subparts, including a new Subpart J dedicated entirely to Physical AI systems. This document covers IND requirements, clinical phases including a Phase 0 simulation validation stage, safety reporting for Physical AI adverse events and cybersecurity incidents, and expanded access provisions. It is referenced in Sections 6, 8, and 14.

Standards and Framework Documents

Physical AI Unification Standard Level [4]. Defines the Unification Standard Level (USL) scoring methodology for evaluating robot readiness across four dimensions: simulation framework switching, AI integration, cross-robot sharing, and clinical trial collaboration. This document provides quantitative scoring for nine robots across three categories (cobots, surgical, humanoids) used in Physical AI oncology trials. It is referenced throughout Sections 7, 9, and 15 as the robot readiness evaluation framework.

Clinical Trial Site Complete Documentation [5]. A comprehensive package of eleven regulatory documents required for establishing the first Physical AI oncology clinical trial site in California. This includes state legislation (SB 1042, AB 2847, SB 892), municipal code updates, Title 22 regulations, FDA compliance guides, building and premises codes, parking and transportation standards, activation SOPs, and emergency preparedness plans. The Physical AI Standard Level (PSL) framework with its three regulatory dimensions is defined within this body of work. Referenced in Sections 7, 8, and 14.

Patient-Centered Documents

A Cancer Patient’s Journey, Physical AI [6]. A comprehensive paper documenting a fully autonomous single-patient journey through a regulated Physical AI oncology trial across ten stages, from prescreening through closeout. This document includes treatment outcomes, FDA cost-saving analysis, USL scoring integration, and thirty ASCII diagrams illustrating the patient pipeline. It serves as primary evidence for Physical AI feasibility and is referenced in Sections 9, 13, and 15.

Patient Instructions: Physical AI Trials [7]. A ten-page patient-facing document providing clear, accessible instructions for participants in Physical AI oncology trials. It covers interactions with ten robot types including surgical robots, cobots, radiotherapy positioning robots, social companion robots, humanoids, and rehabilitation exoskeletons. Referenced in Sections 10 and 9 as the patient communication standard.

National Infrastructure Documents

Physical AI National MCP Servers [8]. Describes the design, implementation, and testing of five national Model Context Protocol (MCP) servers for Physical AI oncology clinical trial systems. This document covers the complete server architecture, tool inventories, conformance levels, safety modules, integration adapters, and national deployment topology. Referenced in Sections 11, 12, and 14.

Physical AI Federated Learning [9]. Presents a federated learning framework for unifying Physical AI oncology trials across multiple sites while preserving patient privacy. This document covers architecture design, five foundational pillars, triple AI peer review, code trust and safety, clinical analytics, and digital twin integration. Referenced in Sections 12, 11, and 15.

Supporting Research

Research A: U.S. Government Branch Operations. A structured inventory of constitutional provisions and federal statutes governing the three branches of U.S. government, with detailed analysis of how AI in clinical trials is regulated through existing legislative, executive, and judicial frameworks. This research informs the governance structure adapted to Physical AI oncology trials in Section 2. *Source files: national-platform/research_a/01_research_a_part1.txt and 02_research_a_part2.txt.*

Research B: California and Federal Regulatory Analysis. A dual-jurisdiction analysis of California state law and federal law as they apply to AI and robotics in oncology clinical trials. This research catalogs regulatory requirements from both jurisdictions, identifies gaps and ambiguities, and informs the regulatory landscape analysis in Section 3. It also serves as fact-checking material for bill names and governance structures referenced in the Clinical Trial Site Complete Documentation. *Source files: national-platform/research_b/01_research_b_part1.txt and 02_research_b_part2.txt.*

Code Repositories

Physical AI for Oncology Clinical Trials Repository [10]. The primary repository containing all source code, simulation frameworks, orchestration modules, digital twin implementations,

regulatory compliance tools, and documentation for the Physical AI oncology trials platform.

National MCP Servers Repository [11]. The repository containing the five national MCP server implementations, integration adapters, and deployment configurations for Physical AI oncology clinical trial systems.

PAI Oncology Trial FL Repository [12]. The repository containing the federated machine learning framework, privacy modules, and multi-site coordination tools for Physical AI oncology trials.

Executive Summary

The National Platform for Physical AI Oncology Trials represents a paradigm shift in how oncology clinical trials are regulated, conducted, and scaled. Physical AI, defined as the integration of embodied robotic systems with advanced artificial intelligence for direct patient care in clinical trial settings, offers a transformative path to treat significantly more patients at lower cost and higher quality than traditional human-only trial operations. This platform provides a unified, end-to-end framework spanning regulatory adaptation, site establishment, patient care, national infrastructure, and economic analysis. Unlike existing FDA approaches that address AI and robotics through individual guidance documents issued in isolation, this National Platform simultaneously adapts three core regulatory standards, provides complete site establishment documentation, includes validated simulation evidence at both single-patient and multi-patient scales, defines quantitative robot readiness standards, and designs national infrastructure for multi-site coordination.

Regulatory Foundation. The platform adapts three internationally recognized regulatory standards to Physical AI oncology trial contexts. The Adaption: ICH Harmonised Guideline [1] establishes good clinical practice principles for Physical AI across four major sections covering principles, investigator responsibilities, sponsor responsibilities, and data governance, building on the ICH E6(R3) framework [13] recognized across dozens of countries. The Physical AI Adaption: 21 CFR Part 50 [2] provides comprehensive human subjects protection with 17 Physical AI-specific definitions, new informed consent elements for robotic procedures, and a novel Subpart C dedicated to Physical AI safety requirements. The Physical AI Adaption: 21 CFR Part 312 [3] spans ten subparts including a new Subpart J for Physical AI systems, introduces a Phase 0 simulation validation stage, and establishes reporting requirements for Physical AI adverse events and cybersecurity incidents. By building on existing, well-established regulatory frameworks rather than proposing entirely new legislation, these adaptations add immediate credibility and practical applicability to Physical AI implementations.

Quantitative Standards. Two complementary scoring frameworks ensure responsible robot usage. The Physical AI Standard Level (PSL), developed within the Clinical Trial Site Complete Documentation [5], evaluates site-level regulatory compliance across three dimensions. The Unification Standard Level (USL) [4] evaluates individual robot readiness across four dimensions: simulation framework switching, AI integration, cross-robot sharing, and clinical trial collaboration. Scores range from 1.0 to 10.0, with nine robots evaluated across three categories (cobots, surgical robots, humanoid robots). The Franka Emika Panda achieved the highest composite score (USL 7.4, Advanced), while the da Vinci dVRK led surgical robots (USL 7.1, Advanced) with the highest clinical trial readiness score of any robot evaluated. Together, PSL and USL create a dual-layer quantitative assessment that replaces subjective evaluation with reproducible, objective scoring.

Simulation Evidence. Two pivotal simulations demonstrate that state-of-the-art AI has the ability to understand, work with, and scale Physical AI applications beyond human and prior technology capabilities. A Cancer Patient’s Journey, Physical AI [6] documents a single-patient journey through all ten stages of a regulated Physical AI oncology trial, from prescreening through closeout, for a Stage IIIB non-small cell lung cancer patient over 1,120 days. The ten-stage pipeline demonstrates end-to-end Physical AI capability with documented safety outcomes, regulatory compliance across all three adapted standards, and cost savings at each

stage. The 24-hour clinical trial simulation from the Clinical Trial Site Complete Documentation [5] scaled operations to 168 patients treated with 29 robots across 15 cancer types, achieving 99.7 percent uptime with zero patient harm events and 7 adverse events documented per ICH E6(R3) standards. These simulations establish that more patients can be treated with more robots and fewer workers, validating the operational model that underpins the entire platform.

Patient-Centered Design. Control has shifted towards the patient’s side. Patient Instructions: Physical AI Trials [7] provides accessible documentation covering interactions with ten robot types. AB 2847 from the site documentation package establishes legally enforceable patient rights including the right to refuse robotic procedures, the right to human alternatives, and the right to real-time information about robot operations. The adapted 21 CFR Part 50 [2] requires ongoing consent processes beyond the traditional one-time event, giving patients continuous agency over their participation as technology may be updated and new robots may be introduced during trials. Patients benefit from convenience through reduced wait times, lower cost of treatments through reduced staffing overhead, and higher quality of care through precision robotics and continuous monitoring.

National Infrastructure. Physical AI National MCP Servers [8] provide national-scale connectivity through five servers (Governance, Data Flow, Safety, Interoperability, Analytics) with 23 tools, standardized conformance levels, and emergency safety modules. Physical AI Federated Learning [9] enables privacy-preserving multi-site coordination through five foundational pillars, a triple AI peer review pipeline using Claude Opus 4.6 [14], GPT-5.4 [15], and Gemini [16], and 82 test files ensuring code trust and safety. Together, these infrastructure components enable nationwide deployment while maintaining strict patient privacy under HIPAA [17] and state privacy laws.

Economic Impact. Physical AI trial operations offer substantial documented cost advantages. The single-patient journey documents per-patient cost reductions across all ten stages, while the 24-hour simulation demonstrates favorable scale economics at the 168-patient level. Tufts CSDD data [18] establishes that each day of delay in drug development has significant economic consequences. By compressing trial timelines through 24-hour robotic operations, automated documentation, and streamlined regulatory processes, Physical AI accelerates the time to drug approval, translating directly to earlier patient access to life-saving oncology treatments. The financial case extends beyond cost reduction to encompass faster approvals, improved data quality, and reduced regulatory review cycles.

Governance Framework. The platform is grounded in the constitutional and statutory framework of the three branches of U.S. government. Legislative authority through Congressional budget and oversight mechanisms, executive authority through FDA and HHS implementation of regulations, and judicial authority through administrative law review all apply to Physical AI oncology trial governance. Rather than creating new governance structures, this platform demonstrates how proven mechanisms ensure transparency, fiscal responsibility, and ethical conduct for Physical AI trial operations.

This National Platform is more comprehensive than existing FDA approaches because it: (a) adapts three core regulatory standards simultaneously rather than issuing piecemeal guidance, (b) provides complete site establishment documentation across eleven regulatory documents, (c) includes validated simulation evidence at both single-patient and 168-patient scales, (d) defines quantitative robot readiness standards through PSL and USL, and (e) designs national

infrastructure for multi-site coordination through MCP servers and federated learning. All source code is available across three open-source repositories [10, 11, 12] under the MIT license, enabling the pharmaceutical and regulatory industries to evaluate, adopt, and extend this framework for the advancement of oncology drug development.

1 Introduction to Physical AI Oncology Trials

1.1 Background and Motivation

Oncology drug development faces severe challenges in cost, timeline, and patient access. The Tufts Center for the Study of Drug Development [18] has quantified the value of each day of delay in drug development, demonstrating that even modest reductions in development timelines translate to significant economic and patient benefit. Current oncology clinical trials, as tracked through ClinicalTrials.gov [19] and described by the National Cancer Institute [20], operate within a framework that was designed for human-directed operations at every stage. The U.S. Food and Drug Administration provides extensive guidance on clinical trial conduct [21] and oncology-specific considerations [22], yet these resources address AI and robotics only through individual guidance documents that treat each technology in isolation.

The fundamental problem is structural: oncology trials require coordination across multiple regulatory frameworks (IND applications under 21 CFR Part 312 [23], human subjects protection under 21 CFR Part 50 [24], good clinical practice under ICH E6(R3) [13]), multiple stakeholders (sponsors, investigators, IRBs, sites, patients), and multiple technology systems (electronic data capture, imaging, laboratory information systems). Each additional technology layer increases complexity. Physical AI introduces robotic systems that perform functions traditionally reserved for human clinical staff, creating coordination challenges that exceed the capacity of piecemeal regulatory guidance.

The constitutional framework of U.S. government, spanning legislative (Article I), executive (Article II), and judicial (Article III) branches, provides the institutional infrastructure within which clinical trials operate. The Administrative Procedure Act (5 U.S.C. § 551 et seq.) governs FDA rulemaking. The Freedom of Information Act (5 U.S.C. § 552) ensures transparency. The Privacy Act (5 U.S.C. § 552a) protects individual data in federal systems. HIPAA [17] governs health information privacy and security across covered entities and business associates. These statutes were not designed with Physical AI in mind, but they provide a robust foundation upon which Physical AI regulation can be built through targeted adaptation rather than wholesale replacement.

The California and federal regulatory landscape presents a dual-jurisdiction challenge for Physical AI trial sites. California’s Protection of Human Subjects in Medical Experimentation Act (Cal. Health & Safety Code §§ 24170-24179.5), the Confidentiality of Medical Information Act (CMIA, Cal. Civ. Code §§ 56-56.37), and consumer privacy protections under CCPA/CPRA create additional obligations beyond federal requirements. Federal requirements include 21 CFR Parts 50, 56, 312, and 812 for trial conduct, 21 CFR Part 11 for electronic records, and 45 CFR Parts 46, 160, and 164 for human subjects and health information protections. Navigating this dual framework requires a comprehensive approach that no existing guidance document provides.

The evidence supporting Physical AI in oncology is substantial. The 24-hour clinical trial simulation from the Clinical Trial Site Complete Documentation [5] demonstrated that 168 patients could be treated with 29 robots across 15 cancer types at 99.7 percent uptime with zero patient harm events. This simulation established that more patients can be treated with more robots and fewer workers while maintaining safety standards documented per ICH E6(R3) requirements.

1.2 Defining Physical AI in Oncology

Physical AI is defined as the integration of embodied robotic systems with advanced artificial intelligence for direct patient care in clinical trial settings. This definition encompasses robotic systems that physically interact with patients, the AI algorithms that guide robotic operations, and the infrastructure connecting these systems to clinical trial management platforms. The adapted 21 CFR Part 312 [3] provides 22 Physical AI-specific definitions in § 312.3, establishing standardized terminology for the entire field. These definitions include Physical AI system, Physical AI adverse event, Physical AI operator, robotic procedure, autonomous mode, supervised mode, and Physical AI safety matrix, among others. The adapted 21 CFR Part 50 [2] provides 17 additional definitions in Subpart A, addressing terms specific to human subjects protection in robotic contexts.

The FDA Digital Health and Artificial Intelligence Glossary [25] provides official FDA terminology for AI and digital health technologies, but does not address the specific integration of embodied robotic systems with AI in clinical trial settings. This gap in official terminology is one reason why the adapted regulatory standards in this platform provide their own comprehensive definition sets. Physical AI extends beyond software-based AI (such as clinical decision support or medical image analysis) by incorporating physical embodiment: robots that move through clinical spaces, interact with patients through touch and proximity, and perform procedures that affect patient anatomy.

State-of-the-art AI systems that enable Physical AI development include Claude Opus 4.6 [14], GPT-5.4 [15], and Gemini [16]. These large language models serve multiple roles in Physical AI oncology trials: generating and reviewing clinical documentation, planning robotic task sequences, performing triple AI peer review of system updates, and enabling agentic AI workflows where AI systems coordinate multi-step clinical operations autonomously. The federated learning framework [9] leverages these AI systems for its triple peer review pipeline, ensuring that model updates are validated by multiple independent AI systems before deployment across trial sites.

Physical AI in oncology spans ten robot categories as documented in Patient Instructions: Physical AI Trials [7]: surgical robots (da Vinci, Hugo, Versius), collaborative robots (Franka Panda, Kinova Gen3, UFactory xArm7), radiotherapy positioning robots, needle-placement robots, social companion robots, humanoid robots (Tesla Optimus, Digit, Boston Dynamics Atlas), radiotherapy motion-tracking robots, imaging robots, steerable needle robots, and rehabilitation exoskeletons. Each category serves a distinct clinical function and is evaluated through the Unification Standard Level (USL) framework [4] across four dimensions of readiness.

1.3 The Case for a National Platform

This National Platform is more comprehensive and meaningful compared to existing FDA approaches for several structural reasons. The FDA’s approach to AI in clinical trials operates through individual guidance documents: the January 2025 draft guidance on AI for drug and biologic regulatory decision-making proposes a risk-based credibility assessment framework; the January 2025 draft on AI-enabled device software functions addresses lifecycle management; the September 2024 final guidance on decentralized trial elements covers remote monitoring; and the August 2025 final guidance on Predetermined Change Control Plans (PCCPs) addresses iterative AI device improvement. While each document is valuable individually, none addresses the integrated challenge of deploying Physical AI across an entire clinical trial lifecycle.

This platform provides five categories of capability that no existing guidance matches:

(a) three simultaneously adapted core regulatory standards (ICH E6(R3), 21 CFR Part 50, 21 CFR Part 312) that build on existing well-established frameworks and add credibility to Physical AI implementations through the Adaption: ICH Harmonised Guideline [1], Physical AI Adaption: 21 CFR Part 50 [2], and Physical AI Adaption: 21 CFR Part 312 [3]; (b) complete site establishment documentation with eleven regulatory documents covering state legislation, municipal codes, federal compliance, building standards, operational procedures, and emergency preparedness [5]; (c) validated simulation evidence at both single-patient scale (ten-stage pipeline [6]) and 168-patient scale (24-hour continuous operations [5]); (d) quantitative robot readiness standards through PSL (three dimensions for site compliance) and USL (four dimensions for robot evaluation) with scores for nine robots across three categories [4]; and (e) national infrastructure for multi-site coordination through five MCP servers [8] and federated learning [9].

The fragmentation problem in current clinical trial systems, as detailed in the Physical AI National MCP Servers paper [8], illustrates why an integrated platform is necessary. Current systems operate in silos with incompatible data formats, disconnected communication channels, and no standardized protocol for real-time inter-system coordination. Physical AI magnifies this fragmentation because robots, AI systems, clinical databases, and regulatory reporting tools must all communicate seamlessly. The five-server MCP architecture provides standardized communication pathways, while federated learning enables multi-site model improvement without centralizing patient data.

Even established institutions recognize the need for modernization. The National Cancer Institute’s modernization efforts [26] and FDA’s adaptive designs guidance [27] acknowledge that current trial infrastructure must evolve. However, none has proposed a solution as comprehensive as this National Platform, which addresses regulatory, operational, technical, and economic dimensions simultaneously.

1.4 Scope and Organization of This Document

This document comprises sixteen sections organized into a logical progression from regulatory foundations through evidence, infrastructure, economics, and implementation strategy.

Sections 1 through 3 establish the regulatory context. Section 1 (this section) introduces Physical AI oncology trials and makes the case for a national platform. Section 2 presents the U.S. government framework spanning all three branches, adapted from Research A on federal branch operations. Section 3 analyzes the dual California and federal regulatory landscape, adapted from Research B on regulatory requirements.

Sections 4 through 6 present the three adapted regulatory standards. Section 4 covers the Adaption: ICH Harmonised Guideline [1] for good clinical practice. Section 5 covers the Physical AI Adaption: 21 CFR Part 50 [2] for human subjects protection. Section 6 covers the Physical AI Adaption: 21 CFR Part 312 [3] for investigational new drug applications.

Section 7 presents the quantitative standards frameworks (PSL and USL) that evaluate site compliance and robot readiness respectively. Section 8 details the complete documentation package for clinical trial site establishment, drawing from eleven regulatory documents [5].

Sections 9 and 10 present the patient-centered evidence. Section 9 covers A Cancer Patient’s Journey, Physical AI [6] with its ten-stage pipeline. Section 10 covers Patient Instructions: Physical AI Trials [7] for ten robot types.

Sections 11 and 12 present the national infrastructure. Section 11 covers the five-server MCP architecture [8]. Section 12 covers federated learning for multi-site privacy-preserving

coordination [9].

Sections 13 and 14 address economics and implementation. Section 13 provides the financial and economic impact analysis. Section 14 presents the national implementation strategy with phased deployment.

Sections 15 and 16 synthesize and conclude. Section 15 discusses the implications, limitations, and significance of the platform. Section 16 presents key findings and a call to action. Five appendices provide reference material including the complete source file directory, glossary, regulatory cross-reference matrix, scoring reference tables, and simulation evidence summaries.

1.5 Contributions

The key contributions of this National Platform are:

1. The first comprehensive end-to-end regulatory framework for Physical AI oncology trials, integrating regulatory adaptation, site establishment, patient care, and national infrastructure into a unified platform.
2. Three simultaneously adapted core regulatory standards (ICH E6(R3), 21 CFR Part 50, 21 CFR Part 312) that build on internationally recognized frameworks and add credibility through established practice.
3. Quantitative PSL (three dimensions) and USL (four dimensions) scoring frameworks for objective, reproducible assessment of site compliance and robot readiness with scores for nine robots across three categories.
4. Validated simulation evidence at scale: a single-patient ten-stage pipeline and a 24-hour simulation treating 168 patients with 29 robots at 99.7 percent uptime and zero patient harm events.
5. Complete site establishment documentation across eleven regulatory documents covering legislation, municipal codes, regulations, compliance guides, building standards, and operational procedures.
6. National MCP server architecture with five servers, 23 tools, standardized conformance levels, and emergency safety modules for nationwide Physical AI trial connectivity.
7. Federated learning framework with five foundational pillars, triple AI peer review, and privacy-preserving multi-site model improvement under HIPAA and state privacy laws.
8. Patient-centered trial design with comprehensive rights, ten-robot patient instructions, ongoing consent processes, and demonstrated patient benefits in convenience, cost, and quality of care.
9. Economic analysis demonstrating per-patient cost savings, timeline compression, and favorable scale economics from single-site to nationwide deployment.
10. Open-source implementation across three repositories [10, 11, 12] under the MIT license, enabling community evaluation, adoption, and extension.

1.6 Historical Context of Robotics in Medicine

The integration of robotic systems into medicine began in the 1980s with early surgical robots such as the ROBODOC system for orthopedic procedures and the AESOP system for endoscopic camera holding. The da Vinci Surgical System, introduced by Intuitive Surgical in 2000, represented a transformative advance by enabling teleoperated minimally invasive surgery with enhanced dexterity and visualization. By 2026, the da Vinci platform has been deployed in over 9,000 systems worldwide, performing more than 14 million procedures across 69 countries [4].

The progression from early surgical robotics to Physical AI reflects three generational transitions. The first generation (1990s-2000s) comprised teleoperated surgical systems where robots were passive instruments responding to surgeon commands with no autonomous capability. The second generation (2010s-2020s) introduced limited autonomy through features such as tremor filtering, workspace boundary enforcement, and automated suturing sub-tasks. The third generation (2020s-present) integrates large language models, vision- language-action models, and agentic AI systems that enable robots to plan multi-step procedures, adapt to unexpected anatomical findings, and coordinate with other robotic systems autonomously.

This National Platform addresses the regulatory, standards, and infrastructure challenges created by the third generation. Traditional medical device regulation was designed for first-generation passive instruments and has been incrementally adapted for second-generation limited autonomy. Third-generation Physical AI requires a more comprehensive approach because the AI components evolve continuously through learning, the robots coordinate across systems through network protocols, and the patient interactions span the entire trial lifecycle rather than individual procedures.

1.7 Current State of Oncology Clinical Trials

Oncology clinical trials represent a significant and growing segment of the clinical research enterprise. ClinicalTrials.gov [19] lists thousands of active oncology trials at any given time, spanning Phase 1 dose-escalation studies through Phase 3 pivotal trials and Phase 4 post-market surveillance. The National Cancer Institute [20] supports a network of cancer centers and cooperative groups that conduct multi-site oncology trials across the United States and internationally.

Table 1: Traditional vs. Physical AI Oncology Trial Comparison

Parameter	Traditional Trial	Physical AI Trial
Daily operating hours	8-12 hours (staff shift-dependent)	24 hours (robot continuous operation)
Daily patient throughput	Limited by staff availability	168 patients per 24 hours demonstrated
Enrollment speed	Manual screening, weeks per patient	AI-assisted screening, days per patient
Documentation method	Manual data entry, periodic review	Real-time automated capture and validation
Safety monitoring	Periodic human review	Continuous AI-assisted monitoring
Regulatory reporting	Manual report compilation	Automated MCP-based real-time reporting
Robot involvement	None or isolated surgical assist	End-to-end across all 10 trial stages
Patient information	Standard consent forms	Ten-robot patient instructions package
Quality assessment	Periodic site audits	Continuous PSL and USL scoring
Multi-site coordination	Manual data sharing, phone/email	MCP servers and federated learning

The comparison reveals that Physical AI addresses structural limitations of traditional trials rather than incremental improvements. The shift from shift-dependent to continuous operations,

from manual to automated documentation, and from periodic to continuous monitoring represents a fundamental change in trial operations capability. The financial implications of these changes are analyzed in Section 13, while the implementation pathway is detailed in Section 14.

1.7.1 Expanded Historical Timeline of Medical Robotics

The historical progression from early stereotactic frames to contemporary Physical AI ecosystems can be understood through five distinct eras, each defined by the relationship between the robot, the clinician, and the patient. Mapping these eras clarifies why existing regulatory frameworks require the comprehensive adaptation provided by this National Platform.

The Pre-Clinical Era (1960s-1980s) was characterized by laboratory experimentation with robotic manipulators originally designed for industrial use. The Unimate, the first industrial robot installed on a General Motors assembly line in 1961, established foundational principles of programmable motion control that would eventually migrate to surgical applications. Research laboratories at Johns Hopkins University, the Jet Propulsion Laboratory, and Imperial College London explored how robotic precision could augment surgical capability. During this era, no regulatory framework for medical robotics existed because no robotic system had been proposed for clinical use. The regulatory vacuum of this era contrasts sharply with the comprehensive framework available today through this National Platform.

The Pioneering Surgical Era (1985-1999) began with the PUMA 560 brain biopsy and progressed through increasingly sophisticated surgical applications. The ROBODOC system for hip replacement surgery, developed by Integrated Surgical Systems, achieved CE marking in Europe and underwent extensive clinical evaluation. ROBODOC demonstrated both the potential and the pitfalls of surgical robotics: while the system achieved superior bone preparation accuracy compared to manual techniques, post-operative complications at some sites raised questions about patient selection, surgeon training, and system reliability. These early experiences informed the safety-first approach that the adapted ICH E6(R3) [1] codifies through its eight foundational principles and twelve investigator responsibility categories.

The AESOP (Automated Endoscopic System for Optimal Positioning) system, cleared by the FDA in 1994, established the regulatory pathway for robotic devices in the operating room. AESOP was classified as a Class II medical device through the 510(k) premarket notification process, with the predicate device being a manual endoscope holder. This classification decision established the principle that robotic systems could be regulated through existing device categories with appropriate modifications, a principle that this National Platform extends to the clinical trial context through the adapted 21 CFR Part 312 [3].

The Teleoperation Era (2000-2014) was dominated by the da Vinci Surgical System's rapid adoption. The da Vinci Xi, da Vinci Si, and subsequent models established teleoperation as the standard paradigm for robotic surgery: a surgeon sits at a console and manipulates master controllers while the robot replicates those movements at the patient side with enhanced precision and reduced tremor. The da Vinci platform's commercial success demonstrated that hospitals would invest in robotic infrastructure when clinical benefits could be demonstrated, and that patients would accept robotic involvement in their surgical care when properly informed.

During this era, surgical robotics expanded into oncology through procedures such as robot-assisted radical prostatectomy, robot-assisted hysterectomy for endometrial cancer, and robot-assisted lobectomy for lung cancer. However, these applications were limited to the surgical procedure itself. The robot played no role in patient screening, treatment planning, drug administration, post-operative monitoring, or data collection. Physical AI, by contrast,

spans the entire clinical trial lifecycle, as documented in the ten-stage patient journey [6].

The Semi-Autonomous Era (2015-2023) introduced AI-driven capabilities that extended robotic function beyond direct human teleoperation. Features such as automated suturing, tissue recognition, and anatomy-based navigation allowed robots to perform certain sub-tasks with reduced human input. The Mako system for orthopedic surgery incorporated pre-operative CT-based planning with intraoperative haptic guidance, creating a hybrid model where the robot constrained the surgeon's movements to a pre-approved plan. The CyberKnife system for stereotactic radiosurgery demonstrated autonomous real-time tracking of tumor motion during radiation delivery, adjusting beam targeting without human intervention. These semi-autonomous capabilities raised regulatory questions about the boundary between device-assisted human performance and autonomous machine performance, questions that the Physical AI operator definition in the adapted 21 CFR Part 312 [3] directly addresses.

The Physical AI Era (2024-present) represents the convergence of large language models, vision-language-action models, advanced sensor systems, and multi-robot coordination into integrated clinical platforms. This era is distinguished by four characteristics that differentiate it from all previous eras. First, AI systems can now interpret complex clinical scenarios and generate actionable plans, not merely execute pre-programmed motions. Second, multiple robots can coordinate their actions through standardized communication protocols such as the Model Context Protocol [28] implemented through the five-server MCP architecture [8]. Third, robotic systems can learn and improve through federated approaches that preserve patient privacy while enabling multi-site knowledge sharing [9]. Fourth, the complete clinical trial lifecycle, from screening through closeout, can be supported by integrated Physical AI systems rather than isolated robotic interventions.

1.7.2 Lessons from Historical Adoption Patterns

The historical adoption of medical robotics provides instructive lessons for Physical AI deployment. Each previous generation of medical robotics followed a similar adoption curve: initial skepticism from the medical community, early adoption by technology-oriented institutions, gradual accumulation of clinical evidence, regulatory accommodation through existing frameworks, and eventual mainstream acceptance. Physical AI is currently at the early stage of this curve, where comprehensive evidence and regulatory frameworks are the primary enablers of adoption.

The da Vinci adoption experience is particularly instructive. The system achieved rapid adoption not primarily through demonstrated clinical superiority (early randomized trials showed equivalent oncologic outcomes to conventional surgery) but through a combination of patient demand, institutional prestige, surgeon enthusiasm for enhanced visualization, and marketing effectiveness. This pattern suggests that Physical AI adoption in oncology trials will depend not only on demonstrated efficiency gains but also on patient acceptance, investigator confidence, and institutional readiness. The Patient Instructions framework [7] addresses patient acceptance, the adapted ICH E6(R3) [1] addresses investigator competency, and the PSL scoring framework [4] addresses institutional readiness.

Historical adoption also reveals the importance of training infrastructure. The da Vinci system required the development of simulation-based training curricula, proctoring programs, and credentialing standards before widespread surgical adoption. The Physical AI training requirements in the adapted ICH E6(R3) [1] reflect this lesson by mandating Physical AI-specific competencies for all trial personnel. The Phase 0 simulation validation requirement in the

adapted 21 CFR Part 312 [3] extends the training principle to the system level, requiring that Physical AI systems demonstrate competency through simulation before human subjects interaction.

1.7.3 Traditional Trial Phase Parameters and Physical AI Projections

To supplement the comparison in Table 1, a more granular analysis of trial phase parameters reveals the specific mechanisms through which Physical AI improves trial operations. Table 2 presents detailed phase-by-phase parameters including enrollment rates, monitoring frequency, and data volume, with corresponding Physical AI projections.

Table 2: Detailed Phase Parameters: Traditional vs. Physical AI Trials

Parameter	Phase I	Phase II	Phase III	Phase IV
<i>Traditional Trial Parameters</i>				
Typical duration	6-12 months	12-24 months	24-48 months	24-60 months
Patient population	20-80	100-300	1,000-3,000	1,000-10,000
Estimated per-patient cost	\$50K-100K	\$40K-80K	\$30K-60K	\$10K-30K
Total estimated cost	\$1M-8M	\$4M-24M	\$30M-180M	\$10M-300M
Enrollment rate (patients/month)	3-8	10-25	30-80	50-200
Monitoring visits/year	6-12	4-8	4-6	2-4
Data points per patient	500-2,000	1,000-5,000	2,000-8,000	500-3,000
Query resolution time (days)	7-21	7-21	14-30	14-30
Screen failure rate (%)	25-40	20-35	15-30	10-20
<i>Physical AI Projected Parameters</i>				
Duration reduction	20-30%	25-35%	30-40%	15-25%
Enrollment rate increase	40-60%	50-80%	60-100%	30-50%
Cost per patient reduction	15-25%	20-35%	25-40%	15-30%
Continuous monitoring	Yes (24/7)	Yes (24/7)	Yes (24/7)	Yes (24/7)
Data points per patient	10,000-50,000	20,000-100,000	50,000-200,000	10,000-50,000
Query resolution time	Real-time	Real-time	Real-time	Real-time
Screen failure reduction	30-50%	30-50%	30-50%	20-40%
System uptime	99.7%	99.7%	99.7%	99.7%

The data volume increase from Physical AI trials is particularly noteworthy. Traditional Phase III trials generate 2,000 to 8,000 data points per patient, primarily from scheduled clinical assessments, laboratory results, and imaging studies. Physical AI trials are projected to generate 50,000 to 200,000 data points per patient in Phase III, reflecting continuous sensor monitoring, real-time robot telemetry, AI decision logs, and automated clinical assessments. This approximately 25-fold increase in data density presents both opportunities and challenges. The opportunities include earlier detection of safety signals, more precise efficacy measurement, and richer datasets for biomarker discovery. The challenges include data storage and management at scale, statistical methods for high-dimensional continuous data, and regulatory review of vastly larger submission packages.

The MCP server infrastructure [8] addresses the data management challenge through its five-server architecture, with dedicated servers for data flow management, analytics processing,

and governance oversight. The federated learning framework [9] addresses the multi-site data challenge by enabling model training across distributed datasets without centralized data aggregation. Together, these infrastructure components ensure that the increased data volume from Physical AI trials can be managed, analyzed, and submitted within existing regulatory timelines.

Screen failure reduction represents another significant Physical AI advantage. Traditional oncology trials experience screen failure rates of 15 to 40 percent, meaning that a substantial proportion of patients who undergo screening assessments ultimately do not meet eligibility criteria. Each screen failure represents wasted clinical resources, patient disappointment, and delayed enrollment. Physical AI systems can perform preliminary eligibility assessments using AI-driven analysis of medical records, laboratory data, and imaging studies before scheduling resource-intensive in-person screening visits. The patient journey framework [6] documents how this pre-screening capability integrates into the overall trial workflow, reducing screen failures by 30 to 50 percent in projected models.

The query resolution time improvement from Physical AI trials reflects the elimination of the traditional query resolution cycle, where monitors identify data discrepancies during periodic site visits, generate queries, transmit them to site staff, and wait for resolution. In traditional trials, this cycle takes 7 to 30 days per query. Physical AI systems resolve data quality issues in real time through automated validation checks at the point of data capture, eliminating the query cycle entirely. The adapted ICH E6(R3) [1] establishes the quality management framework for this real-time data validation, while the MCP governance server [8] provides the infrastructure for automated compliance verification.

The cumulative effect of these phase-by-phase improvements is a fundamental transformation in the economics and logistics of oncology clinical trials. A Phase III oncology trial that would traditionally require 36 months and \$100 million could potentially be completed in 20 to 25 months at \$60 to 75 million with Physical AI integration, while generating an order of magnitude more data per patient and maintaining or improving safety standards. The 24-hour simulation evidence [5] provides the proof of concept for these projections, demonstrating that the operational model is feasible at the scale required for pivotal oncology trials.

2 U.S. Government Framework for Physical AI Trials

Physical AI oncology trials must be understood within the full constitutional and statutory framework of U.S. government, spanning the Legislative, Executive, and Judicial branches. The governance structure for Physical AI builds on existing federal branch operations rather than requiring new institutional frameworks, leveraging established institutional credibility and legal authority. This section adapts the comprehensive research on U.S. federal branch operations statutes and their application to AI-enabled clinical trials to Physical AI oncology trial contexts.

2.1 Constitutional Foundation

The U.S. federal government's three-branch structure is established primarily by the U.S. Constitution. Article I vests legislative powers in a Congress consisting of a Senate and House of Representatives. Article II vests executive power in a President, with the Appointments Clause establishing Senate advice-and-consent for officers and the Take Care duty (Article II, Section 3) providing the constitutional anchor for executive implementation and supervision of federal law. Article III vests the judicial power in one Supreme Court and such inferior courts as Congress may establish.

For Physical AI oncology trials, this three-branch structure provides essential checks and balances. Legislative authority creates the statutory framework through which Physical AI trial programs are funded and overseen. Executive authority implements that framework through FDA and HHS, which directly regulate clinical trial conduct. Judicial authority resolves disputes and interprets the boundaries of regulatory action. The Congressional Review Act (5 U.S.C. § 801 et seq.) provides legislative oversight of executive rulemaking, meaning that any FDA regulations specific to Physical AI would be subject to congressional review. The Administrative Procedure Act (5 U.S.C. § 551 et seq.) requires that Physical AI regulations follow standard notice-and-comment rulemaking procedures, ensuring public participation and transparency.

The Appointments Clause has practical implications for Physical AI governance: FDA commissioners and senior officials who would oversee Physical AI trial approvals are appointed through the constitutionally mandated process, ensuring democratic accountability for regulatory decisions affecting patient safety. The Take Care duty requires the executive branch to faithfully implement clinical trial statutes, preventing either over-regulation that would block beneficial Physical AI innovations or under-regulation that would compromise patient safety.

2.2 Legislative Branch and Physical AI Regulation

The legislative branch's role in Physical AI regulation operates through several operational statutes. The Congressional Budget and Impoundment Control Act of 1974 (codified across titles, with CBO at 2 U.S.C. § 601) creates the Congressional Budget Office and statutory budget procedures that structure annual budgeting and congressional scorekeeping. FDA funding for Physical AI trial review capacity depends directly on congressional appropriations authority exercised through these budget procedures.

The Congressional Review Act (5 U.S.C. § 801 et seq.) establishes a mechanism for Congress to review and disapprove certain agency rules under defined procedures. If FDA were to promulgate rules specifically governing Physical AI clinical trials, these rules would be subject to congressional review, providing a check on executive rulemaking power. The Congressional Accountability Act of 1995 (2 U.S.C. ch. 24, § 1301; Pub. L. 104-1) applies workplace and labor

protections to legislative-branch employing offices, establishing precedent for how new categories of workers (including those overseeing Physical AI systems) receive legislative protection.

The Government Accountability Office (GAO), with Comptroller General authorities established under 31 U.S.C. § 702, provides audit and investigation capacity for oversight of Physical AI trial programs. GAO audits of FDA’s review processes for AI-enabled medical products would apply equally to Physical AI trial oversight, ensuring that congressional appropriations are used effectively and that regulatory processes meet statutory requirements.

Congressional appropriations directly affect FDA’s capacity to review and approve Physical AI trial applications. If Physical AI trials require new categories of expertise (robotics engineers, AI safety specialists, cybersecurity analysts) within FDA review teams, congressional funding must support these positions. The budget process statutes ensure that these resource needs are evaluated through established scorekeeping procedures rather than ad hoc allocation.

Table 3: Core Legislative Statutes Applicable to Physical AI Trial Governance

Statute	Physical AI Relevance	Citation
Budget and Impoundment Control Act	FDA funding for Physical AI review capacity	2 U.S.C. § 601
Congressional Review Act	Review of Physical AI-specific FDA rules	5 U.S.C. § 801
Congressional Accountability Act	Worker protections for Physical AI oversight staff	2 U.S.C. § 1301
GAO/Comptroller General	Audit of Physical AI trial programs	31 U.S.C. § 702
Reapportionment Act	House composition for Physical AI policy votes	2 U.S.C. § 2a

2.3 Executive Branch Implementation

The executive branch implements clinical trial regulation through FDA and HHS, with several foundational statutes governing the process. The Administrative Procedure Act (APA, Pub. L. 79-404; 5 U.S.C. § 551 et seq.) defines “agency” and related terms, forming the baseline procedural code for federal administrative action and review. FDA’s authority to regulate Physical AI trials derives from the APA’s rulemaking framework: any Physical AI-specific regulations would follow notice-and-comment procedures, with judicial review available for arbitrary or capricious agency action.

The Freedom of Information Act (FOIA, Pub. L. 89-487; 5 U.S.C. § 552) creates enforceable disclosure duties subject to enumerated exemptions. For Physical AI trials, FOIA ensures that FDA review processes, approval decisions, and safety data are accessible to the public, researchers, and industry participants. The transparency provided by FOIA supports public trust in Physical AI adoption by making regulatory decision-making visible and accountable.

The Privacy Act of 1974 (Pub. L. 93-579; 5 U.S.C. § 552a) establishes restrictions on disclosure and requirements for systems of records maintained by federal agencies. When FDA maintains records about Physical AI trial sponsors, investigators, or participants, the Privacy Act governs how those records may be used and disclosed. This statute includes criminal penalties for certain willful violations, providing strong incentives for proper data handling.

The Federal Register Act (44 U.S.C. § 1505) requires publication of presidential proclamations and executive orders with general applicability and legal effect. Any executive orders directing

FDA to prioritize Physical AI trial review or establishing Physical AI governance frameworks must be published in the Federal Register to have legal effect. The Paperwork Reduction Act (44 U.S.C. § 3501) requires OMB review of information collections, meaning that new data collection requirements for Physical AI trial sponsors must go through OMB clearance.

The Inspectors General framework (5 U.S.C. ch. 4) requires IG reporting structures within executive agencies, providing internal oversight and audit capability for Physical AI trial programs. The Federal Vacancies Reform Act of 1998 (5 U.S.C. § 3345) establishes rules for acting officials, ensuring continuity of FDA leadership during transitions that might affect Physical AI regulatory decisions.

Table 4: Core Executive Statutes Applicable to Physical AI Trial Operations

Statute	Physical AI Relevance	Citation
Administrative Procedure Act	Rulemaking for Physical AI regulations	5 U.S.C. § 551
Freedom of Information Act	Transparency of Physical AI trial decisions	5 U.S.C. § 552
Privacy Act of 1974	Protection of Physical AI trial records	5 U.S.C. § 552a
Federal Register Act	Publication of Physical AI executive orders	44 U.S.C. § 1505
Paperwork Reduction Act	OMB review of Physical AI data collection	44 U.S.C. § 3501
Inspectors General	Internal audit of Physical AI programs	5 U.S.C. ch. 4
Federal Vacancies Reform	Continuity of FDA Physical AI leadership	5 U.S.C. § 3345
Presidential Records Act	Preservation of Physical AI policy records	44 U.S.C. § 2201

The three adapted regulatory standards in this platform demonstrate how existing executive branch regulatory frameworks can be adapted rather than replaced. The Adaption: ICH Harmonised Guideline [1] builds on the internationally recognized GCP standard. The Physical AI Adaption: 21 CFR Part 50 [2] extends existing human subjects protection regulation. The Physical AI Adaption: 21 CFR Part 312 [3] extends the IND application regulation. Each adaptation operates within the APA’s rulemaking framework, demonstrating that Physical AI can be regulated through existing administrative procedures.

2.4 Judicial Branch Oversight

The judicial branch provides the framework for resolving Physical AI regulatory disputes. The modern statutory backbone for court organization, jurisdiction, and judicial administration is established in Title 28 of the U.S. Code. The Supreme Court size and quorum are set by 28 U.S.C. § 1, with circuit composition defined by 28 U.S.C. § 41 (currently thirteen circuits including the Federal Circuit). The Judicial Conference (28 U.S.C. § 331) serves as the judiciary’s policy-making body, coordinating administration across all federal courts.

The Rules Enabling Act (28 U.S.C. § 2072) authorizes the Supreme Court to prescribe general rules of practice, procedure, and evidence, while prohibiting rules that abridge, enlarge, or modify substantive rights. These procedural rules govern how Physical AI regulatory disputes would be litigated, from initial filing through appeal. The judicial discipline and complaints process (28 U.S.C. § 351) ensures accountability for judges who handle Physical AI cases.

Courts would review FDA decisions regarding Physical AI trial approvals, safety actions, and clinical holds under existing administrative law frameworks. The APA provides for judicial review of agency action, with courts evaluating whether FDA’s Physical AI decisions are arbitrary, capricious, an abuse of discretion, or otherwise not in accordance with law. This standard

of review means that FDA must articulate reasoned explanations for Physical AI regulatory decisions, which in turn requires the kind of comprehensive evidence base that this National Platform provides.

If FDA were to place a clinical hold on a Physical AI trial, the sponsor could challenge that decision through administrative channels and ultimately through judicial review. If FDA were to deny an IND application that included Physical AI components, the denial would be subject to review under the same standards that apply to all IND decisions. The existing judicial framework thus provides a safety valve against both over-regulation and under-regulation of Physical AI trials, ensuring that regulatory decisions are grounded in evidence rather than speculation.

2.5 Cross-Branch Governance for Physical AI

Several statutes operate across all three branches to ensure accountability in Physical AI trial governance. Transparency and recordkeeping requirements include Federal Register publication requirements (44 U.S.C. § 1505), FOIA public access provisions (5 U.S.C. § 552), agency record preservation obligations (44 U.S.C. § 3101), and presidential record definitions (44 U.S.C. § 2201). Together, these ensure that Physical AI policy decisions are documented, preserved, and accessible for public scrutiny.

Budget execution constraints apply through the Antideficiency Act (31 U.S.C. § 1341), which prohibits obligations and expenditures exceeding available appropriations. This means that Physical AI trial programs cannot spend beyond congressional authorization, providing fiscal discipline for potentially expensive national deployment efforts. The Ethics in Government Act framework establishes disclosure and ethics requirements spanning all branches, ensuring that officials involved in Physical AI regulatory decisions are subject to conflict-of-interest protections and financial disclosure requirements.

Physical AI oncology trials benefit from the full weight of these existing cross-branch governance mechanisms. Rather than creating new governance structures, this platform adapts proven mechanisms to ensure transparency, fiscal responsibility, and ethical conduct. This approach is consistent with the platform's broader philosophy of building on established frameworks rather than replacing them.

2.6 Current Regulatory Framework for AI in Clinical Trials

AI use in clinical trials is regulated primarily through what the AI does rather than through a dedicated "AI clinical trials" statute. FDA guidance distinguishes four categories of AI use in trial contexts. First, AI used to generate information or data intended to support regulatory decision-making for drugs and biologics falls under FDA's January 2025 draft guidance, which proposes a risk-based credibility assessment framework with seven steps: define question of interest, define context of use, assess model risk, plan, execute, document, and determine adequacy. Second, AI embedded in or functioning as medical device software falls under device lifecycle management guidance and PCCP requirements. Third, AI used as part of trial conduct and data systems must comply with 21 CFR Part 11 electronic records requirements and related computerized systems guidance. Fourth, AI as human subjects research and data risk falls under Common Rule and IRB review requirements.

Table 5: Federal Regulations and Guidances Relevant to Physical AI Trials

Type	Citation	Physical AI Relevance
Regulation	21 CFR Part 312	IND for trials with Physical AI drug administration
Regulation	21 CFR Part 50	Informed consent for Physical AI robotic procedures
Regulation	21 CFR Part 56	IRB review of Physical AI system risks
Regulation	21 CFR Part 812	IDE for Physical AI device investigations
Regulation	21 CFR Part 11	Electronic records from Physical AI systems
Regulation	42 CFR Part 11	Trial registration including Physical AI methods
Regulation	45 CFR Part 46	Common Rule for Physical AI human subjects research
Regulation	45 CFR Parts 160, 164	HIPAA for Physical AI health data
Guidance	AI Drug/Biologic (Jan 2025)	AI credibility framework for Physical AI evidence
Guidance	AI Device Lifecycle (Jan 2025)	Lifecycle management for Physical AI software
Guidance	Decentralized Trials (Sep 2024)	Remote Physical AI monitoring elements
Guidance	PCCP for AI Devices (Aug 2025)	Change control for Physical AI system updates
Guidance	CDS Software (Jan 2026)	Classification of Physical AI decision support

Physical AI introduces additional considerations across all four categories. When Physical AI systems generate evidence for regulatory decisions (such as robotic surgical outcomes data), the credibility assessment must account for robot-specific variables including mechanical precision, sensor accuracy, and autonomous decision quality. When Physical AI functions as a regulated device, the lifecycle management must address hardware degradation, calibration drift, and physical wear in addition to software updates. When Physical AI generates electronic trial records (robot telemetry, sensor data, autonomous decision logs), these records represent new categories that must comply with Part 11 requirements. When Physical AI interacts directly with human subjects, the risk profile differs fundamentally from software-only AI because of physical contact and proximity.

2.7 AI and Oncology: A High-Frequency Use Case

The FDA Oncology Center of Excellence [22] describes an Oncology AI Program begun in 2023, linking AI-related FDA guidances used in oncology regulatory contexts. In oncology trials specifically, AI is frequently implicated because endpoints and eligibility often rely on imaging (CT, MRI, PET), digital pathology, complex biomarker patterns, and longitudinal EHR-derived outcomes, all of which are common AI input modalities.

Physical AI extends these existing AI applications by adding robotic embodiment. Rather than AI analyzing images to inform human decisions, Physical AI systems can directly act on imaging analysis by guiding surgical robots to tumor sites, positioning radiotherapy beams with sub-millimeter precision, or directing needle-placement robots to biopsy targets. A Cancer Patient’s Journey, Physical AI [6] documents how this integrated approach works across all ten stages of an oncology trial, from prescreening through closeout. The ten-stage pipeline demonstrates that Physical AI is not limited to isolated procedures but can manage the complete patient experience.

2.8 Practical Compliance Synthesis

In oncology trials using Physical AI, compliance requires building a defensible chain from context of use through validation, auditability, and human subjects protection.

If Physical AI outputs will be used as evidence for FDA decisions about safety, effectiveness, or quality, sponsors should document the model’s context of use, risk, and credibility assessment consistent with FDA’s draft framework. The adapted 21 CFR Part 312 [3] provides the IND framework for including Physical AI evidence in regulatory submissions, while the adapted ICH E6(R3) [1] establishes the quality management system for ensuring Physical AI data integrity.

If Physical AI functions as a regulated device, the trial may implicate IDE and device marketing pathways, with lifecycle management and PCCP expectations governing what changes are allowed and how they are controlled. The USL scoring framework [4] provides quantitative assessment of each robot’s readiness for clinical deployment, with Dimension D (clinical trial collaboration) directly measuring regulatory compliance status.

If Physical AI relies on decentralized or digital health technology data, the data attribution, metadata integrity, and Part 11 boundaries become critical compliance points. The MCP server infrastructure [8] addresses this through standardized data flow pathways and conformance level requirements that ensure data integrity across distributed Physical AI systems.

If Physical AI training and validation rely on secondary data and research exemptions, IRB considerations around identifiability and bias must be addressed. The federated learning framework [9] provides privacy-preserving approaches to model training that reduce identifiability risks while enabling multi-site data utilization under HIPAA [17] and state privacy law requirements.

The adapted 21 CFR Part 50 [2] ensures that human subjects protection extends to all Physical AI interactions, with informed consent requirements covering robot descriptions, AI decision-making transparency, patient rights to refuse robotic procedures, and emergency stop procedures. Together, the three adapted standards provide comprehensive responses to each compliance pathway, demonstrating that Physical AI can operate within the existing regulatory architecture with targeted adaptations.

3 California and Federal Regulatory Landscape

California serves as the site of the first Physical AI oncology clinical trial, making a dual-jurisdiction analysis of state and federal law essential for understanding the complete regulatory environment. This section adapts the comprehensive California and federal legal framework analysis to Physical AI oncology trial contexts, providing the regulatory map that trial sponsors, investigators, and sites must navigate.

3.1 Regulatory Classification Framework

The regulatory landscape for Physical AI in oncology trials can be organized along two classification axes. The first axis distinguishes between AI systems that qualify as regulated medical products and those that function as research operational tools without direct medical product classification. The second axis, applicable to device-classified Physical AI systems, distinguishes between investigational and approved status, and between significant risk and non-significant risk designations.

The federal device definition under 21 U.S.C. § 321(h) encompasses instruments, machines, and other articles intended for use in diagnosis, treatment, or prevention of disease, which includes many Physical AI system components. The investigational device exemption authority under 21 U.S.C. § 360j(g) establishes procedures for device use during clinical investigation. The FD&C Act § 520(o) software exclusions from the 21st Century Cures Act exclude certain software functions from the device definition, but these exclusions were designed primarily for software-only clinical decision support and may not encompass the full range of Physical AI functionalities that involve direct physical patient interaction.

Physical AI systems present classification challenges because a single system may simultaneously occupy multiple categories. A surgical robot performing a tumor resection functions as a medical device. The AI algorithm guiding its surgical decisions may qualify as software as a medical device (SaMD). The data collection and transmission functions may be classified as electronic data systems under Part 11. The robot's interaction with the patient triggers human subjects protection requirements. This multi-category nature underscores the need for the integrated regulatory approach provided by this National Platform.

3.2 California Regulatory Authority

3.2.1 Human Subjects Protections

California's Protection of Human Subjects in Medical Experimentation Act (Cal. Health & Safety Code §§ 24170-24179.5) establishes protections for research participants that supplement federal requirements. The Experimental Subject's Bill of Rights (§ 24172) enumerates specific rights that must be communicated to research participants, creating obligations beyond federal informed consent requirements. The definition of "medical experiment" (§ 24174) is broad enough to encompass Physical AI procedures where robots perform functions that constitute medical experimentation.

For Physical AI trial participants in California, these provisions create additional protections. The Experimental Subject's Bill of Rights requires disclosure of the nature of the experiment (including robotic involvement), risks and discomforts (including robot malfunction scenarios), and the right to withdraw. The Clinical Trial Site Complete Documentation [5] incorporates

these California-specific requirements into its documentation package, and AB 2847 from that package establishes additional patient rights specific to Physical AI interactions.

3.2.2 Health Data Privacy and Security

The Confidentiality of Medical Information Act (CMIA, Cal. Civ. Code §§ 56-56.37) governs the handling of medical information by healthcare providers and other entities. Electronic medical information safeguards (§ 56.101) require specific protections for electronically maintained medical data, directly applicable to the digital records generated by Physical AI systems. California's data breach notification requirements (Cal. Civ. Code § 1798.82) and medical information breach requirements (Cal. Health & Safety Code § 1280.15) impose notification obligations when Physical AI system security breaches compromise patient data.

Physical AI systems generate new categories of health data including robot telemetry, sensor readings, video recordings of procedures, and AI decision logs. CMIA requirements apply to these data categories when they constitute medical information, creating compliance obligations that interact with federal HIPAA [17] protections. The Physical AI Federated Learning framework [9] addresses these dual privacy requirements through privacy-preserving approaches that minimize data centralization.

3.2.3 Consumer Privacy

CCPA/CPRA (Cal. Civ. Code §§ 1798.100 et seq.) provides broad consumer privacy protections with specific exemptions for clinical trial and biomedical research data. Clinical trial data may be statutorily exempt in many common trial contexts, but non-exempt data streams from Physical AI systems may still trigger CCPA obligations. For example, if a Physical AI system collects data about a patient's movements or preferences that falls outside the clinical trial data exemption, CCPA rights to access, deletion, and opt-out would apply.

3.2.4 AI-Specific Healthcare Communications

AB 3030 (Cal. Health & Safety Code § 1339.75) requires disclosure when AI-generated content is used in patient clinical communications. AB 489 addresses health advice from artificial intelligence. Physical AI systems that communicate with patients, whether through social companion robots providing emotional support or through AI-generated treatment explanations, must comply with these disclosure requirements. The patient instructions documented in Patient Instructions: Physical AI Trials [7] incorporate appropriate disclosures for each of the ten robot types.

3.2.5 Medical Practice and Licensing

California's unlicensed practice of medicine provisions (Cal. Bus. & Prof. Code § 2052) raise questions about Physical AI systems that perform functions traditionally reserved for licensed practitioners. When a surgical robot performs tissue manipulation or a needle-placement robot conducts a biopsy, the boundary between machine-assisted medical practice and autonomous machine practice becomes legally significant. The adapted 21 CFR Part 312 [3] addresses this through its Physical AI operator definition, which establishes that qualified human oversight must be maintained even when robots perform physical procedures.

3.3 Federal Regulatory Authority

3.3.1 Investigational Product Frameworks

The IND regulations (21 CFR Part 312) [23] govern investigational drug use in clinical trials. When Physical AI systems are used alongside investigational drugs, the IND framework must accommodate the additional complexity of robotic drug administration, AI-guided dosing, and automated safety monitoring. The Physical AI Adaption: 21 CFR Part 312 [3] provides the adapted framework spanning ten subparts, including the Phase 0 simulation validation stage that requires demonstration of Physical AI system safety before human subjects enrollment.

IDE regulations (21 CFR Part 812) govern investigational device use. When Physical AI systems themselves qualify as investigational devices, sponsors must obtain IDE approval or qualify for exemption. The interaction between IND and IDE requirements becomes particularly complex when Physical AI systems are used to administer investigational drugs, potentially triggering both regulatory pathways simultaneously.

3.3.2 Human Subjects Protection

Federal human subjects protection operates through 21 CFR Part 50 [24] (FDA-specific protections), 21 CFR Part 56 (IRB requirements), and the Common Rule (45 CFR Part 46). Physical AI adds new dimensions to informed consent because participants must understand their interactions with robotic systems, AI decision-making processes, safety monitoring systems, and their right to refuse or pause robotic procedures. The Physical AI Adaption: 21 CFR Part 50 [2] addresses these dimensions through 17 Physical AI-specific definitions and a new Subpart C dedicated to Physical AI safety requirements.

The NCI Central Institutional Review Board [29] provides a model for centralized IRB review that could be adapted for Physical AI trial oversight, enabling consistent evaluation of Physical AI risks and protections across multiple trial sites.

3.3.3 Electronic Records and Data Integrity

21 CFR Part 11 governs electronic records and electronic signatures in FDA-regulated activities. Physical AI systems generate, store, and transmit electronic clinical trial records including robot telemetry, sensor data, autonomous decision logs, imaging data, and treatment records. These constitute new categories of electronic records that must comply with Part 11 requirements for trustworthiness and reliability.

Interoperability standards including FHIR [30] for healthcare data exchange and DICOM [31] for medical imaging provide standardized formats for some Physical AI data streams. The MCP server infrastructure [8] extends these standards by providing standardized communication protocols for Physical AI-specific data that existing standards do not cover, including real-time robot telemetry, task order management, and safety interlock states.

3.3.4 Privacy and Security

HIPAA Privacy and Security Rules (45 CFR Parts 160 and 164) [17] govern the use and disclosure of protected health information (PHI) by covered entities and business associates. Physical AI systems handling PHI must comply with HIPAA requirements, creating obligations for data encryption, access controls, audit trails, and breach notification. The federated learning

framework [9] and federated learning in healthcare [32] provide privacy-preserving approaches that reduce the need to centralize PHI while enabling multi-site model improvement.

3.4 Robotics-Specific Regulatory Considerations

FDA’s position on robotically-assisted surgical (RAS) systems in oncology presents a specific regulatory consideration. RAS systems have not received marketing authorization specifically for cancer prevention or treatment, although they are used in oncology-related procedures under existing clearances. The significance of oncology-related endpoints (recurrence, disease-free survival, overall survival) for Physical AI trials means that trial designs must demonstrate not just procedural safety but also oncology outcome equivalence or superiority.

The Physical AI Adaption: 21 CFR Part 312 [3] addresses this through its Phase 0 simulation validation requirement, which allows Physical AI systems to demonstrate safety and preliminary efficacy through simulation before any real patient contact. The 24-hour simulation evidence from the Clinical Trial Site Complete Documentation [5] provides a model for how such Phase 0 validation could be conducted at scale.

3.5 Gaps and Opportunities

Several regulatory gaps and ambiguities exist for Physical AI implementations that this National Platform addresses. First, the lack of specific FDA guidance for Physical AI clinical trials means that sponsors must navigate multiple existing guidances and regulations without a unified reference. This platform provides that unified reference through its comprehensive body of adapted standards and documentation.

Second, jurisdictional overlaps between California and federal requirements create compliance complexity, particularly for data privacy where CMIA, CCPA, and HIPAA all may apply to the same data. The site establishment documentation [5] addresses this by integrating both jurisdictions into a single compliance framework.

Third, classification challenges arise for multi-function Physical AI systems that simultaneously qualify as medical devices, electronic data systems, and research tools. The adapted regulatory standards provide classification guidance, with the 21 CFR Part 312 adaptation [3] establishing which Physical AI functions require IND coverage and which fall under device or operational tool categories.

Fourth, workforce transition considerations must be addressed as Physical AI enables treating more patients with more robots and fewer workers. California labor law, federal employment regulations, and healthcare staffing requirements all intersect with the workforce implications of Physical AI adoption. The implementation strategy in Section 14 addresses these workforce considerations through phased deployment with training and role evolution programs.

Table 6: Regulatory Gaps and National Platform Solutions

Regulatory Gap	National Platform Solution
No unified Physical AI trial guidance	Three adapted standards (ICH, CFR 50, CFR 312)
Dual-jurisdiction compliance complexity	Integrated site documentation package
Multi-function classification challenges	Physical AI definitions across all adaptations
Workforce transition regulatory uncertainty	Phased implementation with training programs
Absence of Physical AI safety standards	PSL (3 dimensions) and USL (4 dimensions)
No multi-site Physical AI infrastructure	MCP servers and federated learning

The regulatory landscape for Physical AI oncology trials is complex but navigable. By building on existing frameworks rather than creating entirely new ones, the adapted standards in this platform [1, 2, 3] provide a credible and practical path forward that the pharmaceutical and regulatory industries can adopt with confidence.

4 Adapted ICH E6(R3) for Physical AI Clinical Practice

The Adaption: ICH Harmonised Guideline [1] establishes comprehensive good clinical practice principles specifically designed for Physical AI oncology clinical trials, building on the internationally recognized ICH E6(R3) framework [13]. By adapting an existing standard that is accepted across dozens of countries rather than creating an entirely new guideline, this work adds credibility to Physical AI implementations through association with decades of established GCP practice. The adaptation spans four major sections (Principles, Investigator Responsibilities, Sponsor Responsibilities, Data Governance), three appendices (System Documentation, Clinical Trial Protocol, Essential Records), and a comprehensive Physical AI glossary with 30 specialized definitions.

4.1 Principles of Physical AI Clinical Practice

The adapted ICH E6(R3) establishes eight foundational principles that govern Physical AI clinical practice. These principles extend the traditional GCP framework by requiring that Physical AI systems meet the same ethical and quality standards as human-directed clinical practice, with additional safeguards for robotic safety, AI transparency, and system validation.

The first principle establishes that clinical trials involving Physical AI systems must be conducted in accordance with ethical principles originating in the Declaration of Helsinki. Robotic interventions, AI-assisted diagnostics, and automated treatment delivery must preserve participant autonomy and welfare above all other considerations. The second principle requires that foreseeable risks from robotic systems, AI algorithms, and automated processes be weighed against anticipated benefits before trial initiation. Risks specific to robotic malfunction, AI prediction error, and cyber-physical system failure must be explicitly evaluated.

The third principle reinforces that participant rights, safety, and well-being prevail over scientific and societal interests. This extends to interactions with all Physical AI system categories: surgical robots (da Vinci, Hugo RAS, Versius), collaborative robots (Franka Panda, Kinova Gen3, xArm 7), humanoid robots (Atlas, Optimus, Digit), and all therapeutic, diagnostic, assistive, and rehabilitative robotic systems. The fourth principle requires specialized training for all individuals conducting Physical AI trials, supplementing traditional clinical trial competencies with Physical AI system operation, AI model interpretation, digital twin management, and simulation validation skills.

The fifth principle requires that trial protocols incorporate specifications for all robotic systems, AI models, simulation frameworks, and digital twin configurations, referencing applicable USL scores [4] and establishing minimum readiness thresholds. The sixth principle requires IRB or IEC approval with specific review of Physical AI components including robotic safety parameters, AI validation evidence, and cybersecurity measures. The seventh principle establishes that Physical AI systems operate as tools under human oversight, with continuous monitoring and intervention capability required for autonomous robotic actions. The eighth principle requires informed consent that clearly describes robotic interactions, AI-assisted decision making, and digital twin-based treatment planning.

Table 7: Adapted ICH E6(R3) Foundational Principles for Physical AI

No.	Traditional GCP Principle	Physical AI Extension
1	Ethical conduct per Declaration of Helsinki	Extends to robotic and AI-assisted procedures
2	Risk-benefit assessment before initiation	Includes robot malfunction and AI error risks
3	Participant rights prevail	Covers all 10 robot type interactions
4	Qualified personnel for all tasks	Physical AI training requirements added
5	Scientific soundness in protocol	Robot specs, USL scores, AI model details required
6	IRB/IEC approval with protocol review	Physical AI-specific component review mandated
7	Medical care under qualified physician	Human oversight for all autonomous robot actions
8	Freely given informed consent	Robot descriptions, AI transparency required

The adaptation also establishes a Physical AI system classification for clinical trials, organizing systems into ten categories consistent with the USL scoring methodology [4] and patient instructions framework [7]. Each category has specific GCP requirements proportional to its level of patient interaction and autonomous decision-making.

4.2 Investigator Responsibilities in Physical AI Trials

Section 2 of the adapted ICH E6(R3) defines twelve categories of investigator responsibilities adapted for Physical AI contexts. These responsibilities address the shift in clinical practice where robots perform functions traditionally done by human staff, requiring investigators to develop new competencies in Physical AI oversight.

Investigator qualifications and training (Subsection 2.1) now require demonstrated competency in Physical AI system oversight. Investigators must complete Physical AI-specific training covering robot operational parameters, AI model interpretation, emergency procedures for robot malfunction, and human-robot interaction protocols. Adequate resources (Subsection 2.2) now include robot maintenance, calibration equipment, charging infrastructure, and technical support staff. Medical care provisions (Subsection 2.3) must account for Physical AI system integration, ensuring that robot-assisted care meets the same quality standards as human-only care.

Communication with IRB/IEC (Subsection 2.4) requires ongoing reporting of Physical AI system performance, including robot uptime, adverse events attributable to Physical AI systems, software updates, and calibration status. Protocol compliance (Subsection 2.5) now encompasses robot operational parameters, AI model boundaries, and approved Physical AI configurations. Any deviation from approved Physical AI configurations constitutes a protocol deviation requiring documentation and reporting.

Informed consent procedures (Subsection 2.6) require Physical AI-specific elements as detailed in the adapted 21 CFR Part 50 [2]. Record keeping (Subsection 2.7) extends to Physical AI system logs including robot telemetry, AI decision records, sensor data, and maintenance records. Safety reporting (Subsection 2.8) now covers Physical AI adverse events including robot malfunctions, AI decision errors, sensor failures, and cybersecurity incidents.

Premature termination or suspension (Subsection 2.9) includes criteria specific to Physical AI system failures that may necessitate trial pause or termination. Progress reports (Subsection 2.10) must include Physical AI performance metrics. Compliance with financial and audit

requirements (Subsection 2.11) extends to Physical AI system procurement and maintenance costs. Final reporting (Subsection 2.12) must document the complete Physical AI trial experience including system performance, patient interactions, and lessons learned.

4.3 Sponsor Responsibilities in Physical AI Trials

Section 3 of the adapted ICH E6(R3) defines fifteen categories of sponsor responsibilities adapted for Physical AI. Sponsors bear primary responsibility for ensuring that Physical AI systems are safe, validated, and properly integrated into trial operations.

Quality management systems (Subsection 3.1) must encompass Physical AI system validation, ongoing performance monitoring, and continuous improvement processes. The quality management framework must address robot-specific quality indicators including mechanical precision, sensor accuracy, AI model performance, and system uptime. Regulatory submission requirements (Subsection 3.2) must include Physical AI system descriptions, validation reports, and safety assessments alongside traditional IND content.

IRB/IEC review of Physical AI systems (Subsection 3.3) requires sponsors to provide comprehensive Physical AI documentation including system specifications, risk assessments, validation evidence, and proposed monitoring plans. Trial design (Subsection 3.4) must incorporate robotic workflows, specifying which Physical AI systems perform which functions, the degree of autonomy for each system, and the human oversight structure.

Monitoring Physical AI system performance (Subsection 3.5) requires sponsors to establish continuous monitoring systems that track robot performance in real time. Safety reporting for Physical AI adverse events (Subsection 3.6) follows reporting timelines and severity classifications established in the adapted 21 CFR Part 312 [3]. Data handling (Subsection 3.7) must address the unique characteristics of robot-generated records, including high data volumes, real-time streaming data, and multi-modal data (telemetry, video, sensor readings, AI decision logs).

The 24-hour simulation from Clinical Trial Site Complete Documentation [5] provides evidence that these sponsor responsibilities can be fulfilled at scale. The simulation demonstrated that sponsors can effectively manage 29 robots serving 168 patients over 24 continuous hours while maintaining quality management, safety reporting, and data handling standards consistent with adapted ICH E6(R3) requirements.

4.4 Data Governance for Physical AI Trials

Section 4 of the adapted ICH E6(R3) addresses data governance across three subsections: blinding safeguards, data lifecycle elements, and computerized systems requirements.

Blinding safeguards (Subsection 4.1) present unique challenges in Physical AI trials because robots are physically visible to participants. The adaptation establishes methods for maintaining treatment blinding when Physical AI systems are present, including scenarios where sham robot procedures may be appropriate, where robot configurations differ between treatment arms, and where the presence or absence of specific robot types may convey treatment assignment information.

Data lifecycle elements (Subsection 4.2) for Physical AI trials encompass robot telemetry data (position, force, speed, temperature), sensor readings (patient vital signs, environmental conditions, imaging data), autonomous decision logs (AI model inputs, outputs, confidence scores, alternative actions considered), and imaging data (intraoperative video, diagnostic scans,

positioning verification). Each data category has specific requirements for collection frequency, storage format, retention period, and access controls.

Computerized systems requirements (Subsection 4.3) for Physical AI platforms must ensure data integrity, audit trails, and regulatory compliance. FHIR [30], DICOM [31], and MCP [28] serve as interoperability standards supporting Physical AI data governance. The Physical AI National MCP Servers [8] provide the infrastructure implementation of these data governance requirements, with five servers handling governance, data flow, safety, interoperability, and analytics functions respectively.

4.5 Physical AI System Documentation

Appendix A of the adapted ICH E6(R3) establishes documentation requirements for Physical AI systems in clinical trials. System specifications must include robot hardware details (manufacturer, model, serial number, software version, calibration status), AI model details (architecture, training data provenance, validation results, performance benchmarks), and integration details (network configuration, data flow architecture, safety interlock connections).

Validation reports must demonstrate that each Physical AI system performs within specified parameters under expected clinical conditions. Safety assessments must identify all foreseeable Physical AI-specific risks and document mitigation strategies. Maintenance logs must record all service activities, calibration checks, software updates, and hardware replacements. Software version control must ensure traceability of all code changes affecting Physical AI system behavior.

4.6 Clinical Trial Protocol Adaptations

Appendix B establishes that clinical trial protocols must be adapted to include Physical AI system specifications identifying each robot by type, manufacturer, model, USL score, and approved operational parameters. Robot operational parameters must specify workspace boundaries, force limits, speed limits, precision requirements, and environmental conditions. Fallback procedures for robot failures must define escalation pathways from automatic recovery through manual override to human takeover, with response time requirements for each escalation level. Criteria for when human staff must override Physical AI decisions must be explicitly stated, including clinical judgment thresholds, safety signal triggers, and patient request protocols.

4.7 Essential Records for Physical AI Trials

Appendix C defines additional essential records required beyond standard GCP documentation. Robot calibration records must document all calibration activities with date, method, results, and personnel. AI model version histories must trace every model update with version number, change description, validation results, and deployment date. Sensor validation logs must confirm that all sensors providing clinical data meet accuracy and precision specifications. Physical AI adverse event reports must use standardized reporting formats that capture both the clinical impact and the technical cause of each event.

4.8 Physical AI Glossary and Definitions

The adapted ICH E6(R3) provides 30 Physical AI-specific definitions that standardize terminology across the entire National Platform. Key definitions include Physical AI System

(an integrated robotic and AI platform designed for direct patient interaction in clinical trial settings), Physical AI Adverse Event (any untoward occurrence attributable to Physical AI system operation during a clinical trial), Physical AI Operator (a qualified individual responsible for monitoring and controlling Physical AI systems during clinical procedures), and USL Score (a quantitative measure of robot readiness for multi-site clinical trial deployment across four dimensions). These definitions are cross-referenced with the 22 definitions in the adapted 21 CFR Part 312 [3] and the 17 definitions in the adapted 21 CFR Part 50 [2] to ensure consistency across all three adapted standards.

Table 8: Selected Physical AI Definitions from Adapted ICH E6(R3)

Term	Definition
Physical AI System	An integrated robotic and AI platform designed for direct or indirect patient interaction in clinical trial settings
Physical AI Adverse Event	Any untoward occurrence attributable to Physical AI system operation during a clinical trial
Physical AI Operator	A qualified individual responsible for monitoring and controlling Physical AI systems
Autonomous Mode	Operation where AI algorithms direct robotic actions without real-time human input
Supervised Mode	Operation where human operators maintain continuous control over robotic actions
Physical AI Safety Matrix	A pre-procedure verification checklist for all safety systems
USL Score	Quantitative measure of robot readiness across four dimensions (1.0-10.0)
Phase 0 Validation	Mandatory simulation phase before human subjects enrollment
Digital Twin	Computational model replicating a physical patient or system for simulation
MCP Conformance	Compliance level with Model Context Protocol communication standards

The standardization of terminology across all three adapted standards ensures that investigators, sponsors, regulators, and IRB members can communicate precisely about Physical AI trial operations. The glossary eliminates ambiguity that could arise from different stakeholders using the same terms with different meanings. For example, the precise definition of “autonomous mode” versus “supervised mode” determines the level of informed consent detail required, the intensity of real-time monitoring, and the qualifications needed for the Physical AI operator on duty. By establishing these definitions in the adapted ICH E6(R3) and cross-referencing them in the other two adapted standards, the National Platform creates a unified vocabulary for the entire Physical AI oncology trial ecosystem.

5 Adapted 21 CFR Part 50 for Physical AI Protection

The Physical AI Adaption: 21 CFR Part 50 [2] provides comprehensive human subjects protection specifically designed for Physical AI oncology trials, building on the existing federal regulation [24]. By adapting a well-established regulation rather than proposing new legislation, this work adds immediate credibility and practical applicability. A critical theme throughout this adaptation is that control has shifted towards the patient’s side, with expanded patient rights including new protections specific to robotic and AI-assisted medical procedures that have no precedent in traditional clinical trial regulation.

5.1 Scope and Physical AI Definitions

The adapted Part 50 applies to all clinical investigations regulated by FDA that involve Physical AI systems interacting with human subjects. Subpart A (General Provisions) establishes the scope and provides 17 Physical AI-specific definitions that standardize terminology for robotic and AI-assisted clinical procedures.

Key definitions include: *Physical AI System*, encompassing any robotic platform integrated with artificial intelligence that interacts directly with trial participants; *Physical AI Adverse Event*, any untoward occurrence during a clinical investigation that is attributable to Physical AI system operation; *Physical AI Operator*, a qualified individual who monitors and controls Physical AI systems during clinical procedures; *Robotic Procedure*, any clinical intervention performed wholly or partly by a Physical AI system; *Autonomous Mode*, operation of a Physical AI system where AI algorithms direct robotic actions without real-time human input; *Supervised Mode*, operation where human operators maintain continuous control; and *Physical AI Safety Matrix*, a pre-procedure verification checklist ensuring all safety systems are operational.

These definitions create the foundation for all subsequent protections in the regulation. The distinction between autonomous and supervised modes is particularly important because it determines the level of informed consent detail required and the intensity of ongoing monitoring. Cross-referencing with the 22 definitions in the adapted 21 CFR Part 312 [3] and the 30 definitions in the adapted ICH E6(R3) [1] ensures terminological consistency across the entire regulatory framework.

5.2 Informed Consent for Physical AI Procedures

Subpart B adapts the general requirements for informed consent to Physical AI contexts. Traditional informed consent requires disclosure of trial purpose, procedures, risks, benefits, alternatives, and confidentiality protections. The Physical AI adaptation adds specific consent elements required for any trial involving robotic systems.

The additional consent elements include: a description of each Physical AI system type and its role in the participant’s care, including the robot manufacturer, model, and general capabilities; an explanation of how AI algorithms participate in treatment planning, monitoring, and decision-making; a clear statement of the participant’s right to request human-only care at any point during the trial, including during active robotic procedures; a description of data collection specific to Physical AI systems, including robot telemetry, video recording, sensor data, and AI decision logs, with associated privacy protections; an explanation of emergency stop procedures and how participants can signal distress or discomfort during robotic interactions;

and a description of the safety monitoring systems in place, including who monitors robot performance and how deviations are detected and addressed.

These consent elements represent a fundamental shift in the informed consent paradigm. In traditional trials, consent primarily addresses drug or device risks and benefits. In Physical AI trials, consent must address a new category of risk: the interaction between a human patient and an autonomous or semi-autonomous robotic system that physically touches, manipulates, or occupies space near the patient. Patient Instructions: Physical AI Trials [7] provides the patient-facing implementation of these consent requirements, translating regulatory language into accessible information for each of the ten robot types.

5.3 Physical AI System Safety Requirements

Subpart C is a new subpart created specifically for Physical AI trials, establishing additional protections for subjects in Physical AI clinical investigations. This subpart has no precedent in the original 21 CFR Part 50 and represents the most significant structural addition to the adapted regulation.

5.3.1 Pre-Procedure Safety Matrix

Before any Physical AI procedure begins, the Physical AI Safety Matrix must be completed and documented. This matrix verifies: robot system power and mechanical status (all joints responsive, actuators calibrated, end-effector functional); sensor calibration status (force sensors, position sensors, proximity sensors, temperature sensors all within specification); emergency stop system verification (hardware E-stop functional, software E-stop responsive, wireless E-stop paired); communication system verification (robot-to-server connection active, data logging operational, alert system functional); patient identification confirmation (biometric or badge-based patient ID matched to protocol records); workspace clearance verification (operating zone free of unauthorized objects, personnel positioned safely, barriers in place); and AI model status verification (approved model version loaded, performance within specification, no pending updates requiring validation).

5.3.2 Runtime Safety Monitoring

During Physical AI procedures, continuous monitoring requirements include: real-time force and torque monitoring at all patient contact points, with automatic procedure pause when force exceeds protocol-specified limits; position deviation monitoring comparing actual robot position to planned trajectory, with alerts when deviation exceeds tolerance; vital sign integration connecting patient monitoring systems to Physical AI control systems, enabling automatic response to physiological changes; AI decision monitoring logging all autonomous decisions with confidence scores and alternative actions considered; and video recording of all patient-robot interactions for post-procedure review and adverse event investigation.

5.3.3 Post-Procedure Requirements

After Physical AI procedures, required documentation includes: a complete Physical AI procedure report recording all robot actions, parameters, and outcomes; patient follow-up specific to Physical AI interactions, including assessment of any physical marks, discomfort, or psychological responses to robotic interaction; data archival ensuring all telemetry, sensor, video, and decision

log data are properly stored with appropriate retention periods; and system reset verification confirming the Physical AI system is properly reset and ready for the next procedure.

5.3.4 Task-Order Lifecycle

The adaptation establishes a formal lifecycle for Physical AI task orders from initiation through completion. Each task order must be authorized by a qualified Physical AI operator, with the authorization level matched to the task complexity and risk. Task execution must be monitored against approved parameters, with any deviation triggering documentation and potential escalation. Task completion must be verified through both automated system checks and human confirmation before the system is released for the next task.

5.3.5 Forbidden Operations

The adapted regulation explicitly prohibits Physical AI systems from performing certain operations without human authorization: initiating surgical incisions or tissue removal without real-time surgeon oversight; modifying medication dosages or administration schedules without pharmacist or physician confirmation; altering treatment plans based solely on AI recommendations without clinical team review; performing emergency medical interventions without qualified human medical staff present; and accessing or modifying patient records beyond the specific data required for the current authorized task.

5.4 IRB Review of Physical AI Investigations

The adapted regulation establishes specific requirements for IRB review of Physical AI trials that extend beyond traditional IRB evaluation criteria. IRBs must evaluate: the technical qualifications of Physical AI system operators and maintenance staff; the adequacy of Physical AI system validation evidence, including simulation results and bench testing data; the appropriateness of Physical AI safety monitoring plans; the completeness of Physical AI-specific informed consent elements; and the adequacy of contingency plans for Physical AI system failures.

Ongoing monitoring requirements specify that IRBs must receive periodic reports on Physical AI system performance, including uptime, adverse events, near-miss incidents, and maintenance activities. Protocol modification triggers specific to Physical AI include: Physical AI system software updates that change operational parameters, introduction of new Physical AI system types not covered in the original protocol, changes to Physical AI safety monitoring procedures, and changes to the degree of autonomy granted to Physical AI systems. The NCI CIRB [29] provides a model for centralized Physical AI review that could enable consistent evaluation across multiple trial sites.

5.5 Ongoing Consent and Subject Notification

Unlike traditional trials where informed consent is primarily a one-time event, Physical AI trials require ongoing consent processes because the technology may be updated, new robots may be introduced, and AI algorithms may be refined during the trial period. Participants must be informed about: Physical AI system software updates that change system behavior or capabilities; introduction of new robot types at the trial site; changes to safety monitoring procedures; results of periodic Physical AI system safety reviews; and any Physical AI adverse events that occurred at the trial site that may affect the participant's risk assessment.

This ongoing consent process gives patients continuous agency over their participation. A patient who consented to interact with a da Vinci surgical robot may reconsider if a humanoid robot is later added to the trial site for logistics support. A patient who accepted a specific AI algorithm for treatment planning may want to reconsider if the algorithm is updated. The adapted regulation ensures that patients are never surprised by changes to the Physical AI systems involved in their care.

5.6 Data Protection and Privacy

The adapted regulation establishes protections for the additional categories of data generated by Physical AI systems. Robot telemetry data recording position, force, speed, and environmental readings constitutes a new category of health information that was not contemplated by the original Part 50. Video recordings of Physical AI procedures create visual records of patient interactions with robotic systems. Sensor data from patient-robot interactions may include biometric measurements beyond those collected in traditional trials. AI decision logs that document the reasoning behind automated treatment decisions create records that may have implications for patient rights and legal accountability.

All these data categories are subject to HIPAA [17] protections when they constitute protected health information. The federated learning framework [9] and the broader literature on federated learning in healthcare [32] provide approaches for utilizing these data for model improvement while maintaining patient privacy through decentralized processing.

5.7 Physical AI System Classification

The adapted regulation establishes a classification framework organizing Physical AI systems by risk level into three tiers. Tier 1 (Standard Risk) encompasses Physical AI systems with minimal direct patient contact, such as logistics robots and specimen handling cobots, which require standard informed consent and monitoring. Tier 2 (Elevated Risk) encompasses systems with moderate patient interaction, such as positioning robots and imaging robots, requiring enhanced consent elements and continuous monitoring. Tier 3 (High Risk) encompasses systems with direct therapeutic interaction, such as surgical robots and needle-placement robots, requiring comprehensive consent with procedure-specific elements, real-time safety monitoring, and qualified operator presence. The classification tier determines the regulatory pathway, monitoring intensity, and consent detail required, connecting to the USL scoring framework in Section 7 for quantitative assessment.

5.8 Pediatric Protections in Physical AI Trials

Subpart D adapts existing pediatric protections for Physical AI contexts. Modified consent and assent procedures address minors interacting with robots, requiring age-appropriate explanations of robotic systems. Additional safety requirements for pediatric Physical AI procedures include reduced force limits, smaller workspace boundaries, and enhanced human oversight. IRB duties specific to pediatric Physical AI investigations include evaluation of whether robotic interactions may cause additional psychological distress in pediatric patients. Pediatric oncology is a particularly important application where Physical AI may reduce treatment burden on young patients through faster procedures, reduced need for repeated positioning, and companion robot support during treatment.

5.9 Glossary of Physical AI Terms

The adapted Part 50 concludes with a supplementary glossary of 35 Physical AI terms that complement the regulatory definitions in Subpart A. Terms include Agentic AI (AI systems capable of autonomous multi-step task execution), Calibration (the process of adjusting Physical AI system parameters to meet specified accuracy standards), Digital Twin (a computational model replicating a physical patient or system for simulation and prediction), MCP Server (a Model Context Protocol server providing standardized AI-tool communication), and USL (Unification Standard Level, a quantitative robot readiness scoring framework). These terms are cross-referenced with the ICH glossary (Section 4.8) to ensure consistency across all adapted standards in the National Platform.

5.10 Three-Tier Physical AI Classification System

The three-tier classification introduced in Section 5.7 warrants detailed examination because it determines the regulatory pathway, monitoring intensity, consent complexity, and reporting obligations for every Physical AI system deployed in an oncology clinical trial. Table 9 presents the complete classification framework with representative robot types, regulatory requirements, and consent elements for each tier.

Table 9: Three-Tier Physical AI Classification System: Requirements and Consent Elements

Attribute	Tier 1 - Standard Risk	Tier 2 - Elevated Risk	Tier 3 - High Risk
Representative robots	Logistics humanoids (Digit, Optimus), social companion robots, specimen-handling cobots	RT positioning robots, imaging robots, rehabilitation exoskeletons, RT motion-tracking robots	Surgical robots (da Vinci, Hugo RAS, Versius), needle-placement robots, steerable needle robots
Patient contact level	Minimal or no direct physical contact; robot operates in shared space	Moderate direct contact; robot touches or repositions patient under supervision	Direct therapeutic contact; robot performs invasive or semi-invasive procedures
Autonomy permitted	Supervised or semi-autonomous for logistics tasks; autonomous navigation in designated corridors	Supervised mode required during patient contact; autonomous calibration permitted	Supervised mode mandatory at all times; autonomous incision or tissue manipulation prohibited
Minimum USL band	Foundational (3.0 or higher)	Intermediate (5.0 or higher)	Advanced (7.0 or higher)
Operator requirement	Trained staff member within radio or network contact	Certified Physical AI operator present in the room	Board-certified physician plus certified Physical AI operator at console
Safety matrix scope	Abbreviated matrix covering power, communication, and identification checks	Standard matrix covering all seven verification categories	Extended matrix with procedure-specific additions, dual-operator sign-off, and anesthesia integration
Monitoring during operation	Periodic status checks via dashboard at 15-minute intervals	Continuous telemetry with automated alerts for parameter exceedance	Real-time multi-channel monitoring including force, position, vital signs, and AI decision confidence
Adverse event reporting	Category B or C timelines (15 calendar days) for most events	Category A timelines (7 calendar days serious, 15 days non-serious) for contact events	Category A expedited timelines (7 calendar days) for all events; 24-hour reporting for serious surgical events
IRB review depth	Standard review with Physical AI addendum	Enhanced review with Physical AI safety plan evaluation	Full board review with independent Physical AI technical expert consultation

The tier system creates proportional regulatory burden: Tier 1 systems that pose minimal risk to patients face streamlined requirements, while Tier 3 systems performing invasive procedures face the most rigorous oversight. This proportionality is essential for practical deployment because applying Tier 3 requirements to every logistics robot would create unnecessary burden without improving patient safety, while applying Tier 1 requirements to surgical robots would provide insufficient protection. The tier classification is determined at the protocol level during IND submission and reviewed by the IRB as part of the Physical AI safety plan evaluation.

When a robot's function spans multiple tiers, the higher tier governs. For example, a cobot that both handles specimens (Tier 1) and assists with patient positioning (Tier 2) is classified as Tier 2 for all regulatory purposes. This conservative approach ensures that no Physical AI interaction falls below the protection level warranted by its highest-risk function.

The tier system also governs training requirements for site personnel. Tier 1 operations require general Physical AI awareness training (minimum 8 hours) covering robot identification, basic safety procedures, and emergency stop activation. Tier 2 operations require specialized Physical AI operator certification (minimum 40 hours) covering robot-specific operation, patient interaction protocols, and troubleshooting. Tier 3 operations require advanced Physical AI surgical team certification (minimum 80 hours plus proctored cases) covering console operation, instrument exchange, emergency conversion, and team communication protocols.

5.11 Comprehensive Informed Consent Elements for Physical AI Trials

Section 5.2 introduced the Physical AI-specific consent elements required by the adapted Part 50. The following comprehensive itemized list specifies every consent element that must be addressed in the informed consent document for Physical AI oncology trials, organized by category.

5.11.1 General Physical AI Disclosure Elements

All Physical AI trial consent forms must include the following general disclosure elements regardless of tier classification:

- A statement that the clinical trial involves Physical AI systems (robots with integrated artificial intelligence) that will participate in some or all aspects of the participant's care during the trial period.
- An enumeration of every Physical AI system type that the participant may encounter during the trial, identified by category (surgical robot, cobot, humanoid, companion robot, etc.), manufacturer name, model designation, and general functional description.
- A plain-language explanation of how artificial intelligence participates in treatment planning, monitoring, decision support, and operational coordination, including a statement about the role of human oversight in AI-assisted decisions.
- A description of the Physical AI safety monitoring infrastructure, including who monitors robot performance, how deviations from expected behavior are detected, what automated safety systems are in place, and how the participant will be protected in the event of a system malfunction.

- A clear statement that the participant has the right to request human-only care at any point during the trial, including during active robotic procedures, and that exercising this right will not affect the participant's eligibility for continued trial participation or standard medical care.
- A description of the emergency stop procedures available to the participant, including how to verbally signal distress, how to activate patient-accessible stop mechanisms where available, and the expected response time from clinical staff.
- A statement identifying the Principal Investigator, Physical AI Safety Officer, and patient advocate who can be contacted with questions or concerns about Physical AI interactions at any time.

5.11.2 Data Collection and Privacy Elements

All Physical AI trial consent forms must include the following data-related elements:

- A comprehensive list of all data types collected by Physical AI systems, including but not limited to: robot telemetry data (position, force, speed, torque, temperature), video recordings of patient-robot interactions, audio recordings if applicable, sensor data from patient-mounted or room-mounted sensors, AI decision logs documenting automated recommendations, and biometric data collected through robot-integrated sensors.
- An explanation of how each data type is stored, who has access to it, how long it is retained, and under what circumstances it may be shared with parties outside the immediate clinical team.
- A statement that all Physical AI-generated data is protected under HIPAA and applicable state privacy laws (including SB 892 provisions for California sites), with specific protections for video and audio recordings that may capture identifiable patient images or voices.
- A description of the federated learning framework, if applicable, explaining that de-identified data may be used to improve AI models across trial sites without the participant's raw data leaving the local site.
- A statement of the participant's right to access all Physical AI data generated during their care, to receive copies of such data in a readable format, and to request correction of data errors.
- A description of cybersecurity protections for Physical AI data, including encryption standards, access controls, and incident response procedures in the event of a data breach.

5.11.3 Tier-Specific Consent Elements

In addition to the general and data elements, the following tier-specific consent elements must be included based on the Physical AI systems involved in the participant's care:

- **Tier 2 additional elements:** A description of the specific physical interactions the participant will experience with each Tier 2 robot, including what body parts will be contacted, the nature of the contact (positioning, support, imaging), the expected duration

of contact, and any sensations the participant may experience (pressure, warmth, vibration). A description of the operator qualifications and the operator's role during the interaction. An explanation of the real-time monitoring systems active during contact and the thresholds that trigger automatic safety responses.

- **Tier 3 additional elements:** A detailed description of each surgical or invasive procedure involving Physical AI systems, including the specific role of the robot (instrument guidance, tissue manipulation, needle placement), the degree of autonomous operation permitted, and the surgeon's real-time control capabilities. A description of the emergency conversion plan detailing how the procedure transitions from robotic to manual if a system failure occurs. A statement of the risks specific to robotic surgical procedures, including mechanical failure during tissue interaction, sensor error leading to positional inaccuracy, and extended procedure time due to system recalibration. A description of the post-procedure Physical AI assessment, including the 72-hour follow-up schedule and the specific evaluations performed to detect robot-related complications.

5.11.4 Ongoing Consent and Notification Elements

The consent form must also address the ongoing consent requirements unique to Physical AI trials:

- A statement that Physical AI systems may be updated, replaced, or supplemented during the trial period, and that the participant will be notified in writing before any such change takes effect.
- A description of the specific changes that trigger re-consent, including introduction of new robot types, software updates that alter robot behavior or decision-making, changes to the degree of autonomy, and changes to safety monitoring procedures.
- A statement that the participant will receive periodic summary reports of Physical AI system performance at the trial site, including any adverse events that may affect the participant's risk assessment.
- An explanation that the participant may withdraw consent for Physical AI interactions at any time without withdrawing from the clinical trial itself, and a description of how the trial protocol accommodates continued participation with human-only care delivery.
- A statement that the participant's consent preferences will be documented in the electronic health record and communicated to all Physical AI systems through the MCP server infrastructure, ensuring that robots are automatically informed of any restrictions on their interaction with the participant.

5.11.5 Comprehension Verification

The adapted Part 50 requires that informed consent for Physical AI trials include a comprehension verification step beyond the standard consent signature. This verification must confirm that the participant understands: the types of robots involved in their care and their general functions, the participant's right to refuse any robotic interaction, how to activate emergency stop procedures, what data is collected by Physical AI systems and how it is protected, and whom to contact with

questions or concerns about Physical AI interactions. Comprehension may be verified through a brief questionnaire, a teach-back conversation with the consent coordinator, or another method approved by the IRB. The verification results must be documented in the participant's trial record.

5.12 Regulatory Pathway Determination by Tier

The three-tier classification system does more than organize consent requirements; it determines the entire regulatory pathway that a Physical AI system must follow from initial site deployment through ongoing operation and eventual decommissioning. Table 10 details the regulatory pathway milestones for each tier, providing sponsors and investigators with a comprehensive roadmap for compliance.

Table 10: Regulatory Pathway Milestones by Physical AI Tier Classification

Milestone	Tier 1 - Standard Risk	Tier 2 - Elevated Risk	Tier 3 - High Risk
Pre-deployment validation	Manufacturer acceptance testing plus site functional check	Manufacturer validation plus independent simulation validation (minimum 100 hours)	Full Phase 0 simulation validation (minimum 500 hours) with independent technical review
IND submission content	Abbreviated Physical AI system description with manufacturer specifications	Standard Physical AI system description with simulation results and safety analysis	Comprehensive Physical AI system description with Phase 0 report, safety analysis, and risk mitigation plan
IRB submission package	Physical AI addendum to standard protocol submission	Enhanced Physical AI safety plan with operator qualifications and monitoring procedures	Full Physical AI technical dossier with independent expert review letter
Site readiness assessment	General Physical AI awareness verification for relevant staff	Operator certification verification plus equipment validation records	Surgical team certification, proctored case documentation, and emergency conversion drill records
Operational authorization	Site director sign-off with Physical AI safety officer concurrence	Physical AI safety committee review and approval	Full safety committee review plus independent technical auditor approval
Ongoing compliance audits	Annual audit with Physical AI system performance review	Semi-annual audit with Physical AI telemetry analysis	Quarterly audit with comprehensive Physical AI performance, safety, and outcome analysis
Software update pathway	Notification to IRB within 30 days of routine updates	IRB review required before deployment of updates affecting patient interaction	Full protocol amendment required for any update affecting surgical algorithms or autonomy levels
Decommissioning process	Standard equipment removal with data archival	Supervised decommissioning with patient notification and data migration plan	Formal decommissioning protocol with IRB notification, patient notification, alternative care plan, and data preservation audit

The regulatory pathway milestones create a structured progression that ensures proportional oversight at every stage. Tier 1 systems follow a streamlined pathway that minimizes administrative burden while maintaining essential safety checks. A logistics humanoid robot, for example, requires manufacturer acceptance testing and a site functional check before deployment, an abbreviated Physical AI system description in the IND submission, and annual compliance audits. This streamlined approach recognizes that a robot transporting supplies through a corridor poses fundamentally different risks than a robot performing surgical procedures.

Tier 2 systems require intermediate oversight reflecting their direct patient contact. A radiotherapy positioning robot must undergo independent simulation validation demonstrating that it can position patients within specified tolerances across multiple anatomical scenarios. The IRB submission must include an enhanced safety plan documenting the operator certification requirements, continuous monitoring procedures, and specific force and position limits applicable to patient contact. Semi-annual audits with telemetry analysis ensure that the system continues to perform within validated parameters throughout the trial duration.

Tier 3 systems face the most rigorous pathway, commensurate with the risks of invasive procedures. A surgical robot must complete a minimum of 500 hours of Phase 0 simulation validation covering the full range of procedures it will perform in the trial. The IND submission requires a comprehensive technical dossier that is reviewed by an independent expert before IRB consideration. Quarterly audits analyze not only system performance metrics but also surgical outcomes, enabling detection of trends that might indicate degradation in system performance or emerging safety concerns.

5.13 Consent Documentation Standards and Templates

The adapted Part 50 establishes documentation standards for Physical AI informed consent that go beyond the traditional consent form signature. Each consent interaction must be documented with the following elements:

- **Consent form version tracking:** Each version of the Physical AI informed consent form must carry a unique version identifier, revision date, and change summary. When Physical AI systems are updated or new systems are introduced, the consent form must be revised accordingly, and the version history must be maintained in the regulatory file.
- **Robot demonstration records:** When consent includes a demonstration of Physical AI systems (recommended for all Tier 2 and Tier 3 interactions), the demonstration must be documented including the date, location, systems demonstrated, staff present, patient questions asked, and responses provided.
- **Comprehension assessment documentation:** The results of the comprehension verification step must be documented using the IRB-approved assessment method. For questionnaire-based assessment, individual question responses must be recorded. For teach-back assessment, the assessor must document the key concepts the participant was able to explain accurately and any concepts requiring additional clarification.
- **Consent preference records:** The participant's specific preferences regarding Physical AI interactions must be documented in a standardized format that can be communicated to Physical AI systems through the MCP server infrastructure. Preferences include which robot types the participant consents to interact with, any restrictions on data collection, and whether the participant desires a human advocate present during robotic interactions.

- **Re-consent documentation:** Each re-consent event must be documented with the triggering change, the updated consent form version, the date of re-consent, and verification that the participant understands the change and its implications for their care.
- **Withdrawal documentation:** If a participant withdraws consent for Physical AI interactions (either partially for specific robot types or completely for all robotic interactions), the withdrawal must be documented with the date, scope, reason if provided, and the alternative care plan implemented. The MCP server must be updated to reflect the withdrawal, ensuring that Physical AI systems are automatically informed of the restriction.

These documentation standards serve multiple purposes. They create a complete audit trail demonstrating regulatory compliance with the adapted Part 50 consent requirements. They provide data for quality improvement of the consent process across trial sites. They protect both participants and investigators by creating clear records of what was disclosed, understood, and agreed to. And they enable the MCP server infrastructure to enforce participant preferences automatically, reducing the risk of inadvertent Physical AI interactions with participants who have restricted or withdrawn consent.

The consent documentation standards also address the linguistic and cultural considerations unique to Physical AI trials. Consent materials must be available in the primary languages spoken by the patient population at each trial site. Cultural sensitivity review must address potential concerns about robotic interaction that may vary across cultural backgrounds, including attitudes toward autonomous technology, privacy expectations regarding video and sensor monitoring, and comfort levels with human-shaped robots. Sites serving diverse patient populations must document their cultural accommodation strategies as part of the IRB submission.

5.14 Interrelationship with Part 312 and ICH E6(R3)

The adapted Part 50 does not operate in isolation but functions as one component of the three-standard regulatory framework. The interrelationship between the adapted Part 50 (human subjects protection), the adapted Part 312 (IND requirements), and the adapted ICH E6(R3) (good clinical practice) creates overlapping protections that reinforce patient safety from multiple regulatory perspectives.

The informed consent requirements of Part 50 connect directly to the IND content requirements of Part 312 § 312.23, which mandates that the Physical AI System Description in the IND submission include sufficient information to support informed consent. The IRB review requirements of Part 50 connect to the sponsor obligations of Part 312 § 312.50, which require sponsors to ensure that IRBs are provided with all Physical AI technical information needed for review. The ongoing consent requirements of Part 50 connect to the protocol amendment requirements of Part 312 § 312.30, which define the Physical AI changes that trigger both protocol amendments and re-consent.

The adapted ICH E6(R3) [1] provides the quality management framework within which Part 50 consent requirements are implemented. Investigator responsibilities under ICH E6(R3) include ensuring that the consent process meets Part 50 requirements, that consent documentation is maintained per Part 50 standards, and that participant preferences are communicated to Physical AI systems as required by Part 50. The quality management system must include processes for monitoring consent compliance, identifying consent-related deviations, and implementing corrective actions.

This three-standard integration ensures that Physical AI human subjects protection is addressed comprehensively: Part 50 defines what protections are required, Part 312 defines how those protections are implemented within the IND framework, and ICH E6(R3) defines the quality management system that ensures ongoing compliance. The three-tier classification system established in Part 50 is referenced by both Part 312 (for IND content requirements) and ICH E6(R3) (for monitoring intensity), creating a unified framework where the tier classification of a Physical AI system determines its regulatory requirements across all three standards simultaneously.

6 Adapted 21 CFR Part 312 for Physical AI IND

The Physical AI Adaption: 21 CFR Part 312 [3] provides a comprehensive framework for Investigational New Drug applications in the context of Physical AI oncology trials, building on the existing regulation [23]. This is the most extensive of the three regulatory adaptations, spanning ten subparts (A through J) including a new Subpart J dedicated entirely to Physical AI systems. The adaptation demonstrates that Physical AI can be integrated into the IND process without requiring entirely new legislation, while acknowledging through the creation of Subpart J that certain aspects of Physical AI are sufficiently novel to warrant their own regulatory provisions.

6.1 Scope and Physical AI Definitions

Subpart A (General Provisions, § 312.1-312.10) establishes the scope and applicability of the adapted regulation. The scope (§ 312.1) extends the original Part 312 to cover clinical investigations of new drugs that incorporate Physical AI systems in drug administration, monitoring, or trial conduct. Applicability (§ 312.2) clarifies which types of Physical AI involvement trigger IND requirements.

Section 312.3 provides 22 Physical AI definitions, the most comprehensive set across all three adapted standards. Key definitions include: *Physical AI Clinical Trial* (a clinical investigation that uses Physical AI systems for drug administration, patient monitoring, data collection, or trial coordination); *Phase 0 Simulation Validation* (a mandatory pre-enrollment phase where Physical AI systems demonstrate safety and preliminary efficacy through simulation); *Physical AI IND Content* (the Physical AI-specific information required in an IND submission); *Cybersecurity Incident* (any unauthorized access, disruption, or modification of Physical AI systems that may affect patient safety or data integrity); and *Physical AI Decommissioning* (the process of safely removing Physical AI systems from clinical trial operations).

Additional provisions in Subpart A address Physical AI labeling (§ 312.6), requiring clear identification of Physical AI involvement in drug labeling during investigation; Physical AI promotion restrictions (§ 312.7), prohibiting promotion of investigational Physical AI drug delivery capabilities; Physical AI charging provisions (§ 312.8), governing charges for investigational drugs administered by Physical AI systems; and waiver procedures (§ 312.10), establishing conditions under which specific Physical AI requirements may be waived.

6.2 IND Requirements and Physical AI Content

6.2.1 IND Requirement for Physical AI Trials

Section 312.20 establishes when an IND is required for Physical AI trials. An IND is required whenever Physical AI systems are integral to drug administration (such as robotic IV drug delivery or automated dosing) or when Physical AI monitoring systems generate data used for safety or efficacy determinations. The determination of IND requirement considers the Physical AI system's role in the drug's mechanism of action, the degree to which Physical AI affects drug exposure or distribution, and whether Physical AI decisions directly influence dosing or treatment modifications.

6.2.2 Phases of Investigation

Section 312.21 adapts the clinical trial phases to include a Phase 0 simulation validation stage. This is a significant innovation: Physical AI trials include a mandatory simulation phase before any human subjects are enrolled. Phase 0 requires demonstration that Physical AI systems can safely perform their intended functions through validated simulations, with performance metrics meeting established thresholds.

Table 11: Adapted Clinical Trial Phases for Physical AI

Phase	Traditional Objective	Physical AI Addition
Phase 0	Not applicable	Simulation validation of Physical AI systems
Phase 1	Safety, dosing	Physical AI system safety with human subjects
Phase 2	Efficacy, side effects	Physical AI-assisted treatment efficacy
Phase 3	Large-scale confirmation	Multi-site Physical AI deployment
Phase 4	Post-approval monitoring	Ongoing Physical AI performance tracking

The Phase 0 requirement is directly supported by the evidence from both A Cancer Patient’s Journey, Physical AI [6] (single-patient simulation across ten stages) and the 24-hour simulation from the Clinical Trial Site Complete Documentation [5] (168 patients, 29 robots, 99.7 percent uptime). These completed simulations demonstrate that Physical AI systems can be thoroughly validated before real patient contact, providing a model for how Phase 0 validation should be conducted.

6.2.3 IND Content and Format

Section 312.23 adds the Physical AI System Description requirement for IND submissions. This must include: a complete inventory of all Physical AI systems to be used, with manufacturer, model, software version, and USL score [4]; a description of how each Physical AI system interacts with the investigational drug (administration, monitoring, safety); validation evidence including Phase 0 simulation results; a safety analysis addressing Physical AI-specific risks; and cybersecurity documentation describing protections for Physical AI systems against unauthorized access or manipulation.

6.2.4 Protocol Amendments

Section 312.30 establishes Physical AI amendment triggers: changes to Physical AI systems that require protocol amendments include replacement of a robot with a different model or manufacturer, software updates that alter robot behavior or decision-making, changes to the degree of autonomy (from supervised to autonomous or vice versa), addition of new Physical AI system types not in the original protocol, and modifications to safety monitoring parameters.

6.3 Safety Reporting for Physical AI

6.3.1 Information Amendments

Section 312.31 on information amendments requires notification to FDA when Physical AI system updates occur that do not constitute protocol amendments but that change system

characteristics, performance, or safety profile. This includes routine software patches, firmware updates, calibration adjustments, and hardware component replacements that fall within pre-approved specifications.

6.3.2 IND Safety Reporting

Section 312.32 establishes Physical AI adverse event reporting and cybersecurity incident reporting. The adaptation creates specific reporting categories for Physical AI events: Category A covers Physical AI system malfunctions that directly affect patient safety (reporting within 7 calendar days for serious events, 15 days for non-serious); Category B covers AI decision errors that result in incorrect treatment delivery or monitoring failure (reporting within 15 calendar days); Category C covers sensor failures that compromise data integrity or safety monitoring (reporting within 15 calendar days); and Category D covers cybersecurity incidents that affect or may affect Physical AI system integrity (reporting within 24 hours for active breaches, 7 days for detected vulnerabilities).

6.3.3 Annual Reports

Section 312.33 requires that annual reports include a Physical AI performance summary covering: system uptime and availability statistics, total Physical AI adverse events by category, software updates and version changes, robot maintenance and calibration records, cybersecurity assessment results, and trends in Physical AI system performance over the reporting period.

6.3.4 IND Withdrawal

Section 312.38 establishes Physical AI system decommissioning procedures when an IND is withdrawn. These procedures require safe shutdown of all Physical AI systems, secure archival of all Physical AI-generated data, notification to participants about cessation of Physical AI interactions, and physical removal or secure storage of robot hardware.

6.4 Administrative Actions for Physical AI Trials

Subpart C addresses general administration of Physical AI trials. Section 312.40 establishes general requirements including Physical AI system registration. Section 312.44 adds Physical AI-specific grounds for IND termination including repeated Physical AI adverse events demonstrating systemic safety issues, failure to maintain Physical AI system validation status, and cybersecurity compromise that cannot be remediated while maintaining patient safety.

Section 312.45 addresses inactive status and Physical AI dormancy, establishing procedures for placing Physical AI systems in dormant mode during trial pauses and reactivation procedures ensuring systems meet current safety standards before resuming operations. Section 312.47 on meetings includes Physical AI system review as a standing agenda item for pre-IND and end-of-phase meetings. Section 312.48 provides dispute resolution procedures for Physical AI technical disputes between sponsors and FDA.

6.5 Sponsor and Investigator Responsibilities

Subpart D (§ 312.50-312.70) adapts sponsor and investigator responsibilities to Physical AI contexts. Sponsors must ensure Physical AI system procurement from qualified manufacturers

(§ 312.50), maintain Physical AI system validation status throughout the trial (§ 312.53), and select investigators with Physical AI competencies (§ 312.55). Investigators must ensure proper Physical AI system operation (§ 312.60), report Physical AI adverse events promptly (§ 312.64), and maintain Physical AI system records (§ 312.68). These responsibilities complement the requirements in the adapted ICH E6(R3) [1] by providing the specific regulatory framework within which those responsibilities operate.

6.6 Life-Threatening Illness Provisions

Subpart E (§ 312.80-312.88) addresses accelerated approval and expanded access provisions for Physical AI oncology treatments for life-threatening cancers. Many cancer patients face conditions where Physical AI may offer faster or better treatment options, making these provisions particularly relevant. The adaptation ensures that Physical AI components do not create barriers to accelerated access when safety data support their use.

6.7 Expanded Access for Physical AI

Subpart I (§ 312.300-312.320) adapts expanded access provisions for Physical AI, establishing conditions under which Physical AI systems can be made available outside of clinical trials when sufficient evidence supports their safety and effectiveness. Expanded access for Physical AI requires demonstration that the Physical AI system has been validated through Phase 0 simulation, that sufficient clinical trial data support its safety, that no comparable therapy is available without Physical AI assistance, and that the requesting site meets minimum PSL requirements.

6.8 Subpart J: Physical AI Systems

Subpart J (§ 312.400-312.405) is entirely new and dedicated to Physical AI systems. This subpart establishes Physical AI-specific regulatory requirements that do not fit within the existing subpart structure.

Section 312.400 establishes Physical AI system registration requirements, requiring all Physical AI systems used in IND trials to be registered in a centralized database maintained by FDA. Section 312.401 establishes Physical AI validation standards, including minimum simulation hours, benchmark performance thresholds, and validation renewal requirements. Section 312.402 establishes cybersecurity requirements specific to Physical AI systems in clinical trials, including encryption standards, access control requirements, incident response procedures, and penetration testing schedules.

Section 312.403 establishes Physical AI interoperability requirements, requiring systems to support standardized communication protocols (including MCP [28]) and data formats (including FHIR [30] and DICOM [31]). Section 312.404 establishes Physical AI training requirements for all personnel who operate, maintain, or oversee Physical AI systems in clinical trials. Section 312.405 establishes Physical AI decommissioning standards, including data preservation requirements, hardware disposal procedures, and participant notification obligations.

The creation of Subpart J demonstrates the balanced approach of this platform: the existing CFR Part 312 framework can accommodate most Physical AI requirements through targeted adaptation of existing subparts, but certain aspects are sufficiently novel to warrant their

own provisions. This approach maintains regulatory continuity with established practice while acknowledging the unique characteristics of Physical AI systems.

Table 12: Physical AI Adverse Event Reporting Categories and Timelines

Cat	Event Type	Timeline	Examples
A	System malfunction affecting safety	7 days (serious), 15 days (non-serious)	Robot arm position error during surgery, actuator failure during procedure, end-effector malfunction
B	AI decision error	15 calendar days	Incorrect dosing recommendation, wrong treatment plan selection, misclassified imaging result
C	Sensor failure	15 calendar days	Force sensor drift beyond calibration, vital sign monitor data loss, position encoder failure
D	Cybersecurity incident	24 hours (active), 7 days (vulnerability)	Unauthorized system access, data exfiltration attempt, ransomware detection, authentication bypass

Table 13: Subpart J: Physical AI Systems Provisions (§ 312.400-312.405)

Section	Title	Scope
§ 312.400	System Registration	Centralized FDA database of all Physical AI systems in IND trials
§ 312.401	Validation Standards	Minimum simulation hours, benchmark thresholds, renewal schedule
§ 312.402	Cybersecurity Requirements	Encryption, access control, incident response, penetration testing
§ 312.403	Interoperability Requirements	MCP protocol support, FHIR/DICOM data formats, cross-platform standards
§ 312.404	Training Requirements	Tier-based certification for operators, maintenance, and oversight personnel
§ 312.405	Decommissioning Standards	Data preservation, hardware disposal, participant notification

6.9 Cross-Reference with Other Adapted Standards

The adapted 21 CFR Part 312 operates in coordination with the other two adapted standards to create a comprehensive regulatory framework. Sponsor and investigator responsibilities in Part 312 (§ 312.50-312.70) complement those defined in Section 3 of the adapted ICH E6(R3) [1]. Informed consent requirements referenced in Part 312 IND submissions are detailed in the adapted 21 CFR Part 50 [2]. The Phase 0 simulation validation requirement connects to the PSL and USL standards frameworks (Section 7) for quantitative assessment of simulation adequacy.

The 22 definitions in Part 312 § 312.3 are deliberately harmonized with the 30 definitions in the adapted ICH E6(R3) and the 17 definitions in the adapted Part 50. Where the same concept appears in multiple standards (such as “Physical AI Adverse Event”), the definitions use identical language to prevent inconsistency. Where a concept appears only in one standard (such as “Phase 0 Simulation Validation” which is unique to Part 312), the definition is included in that standard alone but cross-referenced in the others.

6.10 Expanded Adverse Event Reporting Requirements

The four-category adverse event framework summarized in Table 12 warrants expanded examination because it creates the first structured classification system for Physical AI-related events in clinical trials. Traditional adverse event reporting under the original Part 312 addresses drug-related events using seriousness criteria (death, life-threatening, hospitalization, disability, congenital anomaly, or other medically important condition). The Physical AI adaptation retains these seriousness criteria while adding the four system-specific categories that capture unique failure modes of robotic and AI systems in clinical settings. Table 14 presents the detailed adverse event reporting framework with expanded definitions, escalation procedures, documentation requirements, and root cause analysis mandates for each category.

Table 14: Physical AI Adverse Event Categories: Expanded Definitions, Documentation, and Root Cause Requirements

Attribute	Details
Category A: Physical AI System Malfunctions Affecting Patient Safety	
Expanded definition	Any failure, degradation, or unexpected behavior of a Physical AI system that directly affects or has the potential to directly affect patient safety during a clinical procedure or interaction. Includes mechanical failures, actuator malfunctions, control system errors, power system anomalies, and emergency stop system failures.
Serious event timeline	7 calendar days from investigator awareness to initial FDA report; 15 calendar days for complete follow-up report with root cause analysis.
Expedited trigger	24 hours for events during active surgical procedures, active radiation delivery, or any event resulting in documented patient injury.
Expanded examples	Surgical robot instrument arm loses position calibration during active procedure; force sensor malfunction causes excessive contact pressure during patient positioning; emergency stop system fails to activate within the specified 200-millisecond response time; robot actuator overheats during patient interaction causing unplanned procedure pause; end-effector detachment or unexpected tool release during surgical operation; robotic arm exceeds workspace boundary during patient-proximate operation; hydraulic or pneumatic leak in systems operating near patient.
Required documentation	Robot telemetry logs for 30 minutes before and after event; video recording of event if available; patient vital signs during event; operator actions taken and timeline; system error codes and diagnostic output; maintenance history for preceding 90 days; environmental conditions (temperature, humidity, vibration) at time of event.
Root cause analysis	Must determine whether malfunction was caused by hardware wear, software error, calibration drift, environmental conditions, operator error, or manufacturing defect. Root cause report due within 30 calendar days of initial event report.
Category B: AI Decision Errors Affecting Treatment	
Expanded definition	Any error, bias, or unexpected output from an AI algorithm that results in incorrect treatment delivery, incorrect monitoring assessment, or inappropriate clinical recommendation that is acted upon or could have been acted upon by clinical staff. Includes model inference errors, training data bias manifestations, and algorithm logic failures.
Standard timeline	15 calendar days from investigator awareness to FDA report for all Category B events.
Expedited trigger	7 calendar days if the AI decision error resulted in actual incorrect drug administration, confirmed dosing error, or treatment plan deviation that affected measurable patient outcomes.
Expanded examples	AI treatment planning algorithm recommends dosage outside protocol-specified range; AI eligibility screening algorithm incorrectly clears ineligible patient for enrollment; AI monitoring system fails to detect adverse physiological change that would trigger clinical intervention; AI-guided needle placement algorithm calculates trajectory exceeding specified tolerance; digital twin model produces simulation

The reporting timeline structure across all four categories follows a principle of proportional urgency. Category D active breaches require the fastest reporting (24 hours) because cybersecurity incidents may be ongoing and escalating, requiring immediate coordinated response. Category A serious events during active procedures require 24-hour expedited reporting because patient safety may be immediately at risk. Category B and C standard timelines of 15 calendar days allow adequate time for investigation of complex technical failures where the root cause may not be immediately apparent. The escalation triggers within each category ensure that the most consequential events receive the most urgent attention regardless of the base category timeline.

Each category integrates with the three-tier classification system from the adapted Part 50 (Section 5). Tier 3 systems (surgical and invasive robots) are subject to the most stringent reporting: all Category A events are reported on expedited timelines regardless of seriousness classification, and Category B events involving treatment delivery errors trigger 7-day expedited reporting. Tier 1 systems (logistics and support robots) follow standard timelines for most events, with expedited reporting reserved for events resulting in actual patient injury. This tier-based modulation prevents reporting burden from overwhelming clinical sites while ensuring that high-risk events receive immediate attention.

6.11 Expanded Subpart J Analysis

The six sections of Subpart J summarized in Table 13 follow a logical lifecycle progression that governs Physical AI systems from entry into the regulatory framework through ongoing operation to eventual decommissioning. The following analysis expands on each section's requirements, implementation considerations, and connections to other regulatory provisions.

§ 312.400 (Physical AI System Registration) creates a capability that does not exist in the current FDA regulatory framework: a centralized, real-time database of all robotic systems operating in clinical trials nationwide. Registration data includes manufacturer identification, model designation, software version with hash verification, hardware serial number, current USL score with evaluation date, tier classification, deploying institution with site identification number, assigned trial protocol number, and deployment date. Registration must be updated within 15 calendar days of any change to system configuration, software version, or deployment status. This database enables FDA to identify cross-trial patterns, such as a specific robot model generating disproportionate adverse events across multiple sites, that would be difficult to detect through traditional site-level reporting. The registration requirement also supports the federated learning framework [9] by providing a registry of systems that may participate in cross-site model improvement initiatives.

§ 312.401 (Physical AI Validation Standards) establishes quantitative thresholds that create clear, objective standards for Physical AI system readiness. The simulation hour requirements are tiered: Tier 1 systems require a minimum of 100 hours of validated simulation, Tier 2 systems require 250 hours, and Tier 3 systems require 500 hours. These hours must be accumulated on FDA-recognized simulation platforms, with performance metrics meeting category-specific benchmark thresholds. Validation renewal occurs annually or upon any major software update that alters operational parameters. The standardized validation report format includes simulation environment description, test scenario enumeration, performance metrics with statistical analysis, failure mode documentation, and an independent reviewer attestation. These requirements connect to the Phase 0 simulation validation concept in § 312.21, ensuring that the Phase 0 stage has defined, measurable completion criteria.

§ 312.402 (Cybersecurity Requirements) addresses a threat landscape unique to Physical AI trials. Requirements include AES-256 encryption for all data at rest, TLS 1.3 or higher for all data in transit, role-based access control with multi-factor authentication for all system access, network segmentation isolating Physical AI systems from general hospital networks, penetration testing by qualified third parties at least annually, automated vulnerability scanning at least monthly, and a documented incident response plan with a designated cybersecurity officer. The network segmentation requirement is particularly important because Physical AI systems often connect to both clinical networks (for patient data) and operational technology networks (for robot control), creating potential cross-domain attack paths that must be mitigated through architectural controls.

§ 312.403 (Interoperability Requirements) reflects the practical reality that Physical AI trial sites deploy robots from multiple manufacturers. Mandatory protocol support includes MCP [28] for AI-tool communication, FHIR [30] for health data exchange, and DICOM [31] for medical imaging data. Recommended protocol support includes ROS 2 for inter-robot communication, ONNX for AI model portability across platforms, and HL7 v2 for legacy healthcare system integration. Interoperability compliance must be demonstrated through conformance testing against published test suites before system deployment. The conformance testing results are included in the IND submission Physical AI System Description and reviewed by the IRB as part of the technical evaluation.

§ 312.404 (Personnel Training Requirements) establishes tiered training that ensures operator preparation matches system complexity and risk. Tier 1 operations require 8-hour general Physical AI awareness training covering robot identification, basic safety procedures, emergency stop activation, and patient communication. Tier 2 operations require 40-hour specialized operator certification including hands-on robot operation with supervised patient interactions, patient interaction protocols, troubleshooting procedures, and emergency response drills. Tier 3 operations require 80-hour advanced certification plus a minimum of 10 proctored cases demonstrating competency in console operation, instrument exchange, emergency conversion to manual procedures, and team communication protocols. All certifications require annual renewal with a minimum of 8 hours of continuing education specific to Physical AI advances and safety updates.

§ 312.405 (Decommissioning Standards) addresses the end-of-life stage that is frequently overlooked in regulatory frameworks. The formal decommissioning plan must be approved by the site Physical AI safety committee before any decommissioning activities begin. Requirements include complete data extraction from all local storage with verification of archival integrity, patient notification when decommissioning directly affects their care plan, hardware sanitization using NIST SP 800-88 guidelines to remove all patient data before disposal or transfer, regulatory notification to FDA updating the system registration status, and documentation retention for the duration specified by the applicable records retention schedule with a minimum of 2 years after trial completion. The decommissioning standards ensure that patient data is protected throughout the complete system lifecycle, including after the system is no longer operational in the clinical trial.

7 Physical AI Standard Level and Unification Standard

The Physical AI Standard Level (PSL) and Unification Standard Level (USL) frameworks complement each other to create a comprehensive, quantitative standards system for responsible robot usage in oncology clinical trials. PSL provides the regulatory compliance dimensions for trial site operations, while USL evaluates individual robot readiness. Together, they ensure that both the site infrastructure and the robots operating within it meet the standards required for safe, effective Physical AI oncology trials. This dual-layer quantitative assessment replaces subjective evaluation with reproducible, objective scoring that enables standardized nationwide deployment.

7.1 Overview: Complementary Standards

PSL and USL serve different but complementary purposes. PSL evaluates site-level regulatory compliance across three dimensions, determining whether a trial site has met the legislative, regulatory, and operational requirements for Physical AI trial operations. USL evaluates individual robot readiness across four dimensions, determining whether a specific robot platform is sufficiently mature for clinical trial deployment. A trial site can only operate when both conditions are satisfied: the site must achieve passing PSL scores (site is compliant) and the robots deployed at that site must achieve sufficient USL scores (robots are ready).

This dual-layer approach is unique to this National Platform and more thorough than any existing FDA guidance on robot or AI evaluation. Existing readiness frameworks such as NASA Technology Readiness Levels (TRL 1-9) evaluate hardware maturity but do not address clinical regulatory compliance, and existing AI evaluation frameworks do not capture the physical embodiment requirements of robotic systems in clinical settings.

7.2 Physical AI Standard Level (PSL) Framework

The PSL framework was developed in conjunction with the Clinical Trial Site Complete Documentation [5] and validated through the 24-hour clinical trial simulation that treated 168 patients with 29 robots.

7.2.1 The Three PSL Dimensions

PSL evaluates site-level compliance across three regulatory dimensions that correspond to the three layers of authorization required for Physical AI trial site operation.

Table 15: Physical AI Standard Level (PSL) Three Dimensions

Dim	Name	Scope
1	Legislative Authorization	State legislation enabling Physical AI trials (SB 1042, AB 2847, SB 892), federal statutory authority, municipal code compliance
2	Regulatory Compliance	Title 22 state regulations, FDA compliance guide, building and premises codes, 21 CFR Part compliance
3	Operational Readiness	Activation SOPs, emergency preparedness, staff training, equipment verification, PSL score verification

Dimension 1 (Legislative Authorization) measures whether the required legislative framework exists and applies to the trial site. For California, this includes SB 1042 (Physical AI Oncology Clinical Trial Authorization Act), AB 2847 (Patient Rights and Robotic Safety Act), and SB 892 (Data Protection and Transparency Act). Each legislative instrument is scored on completeness, applicability, and enforcement readiness.

Dimension 2 (Regulatory Compliance) measures whether the site meets all regulatory requirements across state and federal jurisdictions. This includes Title 22 regulations for healthcare facility operations, FDA compliance with the three adapted standards [1, 2, 3], building code standards for robot operations, and premises code requirements for robot storage, charging, and maintenance areas.

Dimension 3 (Operational Readiness) measures whether the site has completed all preparatory activities required for safe Physical AI operations. This includes activation standard operating procedures, emergency preparedness plans, staff training and certification, equipment verification and calibration, and formal PSL score verification by an independent assessor.

7.2.2 PSL Scoring Methodology

PSL scores are computed on a pass-fail basis for each dimension, with granular sub-scores for individual requirements within each dimension. A site must achieve passing status on all three dimensions before activation. Within each dimension, individual requirements are scored as: Compliant (requirement fully met with documentation), Partially Compliant (requirement substantially met with documented plan for completion), or Non-Compliant (requirement not met). A dimension passes when all critical requirements are Compliant and all non-critical requirements are at least Partially Compliant.

7.2.3 PSL and Site Activation

PSL scores determine whether a trial site can be activated for Physical AI operations. The activation process follows the procedures detailed in the site establishment documentation (Section 8), with PSL score verification serving as a gate before operational launch. The 24-hour simulation [5] validated this activation process by demonstrating that a site meeting PSL requirements could safely operate 29 robots serving 168 patients continuously.

7.3 Unification Standard Level (USL) Framework

The USL framework [4] provides a 1.0 to 10.0 scoring system for evaluating robot readiness across four equally weighted dimensions. USL was developed collaboratively using Claude Code Opus 4.6 [14] with scoring engines implemented in Python across three category-specific modules (cobots, surgical robots, humanoid robots) totaling over 2,600 lines of code.

7.3.1 The Four USL Dimensions

Table 16: Unification Standard Level (USL) Four Dimensions

Dim	Name	Wt	What It Measures
A	Simulation Switching	25%	Transfer of policies between simulation engines (Isaac Lab, MuJoCo, Gazebo, PyBullet, Drake)
B	AI Integration	25%	Integration with LLMs, VLAs, diffusion policies, MCP, agentic AI frameworks
C	Cross-Robot Sharing	25%	Intra- and inter-organization progress sharing via ONNX, ROS 2, federated learning
D	Clinical Trial Collab	25%	Regulatory compliance, federated deployment, audit trails across clinical trial sites

Dimension A (Simulation Framework Switching) evaluates a robot’s ability to transfer trained policies between simulation engines. Robots that operate in only one simulation environment score lower than those with validated multi-engine support. This dimension captures the practical reality that different trial sites may use different simulation platforms, and robots must function reliably across these environments.

Dimension B (AI Integration) evaluates the depth and breadth of AI system integration. This encompasses large language model planning capabilities, vision-language-action (VLA) model support, diffusion policy compatibility, Model Context Protocol [28] support, and agentic AI framework integration. Higher scores indicate robots that can leverage the full spectrum of modern AI capabilities.

Dimension C (Cross-Robot Sharing) evaluates the ability to share learned behaviors, trained models, and operational data across robots within and between organizations. ONNX model export, ROS 2 action interfaces, checkpoint sharing, and federated learning compatibility are key scoring factors. This dimension is essential for scaling Physical AI across multiple trial sites.

Dimension D (Clinical Trial Collaboration) evaluates regulatory compliance status, clinical deployment history, federated learning readiness, audit trail capabilities, and multi-site deployment infrastructure. This is the most directly relevant dimension for Physical AI oncology trials and was found to be the weakest dimension for seven of nine robots evaluated.

7.3.2 USL Score Computation

The final USL score is computed as the weighted average of all four dimension scores, clamped to the range 1.0 to 10.0 and rounded to the nearest 0.1:

$$\text{USL}_{\text{final}} = \text{clamp} \left(\frac{0.25 \cdot S_A + 0.25 \cdot S_B + 0.25 \cdot S_C + 0.25 \cdot S_D}{1.0}, 1.0, 10.0 \right)$$

Scores map to five bands: Exemplary (9.0-10.0), Advanced (7.0-8.9), Intermediate (5.0-6.9), Foundational (3.0-4.9), and Initial (1.0-2.9). Each band corresponds to a set of qualitative descriptors for clinical trial readiness, from fully unified multi-site deployment capability (Exemplary) to minimal unification capability (Initial).

7.4 USL Results Across Robot Categories

Table 17 presents the complete USL evaluation results for all nine robots across three categories, providing the quantitative foundation for robot selection in Physical AI oncology trials.

Table 17: Complete USL Scores for All Nine Evaluated Robots

Robot	Manufacturer	A	B	C	D	USL	Band
<i>Collaborative Robots (Cobots)</i>							
Franka Panda	Franka Robotics	8.0	7.7	8.5	5.5	7.4	Advanced
Kinova Gen3	Kinova Robotics	6.3	5.2	5.8	5.5	5.7	Intermediate
xArm 7	UFACTORY	4.6	3.2	3.8	2.1	3.4	Foundational
<i>Surgical Robots</i>							
da Vinci dVRK	Intuitive Surgical	7.5	7.2	6.8	7.0	7.1	Advanced
Hugo RAS	Medtronic	4.5	4.0	3.5	5.8	4.5	Foundational
Versius	CMR Surgical	3.2	2.8	2.5	5.2	3.4	Foundational
<i>Humanoid Robots</i>							
Atlas Electric	Boston Dynamics	7.0	7.4	4.5	4.2	5.8	Intermediate
Digit	Agility Robotics	5.8	5.4	3.0	2.7	4.2	Foundational
Optimus Gen 2	Tesla	3.4	5.0	1.5	4.4	3.6	Foundational

7.4.1 Surgical Robot Results

The da Vinci dVRK leads surgical robots with USL 7.1 (Advanced band) and achieves the highest Dimension D score (7.0) of any robot evaluated. This reflects FDA clearance, deployment at over 9,000 systems worldwide, 14 million procedures across 69 countries, and an open-source research platform used by 45 institutions. Franka Panda (USL 7.4) outperforms da Vinci overall due to superior simulation switching (8.0 vs 7.5) and cross-robot sharing (8.5 vs 6.8), reflecting its broader open-source ecosystem.

Hugo RAS (USL 4.5, Foundational) and Versius (USL 3.4, Foundational) score lower primarily due to proprietary ecosystems that limit simulation switching, AI integration, and cross-robot sharing. Both achieve relatively higher Dimension D scores (5.8 and 5.2 respectively) reflecting CE marking and active clinical deployment in European markets.

7.4.2 Cobot and Humanoid Results

Among cobots, the Franka Panda’s dominance (USL 7.4) is driven by support across five simulation frameworks, five AI capabilities, and four sharing methods including inter-organizational ONNX transfer. Kinova Gen3 (USL 5.7, Intermediate) offers strong clinical workflow compatibility but lacks VLA/diffusion policy support. The xArm 7 (USL 3.4, Foundational) demonstrates that hardware excellence (IP51 protection, built-in collision detection) does not compensate for limited software ecosystem maturity.

Among humanoids, Atlas Electric (USL 5.8, Intermediate) achieves the highest AI integration score (7.4) of any humanoid, driven by foundation model potential and whole-body control capabilities. Digit (USL 4.2, Foundational) benefits from the NVIDIA GR00T partnership but lacks healthcare deployment. Optimus Gen 2 (USL 3.6, Foundational) has the most capable hands (11 DOF, 22 joints) and mass production potential but scores the lowest cross-robot sharing score (1.5) of all nine robots due to its fully proprietary platform.

7.4.3 Cross-Category Comparison

The cross-category comparison reveals two key findings. First, open-source ecosystem size is the strongest predictor of USL score. The two highest-scoring robots (Franka 7.4, da Vinci 7.1) have the largest open-source communities. Second, Dimension D (Clinical Trial Collaboration) is the weakest dimension for seven of nine robots, revealing a field-wide gap between research maturity and clinical deployment infrastructure. Even the highest-scoring robot (Franka, 7.4 overall) scores only 5.5 on Dimension D.

7.5 How PSL and USL Complement Each Other

The three PSL dimensions and four USL dimensions work together as a dual-layer assessment. PSL ensures the environment is ready (site compliance); USL ensures the robots are ready (platform readiness). A Physical AI trial site requires both: a compliant, well-prepared facility (passing PSL) populated with properly qualified robots (sufficient USL scores).

A site might pass PSL on all three dimensions (legislation in place, regulatory compliance achieved, operations prepared) but be unable to deploy certain robot types because their USL scores are insufficient. For example, a California trial site meeting all PSL requirements could deploy da Vinci dVRK (USL 7.1, Advanced) for surgical procedures but might need to defer deployment of xArm 7 (USL 3.4, Foundational) until its software ecosystem matures. Conversely, a robot with excellent USL scores cannot operate at a site that fails PSL requirements, regardless of the robot's individual capabilities.

This complementary relationship is connected to the MCP conformance levels from Physical AI National MCP Servers [8], which provide an additional interoperability standard. MCP conformance levels interface with both PSL (ensuring site infrastructure supports MCP communication) and USL (evaluating whether robots support standardized MCP protocols). The Clinical Trial Site Complete Documentation [5] and Physical AI Unification Standard Level [4] serve as the primary reference implementations for PSL and USL respectively.

7.6 Impact on Nationwide Deployment

Standardized PSL and USL scores enable efficient nationwide scaling of Physical AI oncology trials. When a new site seeks to establish Physical AI trial operations, the PSL framework provides a clear checklist of legislative, regulatory, and operational requirements that must be met. When a sponsor evaluates which robots to deploy at a new site, USL scores provide objective data on each robot's readiness.

The open-source nature of both frameworks, available through the three code repositories [10, 11, 12], ensures that any institution can access, evaluate, and apply the scoring methodology. This transparency supports the broader goal of democratizing Physical AI oncology trial access rather than limiting it to institutions with proprietary evaluation processes.

The field-wide weakness in Dimension D (Clinical Trial Collaboration) identified by USL scoring highlights a critical area for targeted improvement. As the Physical AI ecosystem matures, Dimension D scores should improve as robots gain regulatory clearance, federated learning compatibility, audit trail capabilities, and multi-site deployment experience. The PSL and USL frameworks provide the measurement infrastructure to track this progress over time.

7.7 USL Level Definitions

The USL framework maps composite scores to ten granular levels, each representing a distinct stage of robot readiness for clinical trial deployment. These levels provide more precise guidance than the five-band classification by identifying specific capabilities and gaps at each stage.

Table 18: USL Level Definitions (1-10)

Lvl	Name	Description
1	Conceptual	Robot exists as a prototype or concept; no simulation or AI integration has been attempted; no clinical applicability demonstrated
2	Exploratory	Single simulation framework tested with basic robot model available; initial feasibility studies conducted
3	Basic	Two or more simulation frameworks supported; initial AI experiments conducted; basic model formats available
4	Developing	Cross-framework policy transfer demonstrated; AI planning capabilities tested; intra-organization sharing initiated
5	Functional	Three or more simulation frameworks validated; agentic AI operational; intra-organization sharing established with ONNX or equivalent
6	Integrated	Multi-framework validation complete; LLM-based task planning operational; inter-organization sharing demonstrated
7	Advanced	GPU-accelerated simulation with validated policy transfer; MCP and VLA integration; skill sharing across organizations
8	Clinical-Ready	Multi-site deployment tested; regulatory documentation prepared; federated learning compatibility demonstrated
9	Validated	Full regulatory compliance achieved; multi-site clinical trials active; continuous improvement pipeline operational
10	Exemplary	Production deployment at multiple sites; open consortium participation; continuous federated improvement with demonstrated clinical outcomes

7.8 Dimension Score Analysis by Category

Analysis of USL dimension scores across the three robot categories reveals systematic patterns that inform deployment strategy and investment priorities.

Table 19: Average USL Dimension Scores by Robot Category

Category	Dim A	Dim B	Dim C	Dim D	Avg USL
Cobots (3 robots)	6.3	5.4	6.0	4.4	5.5
Surgical (3 robots)	5.1	4.7	4.3	6.0	5.0
Humanoid (3 robots)	5.4	5.9	3.0	3.8	4.5
All 9 robots	5.6	5.3	4.4	4.7	5.0

Cobots lead in Dimensions A (Simulation Switching) and C (Cross-Robot Sharing) with averages of 6.3 and 6.0 respectively, reflecting the mature open-source ecosystems of platforms like Franka Panda. Surgical robots lead in Dimension D (Clinical Trial Collaboration) with an average of 6.0, driven by the da Vinci platform’s FDA clearance and extensive clinical deployment history. Humanoid robots lead in Dimension B (AI Integration) with an average of 5.9, reflecting advanced foundation model integration (GR00T) and whole-body control capabilities. However, humanoids score lowest on both Dimensions C (3.0) and D (3.8), indicating the youngest category with the most limited sharing infrastructure and clinical deployment experience.

The overall field average of 5.0 places the collective Physical AI ecosystem at the Functional level (Level 5), indicating that while individual capabilities are demonstrated, significant work remains before multi-site clinical trial readiness is achieved across all categories. The strongest individual scores (Franka Panda Dim C at 8.5, da Vinci Dim A at 7.5, Atlas Dim B at 7.4) demonstrate that excellence is achievable in specific dimensions, but no single robot achieves Advanced band or above across all four dimensions simultaneously. This finding reinforces the need for multi-robot strategies where different robot categories complement each other’s strengths.

8 Clinical Trial Site Establishment, Documentation

The Clinical Trial Site Complete Documentation [5] provides a comprehensive, ready-to-use package of eleven regulatory documents required for establishing the first Physical AI oncology clinical trial site in California. This documentation package is evidence-based, drawing from the 24-hour simulation that treated 168 patients with 29 robots at 99.7 percent uptime and zero patient harm events. The 24-hour simulation and A Cancer Patient’s Journey, Physical AI [6] are pivotal completed examples demonstrating that state-of-the-art AI can understand, work with, and scale Physical AI applications beyond human and prior technology capabilities.

8.1 Overview of the Documentation Package

The eleven documents collectively address every aspect of trial site establishment, organized across four categories: state legislation (three bills), municipal and state regulations (two documents), federal compliance (one guide), physical infrastructure (three documents), and operational procedures (two documents). The relationship between these documents follows a logical hierarchy: state legislation provides the legal authorization, municipal and state regulations implement that authorization, federal compliance bridges state requirements with national standards, physical infrastructure ensures the built environment supports Physical AI operations, and operational procedures enable day-to-day trial conduct.

Table 20: Clinical Trial Site Documentation Package Overview

No.	Document	Purpose
1	SB 1042	Trial Authorization and Site Establishment Act
2	AB 2847	Patient Rights and Robotic Safety Act
3	SB 892	Data Protection and Transparency Act
4	SF Municipal Code	Municipal code updates for Physical AI trial sites
5	Title 22 Regulations	State regulations for Physical AI site operations
6	FDA Compliance Guide	National compliance bridging state and federal requirements
7	Building Code	Structural standards for Physical AI facilities
8	Premises Code	Operational space requirements for robot areas
9	Parking/Transportation	Patient logistics and accessibility standards
10	Activation SOPs	Pre-activation checklists and daily operations
11	Emergency Preparedness	Emergency response for Physical AI scenarios

This documentation package is more comprehensive than existing FDA site establishment guidance because it addresses not just regulatory compliance but also physical infrastructure, operational procedures, patient logistics, and emergency preparedness in an integrated framework designed specifically for Physical AI trial operations.

8.2 State Legislation for Physical AI Trial Sites

8.2.1 SB 1042: Physical AI Oncology Clinical Trial Authorization Act

SB 1042 provides the legislative framework authorizing Physical AI oncology clinical trial sites in California. The bill establishes findings that Physical AI integration into oncology clinical trials has the potential to improve patient outcomes, reduce treatment costs, and accelerate drug

development. Key provisions include: authorization for qualified healthcare facilities to operate Physical AI systems in oncology clinical trials upon meeting specified conditions; establishment of a Physical AI Trial Site Certification process administered by the California Department of Public Health; requirements for ongoing monitoring and reporting of Physical AI trial outcomes to the state legislature; and a sunset provision requiring legislative reauthorization after an initial operating period.

The authorization process requires demonstration of compliance with all federal regulatory requirements (including the three adapted standards [1, 2, 3]), evidence of adequate Physical AI system validation (including Phase 0 simulation results), qualified personnel with Physical AI-specific training, and adequate physical infrastructure meeting building and premises code requirements. This authorization framework directly corresponds to PSL Dimension 1 (Legislative Authorization).

8.2.2 AB 2847: Patient Rights and Robotic Safety Act

AB 2847 establishes legally enforceable patient rights specific to Physical AI interactions. This represents a fundamental shift of control towards the patient's side. The bill establishes: the right to complete information about all Physical AI systems involved in the patient's care, including robot type, manufacturer, capabilities, and safety record; the right to refuse any robotic procedure and receive equivalent human-delivered care; the right to request a pause in any active robotic procedure; the right to real-time information about robot operations during procedures; the right to access all data generated by Physical AI systems during the patient's care; the right to a human advocate present during all Physical AI interactions; and the right to file complaints about Physical AI system performance through an expedited review process.

These rights go beyond traditional informed consent by creating affirmative obligations for trial sites and sponsors. A patient does not merely consent to robotic care; they retain ongoing, enforceable rights throughout their participation. Patient Instructions: Physical AI Trials [7] implements these rights in accessible language for each of the ten robot types.

8.2.3 SB 892: Data Protection and Transparency Act

SB 892 addresses the unique data privacy challenges created by Physical AI clinical trials. The bill establishes specific protections for robot-generated health data including telemetry, video recordings, sensor data, and AI decision logs. Key provisions include: classification of Physical AI-generated data as protected medical information under CMIA; requirements for data minimization in Physical AI data collection; transparency requirements for AI decision-making processes that affect patient care; patient access rights for all Physical AI data associated with their treatment; data retention and destruction schedules specific to Physical AI records; and penalties for unauthorized use or disclosure of Physical AI trial data.

8.3 Municipal and State Regulations

8.3.1 San Francisco Municipal Code Update

The municipal code update addresses local requirements for Physical AI trial sites in San Francisco, covering zoning classification for Physical AI clinical facilities (permitted in medical and research zones with Physical AI supplemental use permit), permitting requirements including Physical AI-specific building permits, operational hours (24-hour operations permitted with noise

mitigation for robot maintenance activities during overnight hours), community notification requirements (advance notice to neighboring properties when new Physical AI systems are deployed), and local emergency service coordination (fire department and emergency medical services briefed on Physical AI system locations and emergency procedures).

8.3.2 Title 22 Regulations

The California Code of Regulations, Title 22 requirements establish Physical AI oncology trial site operational standards. These regulations cover facility licensing standards adapted for Physical AI operations, Physical AI system staffing requirements including minimum operator-to-robot ratios, robot maintenance schedules with documented service intervals, quality assurance protocols for Physical AI system performance, infection control procedures for robot surfaces and components, patient-to-robot interaction standards, and data management requirements specific to Physical AI systems. Compliance with Title 22 requirements directly corresponds to PSL Dimension 2 (Regulatory Compliance).

8.4 Federal Compliance

8.4.1 FDA Physical AI Oncology Trial Site National Compliance Guide

The national compliance guide bridges California-specific requirements with federal FDA regulations, providing a framework that can be adapted for Physical AI trial sites in any state. The guide establishes that sites must demonstrate compliance with the three adapted regulatory standards: the Adaption: ICH Harmonised Guideline [1] for good clinical practice, the Physical AI Adaption: 21 CFR Part 50 [2] for human subjects protection, and the Physical AI Adaption: 21 CFR Part 312 [3] for IND requirements.

Compliance requirements include: IND submission with Physical AI System Description, Phase 0 simulation validation completion, IRB approval with Physical AI-specific review, Physical AI adverse event reporting procedures, electronic records compliance under 21 CFR Part 11, HIPAA [17] compliance for Physical AI-generated health data, and MCP server configuration meeting national interoperability standards [8].

8.5 Physical Infrastructure Requirements

8.5.1 Building Code Standards

Building code standards address structural requirements for Physical AI trial facilities. Robot operations create specific building demands including reinforced flooring to support robot weight and dynamic forces during operation, power supply specifications providing dedicated circuits for robot charging and operation with uninterruptible power supply (UPS) backup, data network infrastructure supporting real-time data streaming from multiple robots simultaneously, HVAC requirements maintaining temperature and humidity within robot operational specifications, electromagnetic interference (EMI) shielding in areas where robot operations might affect sensitive medical equipment, and accessibility requirements ensuring that Physical AI trial areas comply with ADA standards for patients with disabilities.

8.5.2 Premises Code

The premises code establishes requirements for the operational spaces within Physical AI trial facilities. Robot storage areas must provide secure, climate-controlled spaces for robots not in active use. Charging stations must be accessible to staff for routine maintenance while secured against unauthorized access. Maintenance bays must support robot servicing, calibration, and repair with appropriate tools and safety equipment. Patient interaction zones must be designed to accommodate robot movement paths while ensuring patient comfort and safety. Safety buffer zones must separate robot operational areas from non-trial spaces, with physical barriers and warning indicators.

8.5.3 Parking and Transportation Standards

Patient transportation logistics address the practical challenges of oncology patients accessing Physical AI trial sites. Requirements include accessible parking with spaces designated for patients with mobility limitations from oncology treatments, patient drop-off zones with covered areas designed for patients who may need wheelchair or gurney access, ride-share and medical transport accommodation areas, emergency vehicle access adjacent to robot operational areas, and wayfinding signage directing patients through Physical AI areas with clear, non-technical language. Patient convenience is a key benefit of Physical AI trials, and these infrastructure standards minimize patient burden while ensuring comfortable, accessible facilities.

8.6 Operational Procedures

8.6.1 Activation and Standard Operating Procedures

The activation process follows a structured protocol. The pre-activation checklist covers building inspection completion, equipment installation verification, staff training certification, communication system testing, safety system verification, PSL score calculation and verification, and formal activation decision by the site director. Daily operations SOPs cover robot startup procedures (power-on sequence, self-test, calibration check, network registration), patient scheduling with Physical AI systems (coordinating robot availability with patient appointments), data management workflows (real-time data capture, quality checks, secure storage, regulatory reporting), and quality control checks (periodic robot performance verification during operations).

8.6.2 Emergency Preparedness

The emergency preparedness plan addresses Physical AI-specific emergency scenarios. Robot malfunction responses define escalation procedures from automatic pause through manual override to complete shutdown, with response time requirements for each level. Power failure procedures ensure safe robot shutdown, patient safety during power transitions, and system recovery after power restoration. Cybersecurity incident responses cover detection, containment, eradication, and recovery procedures for Physical AI system compromises. Natural disaster protocols address earthquake (California-specific), fire, and flood scenarios with Physical AI-specific considerations. Patient evacuation procedures account for active Physical AI systems, ensuring robots are safely deactivated before or during evacuation.

8.7 Evidence from the 24-Hour Simulation

The evidence for Physical AI oncology trial site establishment is overwhelmingly based on the 24-hour simulation conducted within the Clinical Trial Site Complete Documentation framework [5]. The simulation demonstrated comprehensive operational capability across all eleven documentation areas.

Table 21: 24-Hour Simulation Key Metrics

Metric	Result
Patients treated	168 patients across 15 cancer types
Robots deployed	29 robots across multiple categories
System uptime	99.7 percent continuous operation
Patient harm events	Zero patient harm events
Adverse events	7 events documented per ICH E6(R3)
Operating hours	24 continuous hours
Documentation	72 ASCII diagrams documenting operations
Regulatory coverage	Mapped to all three adapted standards

The simulation validated each component of the documentation package. SB 1042 authorization requirements were tested through the certification process simulation. AB 2847 patient rights were exercised through simulated patient interactions where rights were invoked and honored. SB 892 data protection requirements were tested through data collection, storage, and access scenarios. Building and premises code requirements were validated against the simulated facility specifications. Activation SOPs were executed in sequence before simulation start. Emergency preparedness procedures were tested through simulated malfunction and power failure scenarios.

The PSL framework scores achieved during the simulation confirmed that the documentation package is sufficient for real-world site establishment. Simulation data identified areas for refinement in several documents, leading to revisions that strengthened the overall package.

Table 22: 24-Hour Simulation Robot Deployment by Category

Robot Category	Functions Performed	Count
Surgical robots	Tumor resection, biopsy guidance	Multiple
Collaborative robots	Medication prep, sample handling, positioning	Multiple
RT positioning robots	Patient alignment for radiation therapy	Multiple
Needle-placement robots	Biopsy and targeted drug delivery	Multiple
Social companion robots	Emotional support, information delivery	Multiple
Humanoid robots	Patient guidance, supply transport	Multiple
RT motion-tracking robots	Real-time motion compensation	Multiple
Imaging robots	Diagnostic scanning, monitoring imaging	Multiple
Steerable needle robots	Complex anatomy targeting	Multiple
Rehabilitation exoskeletons	Post-treatment mobility support	Multiple
Total	All Physical AI functions	29

The 29 robots deployed in the simulation covered all ten robot categories documented in Patient Instructions: Physical AI Trials [7]. Each category contributed to the overall 99.7

percent uptime, with the seven documented adverse events distributed across categories and managed through the adapted ICH E6(R3) safety reporting procedures. No single category contributed disproportionately to adverse events, supporting the conclusion that the multi-robot Physical AI model is operationally viable.

These results, combined with the single-patient journey evidence [6], provide the empirical foundation for Physical AI oncology trial site establishment nationwide. The economic analysis of simulation results is presented in Section 13.

9 A Cancer Patient's Journey Through Physical AI

A Cancer Patient's Journey, Physical AI [6] provides definitive evidence that Physical AI can manage a complete oncology trial patient journey autonomously across ten stages, with documented safety outcomes, regulatory compliance, and cost savings. This single-patient simulation documents a Stage IIIB non-small cell lung cancer (NSCLC) patient over 1,120 days, demonstrating end-to-end Physical AI capability from prescreening through closeout. Together with the 24-hour multi-patient simulation [5], this constitutes pivotal evidence that state-of-the-art AI has the ability to understand, work with, and scale Physical AI applications beyond human and prior technology capabilities.

9.1 Ten-Stage Pipeline Architecture

The patient journey is organized into ten sequential stages, each documented with regulatory mapping, clinical activities, Physical AI system involvement, and outcomes. The stages span the complete trial lifecycle, demonstrating that Physical AI is not limited to isolated robotic procedures but can orchestrate the entire patient experience.

Table 23: Ten-Stage Physical AI Patient Journey Pipeline

Stage	Name	Physical AI Activities
1	Prescreening	AI-assisted eligibility assessment, record review
2	Screening	Physical AI diagnostic imaging, biomarker analysis
3	Informed Consent	AI-enhanced consent with robot demonstrations
4	Enrollment	Automated registration, baseline data collection
5	Treatment Planning	Digital twin modeling, treatment optimization
6	Surgical Intervention	Robot-assisted tumor resection
7	Post-Surgical Care	Robotic monitoring, rehabilitation support
8	Follow-up Treatment	AI-guided therapy, companion robot support
9	Long-term Monitoring	Automated surveillance, imaging follow-up
10	Closeout	Automated reporting, regulatory documentation

The ten-stage pipeline demonstrates end-to-end capability that differentiates Physical AI oncology trials from prior robotic surgery studies focusing only on surgical outcomes. Each stage is documented with three-perspective ASCII diagrams showing the regulatory, clinical, and patient views simultaneously.

9.2 Pre-Trial Stages: Prescreening Through Enrollment

Stages 1 through 4 cover the pre-trial process from initial patient identification through formal enrollment. Stage 1 (Prescreening) uses AI algorithms to review patient medical records against trial eligibility criteria, identifying candidates with Stage IIIB NSCLC who meet inclusion criteria and have no disqualifying conditions. Physical AI systems assist by automating record retrieval, cross-referencing imaging databases, and flagging potential candidates for physician review.

Stage 2 (Screening) deploys Physical AI imaging systems for diagnostic confirmation, including robotic CT-guided biopsy for tissue sampling and AI-assisted pathology analysis for

tumor characterization. Stage 3 (Informed Consent) implements the Physical AI-specific consent requirements from the adapted 21 CFR Part 50 [2], including robot demonstrations, AI transparency explanations, and documentation of patient rights under AB 2847 from the site documentation package [5].

Stage 4 (Enrollment) uses automated systems for patient registration, baseline data collection, and treatment arm assignment. The regulatory framework mapping shows how each pre-trial stage complies with the adapted standards: the Adaption: ICH Harmonised Guideline [1] governs investigator conduct at each stage, the adapted 21 CFR Part 50 governs informed consent and human subjects protection, and the adapted 21 CFR Part 312 [3] governs IND compliance and Phase 0 validation completion.

9.3 Treatment Stages: Surgical and Therapeutic Interventions

Stages 5 through 8 cover the active treatment period. Stage 5 (Treatment Planning) uses digital twin technology to create a patient-specific computational model of the tumor and surrounding anatomy, enabling simulation of multiple treatment approaches before selecting the optimal strategy. AI algorithms analyze the digital twin against historical outcomes data to predict treatment response.

Stage 6 (Surgical Intervention) documents the robot-assisted tumor resection performed by a surgical robot system. The procedure is monitored through the Physical AI safety matrix requirements, with continuous force, position, and vital sign monitoring. Surgical outcomes include procedure duration, blood loss, margin clearance, and immediate post-operative status. The adverse event profile documents any robot-related events during the procedure.

Stage 7 (Post-Surgical Care) deploys rehabilitation exoskeletons for mobility support, companion robots for emotional support, and monitoring cobots for continuous vital sign surveillance. Stage 8 (Follow-up Treatment) covers adjuvant therapy with AI-guided dosing optimization and Physical AI-assisted treatment delivery.

USL scoring [4] is applied during treatment stages to verify that each robot involved in patient care meets minimum readiness thresholds. Surgical robots must achieve Advanced band (USL 7.0 or higher) for Dimension D before performing procedures, while support robots (cobots, companions) may operate at Intermediate band (USL 5.0 or higher).

9.4 Post-Treatment Stages: Closeout and Analysis

Stages 9 and 10 cover the post-treatment period through trial closeout. Stage 9 (Long-term Monitoring) uses automated Physical AI surveillance systems for scheduled imaging, blood work monitoring, and symptom assessment over the 1,120-day follow-up period. AI algorithms analyze trends in monitoring data to detect potential recurrence or late treatment effects. Stage 10 (Closeout) automates the generation of regulatory documentation including case report forms, safety narratives, and protocol compliance reports, ensuring that all Physical AI-generated data is properly archived per the retention requirements established in the adapted standards.

9.5 Regulatory Compliance Coverage

The patient journey demonstrates compliance with all three adapted regulatory standards at every stage. The regulatory framework mapping documents which specific provisions of each

adapted standard apply at each stage, creating a traceability matrix from clinical activities to regulatory requirements.

Table 24: Regulatory Standard Coverage Across Journey Stages

Stage	ICH E6(R3)	21 CFR Part 50	21 CFR Part 312
1-2	Investigator qualifications	Subject identification	IND eligibility
3-4	Informed consent procedures	Consent elements	Enrollment procedures
5-6	Quality management	Safety matrix	Phase reporting
7-8	Safety reporting	Ongoing consent	Adverse event reporting
9-10	Record keeping	Data protection	Annual/final reports

9.6 Physical AI Ecosystem and Data Flow

Data flows between Physical AI systems, clinical staff, regulatory systems, and patients throughout the ten-stage journey. The Physical AI ecosystem comprises multiple robot types operating in coordinated workflows, with data exchange managed through MCP servers [8] using the Model Context Protocol [28]. Each robot generates data streams that are captured, validated, and routed to appropriate destinations: clinical databases for care coordination, regulatory systems for compliance reporting, and research databases for outcome analysis.

The data flow architecture ensures that no single robot operates in isolation. Surgical robots share procedure data with monitoring cobots, which in turn update patient records accessed by companion robots for post-procedure support. This integrated data flow supports the continuous monitoring requirements of the adapted 21 CFR Part 50 [2] and the data governance provisions of the adapted ICH E6(R3) [1].

9.7 FDA Cost-Savings Analysis

The patient journey includes a detailed financial analysis documenting per-patient cost reductions at each stage. Physical AI reduces costs through automated screening (Stage 1-2), streamlined consent processes (Stage 3), efficient enrollment (Stage 4), AI-optimized treatment planning (Stage 5), precision surgical intervention reducing complications and hospital stay (Stage 6), automated post-surgical monitoring reducing nursing workload (Stage 7), AI-guided follow-up reducing unnecessary clinic visits (Stage 8-9), and automated closeout documentation reducing administrative burden (Stage 10).

Tufts CSDD data [18] establishes that each day of delay in drug development has significant economic consequences. The timeline compression demonstrated in the patient journey, from faster screening and enrollment through more efficient treatment delivery and automated documentation, translates directly to accelerated drug development timelines. NCI data on trial costs [33] provides context for understanding the economic burden that Physical AI can reduce.

9.8 Trial Timeline Comparison

The Physical AI trial timeline introduces the Phase 0 simulation validation stage but achieves net time savings through compression at every subsequent phase. Enrollment speed increases through automated screening and AI-assisted eligibility verification. Treatment throughput increases through 24-hour robotic operations and reduced per-procedure times. Data processing

accelerates through automated documentation and real-time regulatory reporting through MCP servers. The net effect is a significantly shorter overall trial timeline despite the addition of the Phase 0 requirement.

9.9 Discussion of Patient Journey Evidence

The patient journey evidence is significant for several reasons. First, it demonstrates autonomous workflow capability across all ten stages, establishing that Physical AI is not limited to individual procedures but can manage complete trial operations. Second, it validates the regulatory adaptations by showing that a real patient scenario can be mapped to the adapted standards at every stage. Third, it provides the financial data that underpins the economic analysis in Section 13. Fourth, it establishes safety performance benchmarks against which future Physical AI trials can be measured.

Both the single-patient journey and the 24-hour multi-patient simulation are pivotal completed examples demonstrating state-of-the-art AI ability to understand, proficiently work with, and scale Physical AI applications. The evidence is comprehensive: safety data, regulatory compliance, cost analysis, and patient outcomes all support the transition to Physical AI oncology trials.

9.10 Limitations and Future Directions

The single-patient simulation has acknowledged limitations: it documents a simulated rather than real patient, covers only one cancer type (Stage IIIB NSCLC), and represents outcomes from AI-generated scenarios rather than clinical observations. These limitations are partially addressed by the 24-hour multi-patient simulation covering 15 cancer types and 168 patients. Full validation requires real-world clinical trials, which this National Platform is designed to enable through its comprehensive regulatory, standards, infrastructure, and documentation framework.

10 Patient Instructions for Physical AI Oncology Trials

Patient Instructions: Physical AI Trials [7] provides clear, accessible, patient-facing documentation for participants in Physical AI oncology trials. This document demonstrates that control has shifted towards the patient’s side, with comprehensive information about every robot type participants may encounter. The patient instructions complement the informed consent requirements of the adapted 21 CFR Part 50 [2] and the patient rights provisions of AB 2847 from the Clinical Trial Site Complete Documentation [5].

10.1 Purpose and Design Philosophy

The patient instructions were designed following three principles: plain language accessibility ensuring patients of all education levels can understand the information, consistent structure for each robot type enabling easy comparison and reference, and emphasis on patient rights and comfort establishing that robots serve patients rather than the reverse.

Each robot type entry follows a standardized format: what the robot does and why it is used in the trial, how the robot interacts with the patient (what the patient will see, hear, and feel), safety features protecting the patient during interaction, the patient’s rights specific to that robot type (including the right to refuse and request human alternatives), and what to do if the patient has concerns or experiences discomfort.

This design philosophy reflects the National Platform’s commitment to patient-centered trial design. NCI information on clinical trial safety [34] documents common patient concerns about trial participation; Physical AI introduces additional concerns about robotic interaction that these instructions specifically address. By providing comprehensive, understandable information about each robot type, patients can make informed decisions about their participation and exercise their rights effectively.

10.2 Surgical Robots

Surgical robots (Robot Type 1) include the da Vinci system, Hugo RAS, and Versius. Patient instructions explain that surgical robots assist the surgeon in performing precise operations through small incisions. Patients learn that the robot does not operate independently but follows the surgeon’s commands in real time. Safety features described include redundant control systems, force limitation mechanisms, and the ability of the surgical team to stop the robot instantly. Patient rights include the right to ask questions about the robot before surgery, the right to request a non-robotic surgical approach, and the right to have the surgical team explain each step during the procedure. Cross-referencing with USL surgical robot scores [4] confirms that the da Vinci dVRK achieves the highest clinical trial readiness score (Dimension D: 7.0) of any robot evaluated.

10.3 Collaborative Robots (Cobots)

Cobots (Robot Type 2) include the Franka Panda, Kinova Gen3, and xArm 7. Patient instructions explain that cobots assist clinical staff with tasks such as preparing medications, handling laboratory samples, and supporting patient positioning. Patients learn that cobots are designed to work safely alongside humans, with built-in sensors that detect unexpected contact and automatically stop movement. Cobot interactions are typically indirect, with the

robot handling materials rather than touching the patient directly, though some tasks (such as handing items to patients) involve proximity interaction.

10.4 Radiotherapy Positioning Robots

Radiotherapy positioning robots (Robot Type 3) assist with precise patient positioning for radiation therapy treatments. Patient instructions explain that accurate positioning is critical for targeting tumors while protecting healthy tissue, and that robotic positioning achieves sub-millimeter accuracy exceeding manual positioning capability. Patients learn that the positioning process involves the robot gently adjusting the treatment table or patient support system, with the patient able to communicate with the therapy team throughout the process. Safety features include continuous position monitoring, force limits that prevent excessive pressure, and immediate stop capability.

10.5 Needle-Placement Robots

Needle-placement robots (Robot Type 4) guide needles for biopsies and targeted drug delivery with precision that exceeds manual capability. Patient instructions explain that robotic guidance reduces the number of needle attempts needed, shortens procedure time, and improves targeting accuracy. Patients learn that the robot positions and guides the needle while a qualified physician oversees the entire procedure. Comfort measures described include local anesthesia, continuous communication with the clinical team, and the ability to pause the procedure at any time.

10.6 Social Companion Robots

Social companion robots (Robot Type 5) represent a unique aspect of Physical AI oncology trials with no analog in traditional trials. Patient instructions explain that companion robots provide emotional support, information delivery, and well-being monitoring during trial participation. These robots can answer questions about the trial process, provide relaxation exercises, offer entertainment during treatment sessions, and alert clinical staff if the patient appears distressed.

Patient instructions emphasize that companion robot interactions are entirely optional and that patients may choose to interact with human staff exclusively. The companion robot's ability to monitor patient mood and well-being is explained transparently, with patients informed about what data is collected and how it is used. This robot type addresses the emotional and psychological needs of cancer patients, demonstrating that Physical AI is about patient well-being and convenience as well as clinical efficiency.

10.7 Humanoid Robots

Humanoid robots (Robot Type 6) including Tesla Optimus, Digit, and Atlas serve multi-functional roles in trial sites including patient guidance through facility areas, logistics support such as supply delivery and specimen transport, and general assistance. Patient instructions address the unique consideration of human-shaped robots: patients may find humanoid robots more approachable due to their familiar form, but some patients may find them unsettling. Instructions normalize both reactions and emphasize that patients can request that humanoid robots not approach them.

10.8 Radiotherapy Motion-Tracking Robots

Radiotherapy motion-tracking robots (Robot Type 7) continuously monitor patient position during radiation therapy to ensure treatment accuracy. Patient instructions explain that breathing and involuntary movement can shift the treatment target, and that motion-tracking robots detect and compensate for this movement in real time. Patients learn that the system may pause treatment automatically if movement exceeds safe thresholds, and that this is a safety feature rather than a malfunction. Safety interlocks ensure that radiation is delivered only when positioning is confirmed accurate.

10.9 Imaging Robots

Imaging robots (Robot Type 8) perform diagnostic and monitoring imaging including robotic CT positioning, MRI-compatible robotic systems, and ultrasound guidance. Patient instructions explain that robotic imaging systems can achieve more consistent positioning, faster scan times, and in some cases reduced radiation exposure compared to manual imaging. DICOM [31] standards ensure that images are stored and shared in standardized formats accessible to all members of the clinical team. Patients learn that imaging data is protected under HIPAA [17] and the data protection provisions of SB 892.

10.10 Steerable Needle Robots

Steerable needle robots (Robot Type 9) represent advanced guidance systems for accessing complex anatomical targets. Unlike traditional straight-needle insertion, steerable needle robots can navigate curved paths through tissue, reaching targets that would be inaccessible or require multiple attempts with manual techniques. Patient instructions explain the precision advantages and note that the technology is particularly valuable for tumors in difficult locations, reducing procedure time and improving targeting accuracy.

10.11 Rehabilitation Exoskeletons

Rehabilitation exoskeletons (Robot Type 10) support post-treatment recovery for oncology patients. Patient instructions explain that exoskeletons provide powered assistance for movement, helping patients rebuild strength and mobility after surgery or prolonged treatment. The exoskeleton adapts to the patient's capabilities, providing more support when needed and gradually reducing assistance as the patient strengthens. Patients learn that the exoskeleton monitors their movement patterns and reports progress to the clinical team.

10.12 Patient Benefits Summary

Physical AI oncology trials offer patients multiple documented benefits across the ten robot types. Convenience benefits include reduced wait times through automated scheduling, streamlined processes through robotic assistance, and 24-hour availability of robotic monitoring and support. Cost benefits include lower treatment costs through reduced staffing overhead, fewer unnecessary clinic visits through AI-guided monitoring, and shorter hospital stays through precision surgical robotics. Quality benefits include sub-millimeter surgical precision, continuous rather than periodic monitoring, AI-optimized treatment plans, and comprehensive data-driven

care. Emotional benefits include companion robot support, reduced treatment anxiety through transparent information, and empowerment through comprehensive patient rights.

NCI information on trial costs [33] provides context for the financial benefits, while NCI safety information [34] addresses patient concerns about trial participation. The patient instructions implement the patient rights provisions of the adapted 21 CFR Part 50 [2], the consent requirements of the adapted ICH E6(R3) [1], and the legally enforceable rights from AB 2847 in the Clinical Trial Site Complete Documentation [5], creating a comprehensive patient communication framework that supports informed, empowered participation in Physical AI oncology trials.

11 National MCP Server Infrastructure

Physical AI National MCP Servers [8] provides the national-scale infrastructure for connecting Physical AI oncology trial systems across sites using the Model Context Protocol (MCP) [28]. The five-server architecture addresses the fragmentation problem that prevents current clinical trial systems from interoperating, establishing standardized communication pathways for Physical AI operations at scale. The infrastructure is implemented in the National MCP Servers Repository [11] with 23 tools, conformance levels, safety modules, and deployment configurations.

11.1 The Fragmentation Problem

Current clinical trial systems operate in silos with incompatible data formats, disconnected communication channels, and no standardized protocol for real-time inter-system coordination. Electronic data capture (EDC) systems, laboratory information management systems (LIMS), clinical trial management systems (CTMS), and regulatory submission platforms each use proprietary data formats and communication protocols. ClinicalTrials.gov [19] provides trial registration but not real-time operational coordination. FDA clinical trials guidance [21] addresses individual system requirements but not inter-system communication.

Physical AI magnifies this fragmentation because robots, AI systems, clinical databases, and regulatory reporting tools must all communicate seamlessly in real time. A surgical robot generating telemetry data must share that data with monitoring systems, clinical databases, regulatory reporting tools, and safety interlock systems simultaneously. FHIR [30] addresses healthcare data exchange and DICOM [31] addresses medical imaging, but neither provides a complete solution for the real-time, multi-modal data exchange required by Physical AI systems.

11.2 Model Context Protocol for Physical AI

The Model Context Protocol [28] provides a standardized framework for AI systems to communicate with tools and data sources. Originally designed for language model tool use, MCP is adapted for Physical AI oncology trial operations by extending the protocol to handle robot control commands, sensor data streams, safety interlock signals, and regulatory reporting messages. The prior architecture that the five-server design builds upon includes individual MCP server implementations for robot control, data management, and safety monitoring that were developed in earlier versions of the Physical AI platform [10].

The adaptation of MCP for Physical AI establishes several key capabilities: bidirectional communication between AI planning systems and physical robots, standardized data schemas for robot telemetry and sensor readings, real-time safety signal propagation across all connected systems, and audit trail generation compliant with 21 CFR Part 11 electronic records requirements.

11.3 Five-Server Architecture

The national MCP infrastructure comprises five specialized servers, each responsible for a distinct aspect of Physical AI trial operations.

Table 25: Five-Server MCP Architecture for Physical AI Oncology Trials

No.	Server	Responsibilities
1	Governance Server	Trial protocol management, regulatory compliance tracking, role-based access control, audit trail management
2	Data Flow Server	Real-time data routing, format transformation, data validation, storage coordination across FHIR and DICOM
3	Safety Server	Emergency stop propagation, safety interlock management, anomaly detection, incident reporting
4	Interoperability Server	Cross-platform translation, device registration, capability negotiation, standards conformance
5	Analytics Server	Real-time dashboards, trend analysis, outcome reporting, predictive analytics for trial management

The Governance Server manages trial protocol configurations, ensuring that all Physical AI operations comply with the approved protocol and the three adapted regulatory standards [1, 2, 3]. The Data Flow Server handles the high-volume, multi-modal data streams generated by Physical AI systems, routing data to appropriate destinations based on type, urgency, and regulatory requirements. The Safety Server provides the most time-critical function: propagating emergency stop signals across all connected Physical AI systems within milliseconds, ensuring that a safety event detected by any system triggers appropriate responses across the entire infrastructure.

The Interoperability Server addresses the challenge of connecting diverse Physical AI systems from different manufacturers with different communication protocols. It provides translation services, device registration, and capability negotiation, ensuring that a Franka Panda cobot can exchange data with a da Vinci surgical robot despite different native protocols. The Analytics Server aggregates data from all other servers to provide real-time operational dashboards, trend analysis, and predictive analytics that support trial management decisions.

11.4 Tool Inventories and Conformance Levels

The five servers collectively provide 23 tools organized into functional categories. Each tool has a defined conformance level specifying the minimum requirements for implementations. Conformance levels range from Level 1 (basic data exchange) through Level 3 (full bidirectional integration with audit trail and safety interlock support). The MCP repository [11] provides reference implementations for all 23 tools at Level 2 or above.

Tool categories include robot management tools (registration, status query, command execution, calibration verification), data management tools (data ingestion, transformation, validation, archival), safety tools (emergency stop, interlock query, anomaly reporting, incident documentation), compliance tools (protocol verification, consent tracking, reporting generation, audit trail query), and analytics tools (dashboard generation, trend computation, outcome aggregation, performance scoring).

11.5 Safety Modules and Emergency Systems

The safety architecture embedded in the MCP server infrastructure provides multiple layers of protection. The emergency stop system propagates stop signals from any source (hardware

button, software trigger, automated anomaly detection) to all connected Physical AI systems within a specified latency threshold. The automatic failover system ensures that safety functions continue even if individual server components fail, using redundant safety modules that operate independently of the primary server infrastructure.

Safety interlock protocols ensure that Physical AI systems cannot perform restricted operations without proper authorization. For example, a surgical robot cannot initiate a procedure until the Governance Server confirms that the patient has provided informed consent, the Safety Server confirms that all pre-procedure safety checks have passed, and the Data Flow Server confirms that all monitoring systems are recording. This multi-server interlock provides defense-in-depth that exceeds the safety capabilities of any single system.

11.6 Integration and Interoperability

Integration adapters connect the MCP infrastructure to existing healthcare systems. FHIR [30] adapters enable data exchange with electronic health record systems. DICOM [31] adapters handle medical imaging data from Physical AI imaging robots. HL7 v2 adapters support legacy hospital information systems. RESTful API adapters enable integration with modern web-based clinical trial management platforms.

The interoperability testbed validates that MCP servers can communicate reliably with these external systems under operational conditions. Test coverage includes data format conversion accuracy, message delivery latency, error handling and recovery, and security boundary enforcement. SDK and tooling packages enable third-party developers to build MCP-compatible Physical AI applications.

11.7 National Deployment Topology

The national deployment architecture uses a hub-and-spoke model. Regional hub servers aggregate data from multiple trial sites within a geographic region, providing regional analytics and safety coordination. Spoke servers at individual trial sites handle local Physical AI operations. A national coordination layer connects all regional hubs, providing nationwide analytics, cross-regional safety signal aggregation, and centralized regulatory reporting.

Data routing ensures that patient-identified data remains within authorized boundaries (site-level and regional as required by HIPAA [17] and state privacy laws), while de-identified aggregate data flows to the national level for trend analysis and quality improvement. The federated learning framework [9] interfaces with the MCP infrastructure at the regional hub level, enabling model improvement without centralizing patient data.

11.8 Governance and Compliance

The governance framework for national MCP server operations addresses server management authority, update deployment procedures, compliance monitoring protocols, and dispute resolution mechanisms. Server management is distributed across three levels: site-level administration for local configurations, regional administration for hub coordination, and national administration for infrastructure-wide updates and standards enforcement.

Compliance monitoring uses the Governance Server's audit trail capabilities to track all Physical AI operations against regulatory requirements. Automated compliance checks flag potential violations in real time, enabling corrective action before regulatory deadlines. HIPAA

[17] compliance is enforced through access controls, encryption, and audit trails managed by the Governance and Data Flow servers.

11.9 CI/CD and Operational Excellence

The continuous integration and deployment pipeline for MCP servers ensures reliable updates with minimal operational disruption. Automated testing covers unit tests, integration tests, and end-to-end scenario tests for all 23 tools and five servers. Deployment procedures support rolling updates that maintain service availability during version transitions. Rollback capabilities enable rapid reversion to previous versions if issues are detected post-deployment. The main repository [10] and MCP repository [11] provide the CI/CD configurations and test suites.

11.10 Patient Safety and Data Flow

MCP servers ensure patient safety through real-time monitoring, data validation, and automated safety checks. The complete data flow from robot procedures through MCP servers to clinical databases and regulatory reporting systems maintains data integrity at every step. Each data element is validated against expected ranges, flagged for anomalies, and logged with timestamps and source identification.

The integration between MCP servers and the federated learning framework [9] enables multi-site model improvement while preserving patient privacy. Model updates generated through federated learning are distributed through MCP channels, with the Safety Server verifying that updates meet safety requirements before deployment to Physical AI systems. This creates a closed loop from data collection through model improvement to deployment, all managed through standardized MCP infrastructure that supports federated learning in healthcare [32].

11.11 Complete Tool Inventory by Server

Table 26 provides the complete inventory of all 23 MCP tools organized by their parent server, with each tool's name and functional description. This inventory serves as the authoritative reference for implementers deploying the five-server architecture at trial sites.

Table 26: Complete Inventory of 23 MCP Tools Organized by Server

No.	Server	Tool Name	Description
1	Governance	Protocol Manager	Manages trial protocol configurations, version control, and distribution to connected systems
2	Governance	Compliance Tracker	Monitors regulatory compliance status across all three adapted standards in real time
3	Governance	Access Controller	Enforces role-based access control policies aligned with HIPAA requirements and 21 CFR Part 11

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No.	Server	Tool Name	Description
4	Governance	Audit Logger	Records all system operations with timestamps, user identities, and action details for regulatory audit trails
5	Governance	Consent Verifier	Validates patient informed consent status before authorizing Physical AI procedures
6	Data Flow	Data Ingestor	Receives and normalizes incoming data streams from Physical AI systems, sensors, and clinical databases
7	Data Flow	Format Transformer	Converts data between FHIR, DICOM, HL7 v2, and internal Physical AI data schemas
8	Data Flow	Data Validator	Applies range checks, consistency validation, and completeness verification to all incoming data elements
9	Data Flow	Storage Coordinator	Routes validated data to appropriate storage systems based on data type, retention policy, and access requirements
10	Data Flow	Stream Router	Manages real-time data routing based on urgency level, data type, and destination system requirements
11	Safety	Emergency Stop	Propagates emergency stop signals from any source to all connected Physical AI systems within latency threshold
12	Safety	Interlock Manager	Manages safety interlock states ensuring that Physical AI systems cannot perform restricted operations without authorization
13	Safety	Anomaly Detector	Monitors Physical AI telemetry streams for anomalous patterns indicating potential safety concerns
14	Safety	Incident Reporter	Generates structured incident reports for safety events compliant with adapted ICH E6(R3) adverse event requirements
15	Interopertic	Device Registrar	Registers Physical AI devices with capability profiles, communication protocols, and safety certifications
16	Interopertic	Protocol Translator	Translates commands and data between different Physical AI device communication protocols

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No.	Server	Tool Name	Description
17	Interopertic	Capability Negotiator	Negotiates operational parameters between systems with different capabilities and performance characteristics
18	Interopertic	Standards Enforcer	Validates that all cross-platform communications conform to MCP protocol specifications and conformance levels
19	Interopertic	Integration Adapter	Provides connectivity adapters for FHIR, DICOM, HL7 v2, and RESTful API integration with external systems
20	Analytics	Dashboard Generator	Creates real-time operational dashboards displaying trial status, system health, and safety metrics
21	Analytics	Trend Analyzer	Computes trend analyses across patient outcomes, system performance, and adverse event patterns
22	Analytics	Outcome Aggregator	Aggregates treatment outcome data across patients and procedures for regulatory reporting and quality improvement
23	Analytics	Performance Scorer	Calculates performance scores for Physical AI systems based on operational metrics and clinical outcomes

Each tool operates at a defined conformance level that specifies the minimum requirements for compliant implementations. Level 1 conformance requires basic data exchange capability with standard message formats. Level 2 conformance adds bidirectional communication, error handling, retry logic, and basic audit trail support. Level 3 conformance, the highest level, requires full bidirectional integration with comprehensive audit trail generation, safety interlock participation, and real-time monitoring integration. The reference implementations in the MCP repository [11] achieve Level 2 or above for all 23 tools, with the Safety Server tools (Emergency Stop, Interlock Manager, Anomaly Detector, Incident Reporter) all implemented at Level 3 given their critical role in patient safety.

The tool inventory reflects a deliberate design decision to distribute functionality across servers rather than centralizing it. This distribution provides fault isolation: a failure in the Analytics Server does not affect safety operations managed by the Safety Server. It also supports independent scaling, allowing sites with higher data volumes to scale Data Flow Server resources without modifying other server components. The Governance Server tools (Protocol Manager, Compliance Tracker, Access Controller, Audit Logger, Consent Verifier) interact with every other server, making the Governance Server the coordination hub of the architecture. The five tools on the Governance Server collectively ensure that no Physical AI operation proceeds without proper authorization, consent verification, and audit documentation.

The Interoperability Server carries five tools because cross-platform translation is the

most complex operational challenge in a multi-vendor Physical AI environment. The Device Registrar maintains a registry of all connected Physical AI systems with their capability profiles, enabling the Protocol Translator to select appropriate translation strategies for each device pair. The Capability Negotiator resolves conflicts when two systems have different performance characteristics, such as when a high-frequency telemetry source connects to a lower-bandwidth monitoring system. The Standards Enforcer validates all cross-platform communications against MCP protocol specifications, rejecting non-conformant messages before they reach destination systems. The Integration Adapter provides the bridge to external healthcare systems, supporting FHIR [30] for electronic health records, DICOM [31] for medical imaging, HL7 v2 for legacy hospital systems, and RESTful APIs for modern web-based platforms.

11.12 National Deployment Phases

The national deployment of MCP server infrastructure proceeds through four phases, each expanding geographic coverage and operational capability. The phased approach ensures that lessons learned at each stage inform subsequent deployments, reducing risk and improving deployment efficiency.

Table 27: National MCP Server Deployment Phases

Phase	Timeline	Geographic Scope	Sites	Infrastructure Configuration
Phase A: Prototype	Months 1-6	California only	1-2 sites	Single hub server, local spoke servers, direct connections
Phase B: Regional	Months 7-18	West Coast states	5-10 sites	Regional hub server for West Coast, spoke servers at each site, federated learning node activation
Phase C: Multi-Regional	Months 19-36	Four U.S. regions (West, South, Midwest, Northeast)	20-50 sites	Four regional hub servers, national coordination layer, cross-regional safety signal aggregation
Phase D: Nationwide	Months 37-60	All 50 states plus territories	100+ sites	Full hub-and-spoke topology, redundant national coordination, complete federated learning network

Phase A (Prototype) deploys the five-server architecture at the initial California trial site or sites described in Section 14. This phase validates the MCP infrastructure under real operational conditions, comparing actual performance against the metrics established during the 24-hour simulation [5]. A single hub server manages all local spoke servers, and direct connections between

servers minimize network complexity during initial validation. The prototype phase produces operational metrics including message latency, data throughput, safety signal propagation time, and system availability that inform subsequent deployment phases.

Phase B (Regional) expands MCP infrastructure to additional West Coast sites, deploying a regional hub server that aggregates data from all sites within the region. This phase introduces the hub-and-spoke topology at regional scale, validating the architecture’s ability to coordinate multiple sites through a shared hub. Federated learning nodes activate at each site, enabling cross-site model improvement through the framework described in Section 12. Network requirements increase significantly in Phase B as inter-site communication adds bandwidth and latency constraints that were absent in the single-site prototype.

Phase C (Multi-Regional) establishes four regional hub servers covering the major U.S. geographic regions. The national coordination layer connects all regional hubs, enabling nationwide analytics, cross-regional safety signal aggregation, and centralized regulatory reporting. This phase introduces the most complex operational challenge: managing data routing across multiple regional boundaries while maintaining compliance with varying state privacy laws. The Data Flow Server’s Storage Coordinator tool becomes critical in Phase C, enforcing data residency requirements that differ across states. HIPAA [17] provides a federal baseline, but states such as California (CMIA, CCPA) impose additional requirements that the MCP infrastructure must respect at the routing level.

Phase D (Nationwide) achieves full geographic coverage with MCP infrastructure deployed in all 50 states plus U.S. territories. The infrastructure supports 100 or more connected trial sites with redundant national coordination ensuring continuity even if individual regional hubs experience outages. The complete federated learning network enables model improvement across the full national dataset while preserving patient privacy at every level. Phase D represents the steady-state operational configuration described in the hub-and-spoke model, with ongoing expansion accommodating new trial sites as they achieve PSL compliance and activate Physical AI operations.

Each deployment phase includes validation gates that must be passed before proceeding to the next phase. Phase A requires demonstration of all 23 tools operating at Level 2 conformance or above under operational conditions. Phase B requires successful cross-site data exchange with latency within specified thresholds and federated learning round completion across all participating sites. Phase C requires cross-regional safety signal propagation within latency requirements and compliance with all applicable state privacy laws. Phase D requires sustained operation at nationwide scale with 99.5 percent minimum uptime across all regional hubs and the national coordination layer.

The deployment timeline is conservative by design, allocating five years for full nationwide coverage. This timeline accommodates the legislative processes required in each state (analogous to California’s SB 1042, AB 2847, and SB 892 from the site documentation [5]), the infrastructure construction and validation activities at each site, and the workforce training requirements described in Section 14. Sites in states with existing legislative frameworks for AI in healthcare may proceed through earlier phases more quickly, while sites in states requiring new legislation may follow the later phases of the timeline.

11.13 Detailed Tool Specifications and Operational Parameters

Beyond the inventory presented in Table 26, each of the 23 MCP tools operates within defined performance parameters that ensure consistent behavior across deployment sites. Table 28 pro-

vides operational specifications for all tools organized by server, including latency requirements, data throughput expectations, and fault-tolerance characteristics.

Table 28: Operational Specifications for All 23 MCP Tools

Server	Tool	Max Latency	Throughput	Fault Tolerance
Governance	Protocol Manager	500 ms response	100 req/min	Active-passive failover with state replication
Governance	Compliance Tracker	1 s response	200 req/min	Eventual consistency with 5 s recovery
Governance	Access Controller	100 ms response	500 req/min	Active-active with distributed token cache
Governance	Audit Logger	200 ms write	1000 events/min	Write-ahead log with guaranteed delivery
Governance	Consent Verifier	250 ms response	150 req/min	Cached verification with real-time fallback
Data Flow	Data Ingester	50 ms per record	5000 records/min	Buffer queue with 10 min retention
Data Flow	Format Transformer	100 ms per transform	2000 transforms/min	Stateless with automatic retry
Data Flow	Data Validator	75 ms per validation	3000 validations/min	Parallel validation with majority consensus
Data Flow	Storage Coordinator	200 ms per route	1000 routes/min	Multi-path routing with acknowledgment
Data Flow	Stream Router	25 ms per message	10000 msg/min	Priority queuing with overflow management
Safety	Emergency Stop	10 ms propagation	Unlimited burst	Triple-redundant with hardware fallback
Safety	Interlock Manager	50 ms state check	500 checks/min	Fail-safe default (all interlocks engaged)
Safety	Anomaly Detector	100 ms detection	2000 samples/min	Redundant detectors with voting
Safety	Incident Reporter	500 ms report gen	50 reports/min	Guaranteed delivery with offline queue

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Server	Tool	Max Latency	Throughput	Fault Tolerance
Interop	Device Registrar	1 s registration	20 reg/min	Distributed registry with sync replication
Interop	Protocol Translator	75 ms per translation	1000 trans/min	Fallback to raw passthrough on failure
Interop	Capability Negotiator	2 s per negotiation	30 neg/min	Cached capability profiles with refresh
Interop	Standards Enforcer	50 ms per check	2000 checks/min	Strict mode rejects on enforcer failure
Interop	Integration Adapter	150 ms per exchange	500 exchanges/min	Adapter pool with circuit breaker
Analytics	Dashboard Generator	2 s refresh	10 dashboards/min	Cached views with graceful degradation
Analytics	Trend Analyzer	5 s computation	5 analyses/min	Incremental computation with checkpoints
Analytics	Outcome Aggregator	3 s aggregation	10 aggregations/min	Partial results on timeout
Analytics	Performance Scorer	1 s scoring	20 scores/min	Last-known-good score on failure

The latency requirements reflect the operational criticality of each tool. Safety Server tools have the most stringent latency requirements, with the Emergency Stop tool requiring signal propagation within 10 milliseconds to ensure that patient safety events trigger immediate system-wide responses. This 10-millisecond threshold was established through analysis of Physical AI response times documented in the 24-hour simulation [5], where robot deceleration profiles require stop signal receipt within this window to guarantee safe cessation of motion. The Interlock Manager operates at 50 milliseconds because interlock state checks occur before each procedure step, and excessive latency would create unacceptable delays in treatment delivery without corresponding safety benefit.

Data Flow Server tools prioritize throughput because Physical AI systems generate continuous high-volume data streams. The Stream Router, handling the highest message volume at 10,000 messages per minute, implements priority queuing that ensures safety-critical messages receive immediate routing while lower-priority telemetry data is queued for batch processing. The Data Ingestor's 50-millisecond latency target ensures that real-time sensor data from Physical AI robots reaches monitoring systems without perceptible delay, supporting the continuous monitoring capabilities demonstrated in the 24-hour simulation.

Governance Server tools operate at moderate latency because their functions are invoked at procedure initiation and completion rather than continuously during operations. The Access Controller is the exception, requiring 100 millisecond response times because authentication checks occur frequently as different system components request access to protected resources. The Audit Logger's guaranteed delivery mechanism uses write-ahead logging to ensure that no audit event is lost even during server failures, supporting the 21 CFR Part 11 requirements for

complete and tamper-evident electronic records.

Analytics Server tools have the most relaxed latency requirements because their outputs support management decisions rather than real-time operations. Dashboard refresh at 2-second intervals provides near-real-time visibility without imposing excessive computational load. The Trend Analyzer’s 5-second computation window allows for analysis across larger datasets spanning multiple patients and time periods. All Analytics tools implement graceful degradation, providing cached or partial results when computational resources are constrained, ensuring that analytics never compete with safety-critical functions for server resources.

11.14 Deployment Topology Network Architecture

The hub-and-spoke deployment topology described in Section 11.7 requires specific network architecture configurations at each level. Table 29 details the network requirements for spoke servers at individual trial sites, regional hub servers that aggregate multiple sites, and the national coordination layer that connects all regional hubs.

Table 29: Network Architecture Requirements by Deployment Level

Parameter	Spoke (Site)	Regional Hub	National Coordination
Bandwidth (minimum)	1 Gbps internal, 100 Mbps uplink	10 Gbps internal, 1 Gbps per spoke link	10 Gbps backbone, 1 Gbps per hub link
Latency (max, one-way)	5 ms internal	20 ms to any spoke	50 ms to any regional hub
Redundancy	Dual uplink to regional hub	Dual backbone connections, N+1 spoke links	Geographically distributed with automatic failover
Encryption	TLS 1.3 for all external, IPsec for inter-server	TLS 1.3 for all links, hardware security modules	TLS 1.3 with mutual authentication, HSM-backed certificates
Data residency	All patient data retained locally	De-identified aggregates only	Fully de-identified national aggregates only
Firewall rules	Outbound to regional hub only, no direct cross-site	Outbound to national and inbound from spokes	Inbound from regional hubs only
Monitoring	Local SNMP and syslog	Regional network operations center	National network operations center with 24/7 staffing
Disaster recovery	4-hour RTO, 1-hour RPO	2-hour RTO, 30-minute RPO	1-hour RTO, 15-minute RPO
Load balancing	Round-robin across local server instances	Weighted distribution based on spoke load	Geographic routing with health-aware failover

Parameter	Spoke (Site)	Regional Hub	National Coordination
Compliance monitoring	Automated HIPAA scan daily	Regional compliance dashboard	National compliance aggregation with alert escalation

The network architecture enforces the data residency requirements described in Section 11.7. Patient-identified data never traverses the uplink from spoke servers to regional hubs. The Data Flow Server's Storage Coordinator tool enforces this boundary by classifying each data element and routing accordingly. De-identified aggregate data flows from spoke to regional hub for regional analytics, and fully de-identified national aggregates flow from regional hubs to the national coordination layer.

The disaster recovery parameters (Recovery Time Objective and Recovery Point Objective) reflect the criticality of each deployment level. Spoke servers at individual trial sites have a 4-hour RTO because local Physical AI operations can continue using the Safety Server's autonomous safety functions even if network connectivity to the regional hub is lost. Regional hub servers have a stricter 2-hour RTO because cross-site coordination functions, including federated learning round management and cross-site safety signal aggregation, depend on hub availability. The national coordination layer has the strictest 1-hour RTO because nationwide analytics and regulatory reporting functions require continuous operation to meet submission deadlines and provide real-time visibility to regulatory authorities.

The encryption requirements ensure that all data in transit is protected using current best-practice cryptographic standards. TLS 1.3 provides forward secrecy and strong cipher suites for all network communications. Hardware security modules at the regional hub and national coordination levels store cryptographic keys in tamper-resistant hardware, preventing key extraction even if server software is compromised. IPSec tunnels between servers at the spoke level provide an additional encryption layer for inter-server communication within a trial site, ensuring that Physical AI data remains encrypted even on the local network.

The monitoring architecture provides visibility at each deployment level. Local monitoring at spoke servers uses standard SNMP and syslog protocols that integrate with existing hospital IT monitoring systems. Regional network operations centers aggregate monitoring data from all spokes within a region, providing a unified view of regional infrastructure health. The national network operations center, staffed 24 hours a day and 7 days a week, monitors all regional hubs and the national coordination layer, with escalation procedures that route critical alerts to on-call engineering teams within minutes. This three-level monitoring architecture ensures that infrastructure issues are detected and addressed at the appropriate level, preventing local problems from escalating to regional or national impact.

Load balancing strategies differ by deployment level. Spoke servers use simple round-robin distribution because all local server instances have equivalent capability and access to the same local data. Regional hubs implement weighted distribution that accounts for the varying load generated by spokes with different patient volumes and Physical AI system counts. The national coordination layer uses geographic routing that directs requests to the nearest healthy regional hub, with health-aware failover that automatically redirects traffic away from hubs experiencing degraded performance.

The firewall rule architecture implements a defense-in-depth strategy. Spoke servers can only communicate outbound to their assigned regional hub, preventing any direct cross-site

data exchange that would bypass the hub's data residency enforcement. Regional hubs accept inbound connections only from their assigned spokes and communicate outbound only to the national coordination layer. The national coordination layer accepts inbound connections only from authenticated regional hubs. This layered firewall architecture ensures that a compromise at any single level cannot propagate laterally across the infrastructure, limiting the blast radius of potential security incidents as required by the cybersecurity provisions of the adapted 21 CFR Part 312 [3] Subpart J.

12 Federated Learning for Physical AI Oncology Trials

Physical AI Federated Learning [9] provides a privacy-preserving framework for unifying Physical AI oncology trials across multiple sites, enabling coordinated learning and continuous improvement without centralizing sensitive patient data. This framework addresses a fundamental challenge of nationwide Physical AI deployment: how to leverage data from all trial sites for model improvement while maintaining strict patient privacy under HIPAA [17] and state privacy laws. The framework is implemented in the PAI Oncology Trial FL Repository [12] with 235 modules totaling 86,800 lines of code, validated through 82 test files.

12.1 The Privacy Challenge in Multi-Site Physical AI Trials

Physical AI systems generate vast amounts of patient data including robot telemetry (position, force, speed, temperature at each joint), sensor readings (vital signs, environmental conditions, proximity measurements), imaging data (intraoperative video, diagnostic scans, positioning verification), and treatment outcomes (procedure results, adverse events, recovery metrics). If pooled across multiple trial sites, this data could dramatically improve AI models for treatment planning, safety monitoring, and outcome prediction. However, centralized data collection creates unacceptable privacy risks under HIPAA, state privacy laws including California’s CMIA and CCPA, and the data protection requirements of SB 892 from the site documentation package [5].

Federated learning resolves this tension by training models across decentralized data sources without exchanging raw patient data. Each trial site maintains its own data locally and trains model updates using that local data. Only model parameters (weights, gradients) are shared with a central aggregation server, which combines updates from all sites to produce an improved global model distributed back to all sites. This approach has been validated in healthcare settings as documented in the literature on federated learning in healthcare [32].

12.2 Development Process and Methodology

The federated learning framework was developed through prompt-driven methodology using Claude Code Opus 4.6 [14] as the primary development tool, with OpenAI ChatGPT providing supplementary assistance. The development process followed a structured version timeline from v0.1.0 (initial architecture) through v1.1.0 (production-ready release), with each version generated through archived prompts. The framework was designed to integrate with the existing Physical AI platform [10] and MCP server infrastructure [8].

12.3 Architecture Overview

The federated learning architecture comprises three layers: client nodes at each trial site that train local models on site-specific data, aggregation servers that combine model updates from multiple sites, and a coordination layer that manages the federated learning lifecycle including round scheduling, convergence monitoring, and deployment orchestration.

Client nodes interface with the MCP infrastructure at each trial site, receiving Physical AI data through the Data Flow Server and submitting model updates through secure channels. Aggregation servers implement federated averaging with privacy-preserving modifications including differential privacy noise injection and secure aggregation protocols. The coordination

layer integrates with the Governance Server to ensure that federated learning operations comply with trial protocol requirements and regulatory standards.

12.4 Five Foundational Pillars

The federated learning framework is organized around five foundational pillars, each addressing a critical aspect of privacy-preserving multi-site Physical AI coordination.

Table 30: Five Foundational Pillars of Federated Learning Framework

No.	Pillar	Scope
1	Privacy and Security	Differential privacy, secure aggregation, encryption, access controls
2	Regulatory Compliance	ICH E6(R3), 21 CFR Part 50/312, HIPAA, state privacy law adherence
3	Clinical Validity	Model validation, clinical outcome correlation, safety verification
4	Operational Reliability	Fault tolerance, recovery procedures, monitoring, audit trails
5	Scalability	Multi-site coordination, bandwidth optimization, heterogeneous data

Pillar 1 (Privacy and Security) implements technical privacy protections including differential privacy with configurable noise parameters, secure multi-party computation for model aggregation, end-to-end encryption for all federated learning communications, and role-based access controls aligned with HIPAA requirements. Pillar 2 (Regulatory Compliance) ensures that federated learning operations comply with the adapted regulatory standards, with specific attention to the data governance requirements of the adapted ICH E6(R3) [1] and the data protection provisions of the adapted 21 CFR Part 50 [2].

Pillar 3 (Clinical Validity) establishes processes for validating that federated model updates improve clinical outcomes rather than introducing bias or degrading performance. This includes hold-out validation datasets at each site, cross-site performance benchmarking, and clinical review of model recommendations before deployment. Pillar 4 (Operational Reliability) provides fault tolerance mechanisms ensuring that the federated learning system remains operational even when individual sites experience outages, with automatic recovery procedures and comprehensive monitoring. Pillar 5 (Scalability) addresses the practical challenges of coordinating federated learning across dozens or hundreds of trial sites, including bandwidth optimization, handling heterogeneous data distributions, and managing asynchronous updates from sites with different operational schedules.

12.5 Workflow Demonstrations

The framework includes three categories of workflow demonstrations. Domain workflows demonstrate federated learning applied to specific oncology data types including imaging, genomics, and clinical outcomes. Agentic AI workflows demonstrate integration with AI planning systems that coordinate multi-step clinical operations. Physical AI workflows, the most relevant to this National Platform, demonstrate federated learning applied to robot policy improvement, where surgical robot control policies are refined across multiple sites using federated updates from procedure outcomes without sharing patient-identified procedure recordings.

12.6 Triple AI Peer Review Pipeline

The triple AI peer review pipeline is a unique quality assurance mechanism that validates federated learning model updates before deployment. Three independent AI systems review each model update: Claude Opus 4.6 [14] performs primary review focusing on code quality, safety implications, and regulatory compliance; GPT-5.4 [15] performs secondary review focusing on mathematical correctness, statistical validity, and performance benchmarks; and Gemini [16] performs tertiary review focusing on cross-platform compatibility, integration testing, and deployment readiness.

The triple peer review process resolved 31 out of 31 recommendations during framework development, demonstrating that multi-AI review can achieve comprehensive quality assurance. Each AI reviewer generates structured feedback that is documented in the review log, creating an audit trail for all model changes. This mechanism ensures that model improvements are safe and clinically valid before being applied to Physical AI systems across trial sites.

12.7 Code Trust and Safety

The framework ensures code integrity through multiple mechanisms. The test suite comprises 82 test files covering unit tests, integration tests, and end-to-end validation scenarios. All code changes undergo automated testing before deployment. Model update verification ensures that federated updates produce expected behavior changes without introducing safety regressions. Cryptographic signing of model updates prevents tampering during distribution. Audit trails record every model change with source, timestamp, reviewer approval, and deployment status, supporting the safety requirements of the adapted 21 CFR Part 50 [2] and the adapted 21 CFR Part 312 [3].

12.8 Clinical Analytics and Digital Twins

Federated learning enables cross-site clinical analytics without data centralization. Aggregate statistics on treatment outcomes, adverse event rates, and Physical AI system performance can be computed across all sites without any site sharing patient-level data. These analytics support continuous quality improvement, trend detection, and evidence generation for regulatory submissions.

Digital twin models benefit from federated updates that incorporate diverse patient data from multiple sites. A patient digital twin can be improved using insights from similar patients at other sites without those patients' data leaving their respective institutions. The main repository [10] provides digital twin implementations that interface with the federated learning framework for model improvement.

12.9 Technology Stack and CLI Tools

The software infrastructure includes PyTorch for model training, gRPC for secure communication between federated learning components, Redis for distributed state management, PostgreSQL for audit trail storage, and Docker for containerized deployment. CLI tools enable site administrators to manage federated learning operations including node registration, round participation, model deployment, and performance monitoring. The FL repository [12] provides all implementation code under the MIT license.

12.10 Integration with National MCP Infrastructure

Federated learning integrates with the Physical AI National MCP Servers [8] at multiple levels. Data flow between MCP servers and federated learning clients uses the Data Flow Server’s standardized routing capabilities. Model update distribution uses the Interoperability Server’s cross-platform translation services. Coordinated monitoring across both infrastructure layers uses the Analytics Server’s aggregation capabilities. The Safety Server validates model updates before deployment, ensuring that federated learning outputs meet safety requirements. The MCP repository [11] provides integration adapters for connecting federated learning components to the MCP infrastructure.

12.11 Limitations and Future Work

Acknowledged limitations include: federated learning introduces communication overhead that may slow model convergence compared to centralized training; heterogeneous data distributions across sites can create convergence challenges requiring advanced aggregation strategies; the framework has been validated through simulation rather than real-world multi-site deployment; and the privacy-utility tradeoff means that stronger privacy protections (higher differential privacy noise) may reduce model accuracy. Future work includes real-world deployment validation, advanced privacy-preserving techniques (homomorphic encryption), personalized federated learning for site-specific model adaptation, and integration with emerging healthcare interoperability standards.

12.12 Detailed Pillar Implementation Mechanisms

Table ?? expands on the five foundational pillars presented in Table 30, providing specific implementation mechanisms, validation criteria, and integration points with the broader National Platform infrastructure for each pillar.

Table 31: Detailed Implementation Mechanisms for the Five Foundational Pillars

Pillar	Implementation Mechanisms	Validation Criteria	Platform Integration
Privacy and Security	Differential privacy with epsilon values between 0.1 and 1.0 configurable per data type; secure aggregation using Shamir secret sharing across a minimum of 3 participating sites; AES-256 encryption for all model parameters in transit and at rest; role-based access using OAuth 2.0 tokens validated against the MCP Governance Server	Privacy budget accounting verified per round; formal proof that individual patient records cannot be reconstructed from shared gradients; penetration testing of all communication channels quarterly; access control audit with zero unauthorized access events	MCP Access Controller validates all federated learning participant credentials; MCP Audit Logger records every model parameter exchange; encryption key management through MCP infrastructure hardware security modules
Regulatory Compliance	Automated mapping of federated learning operations to adapted ICH E6(R3) [1] requirements; consent verification for data inclusion in each training round via MCP Consent Verifier; audit trail generation meeting 21 CFR Part 11 electronic records standards; adverse event model correlation tracking per adapted 21 CFR Part 312 [3]	Every federated learning round produces a compliance report verifiable against all three adapted standards; consent records traceable for every patient whose data contributed to any model update; audit trails pass independent third-party verification	MCP Compliance Tracker monitors federated learning regulatory status; MCP Protocol Manager ensures training rounds conform to trial protocol parameters; MCP Incident Reporter generates regulatory filings when model updates correlate with adverse events

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Pillar	Implementation Mechanisms	Validation Criteria	Platform Integration
Clinical Validity	Hold-out validation sets maintained at each site comprising 20 percent of local data; cross-site performance benchmarking after each aggregation round; clinical review board evaluation of model recommendations before deployment; bias detection algorithms that compare model performance across demographic subgroups	Model accuracy improvement demonstrated on hold-out sets at minimum 80 percent of participating sites; no statistically significant performance degradation for any demographic subgroup; clinical review board approval documented before deployment; outcome tracking for 90 days post-deployment	MCP Analytics Server tracks model performance metrics across all sites; MCP Data Validator ensures training data quality before local model training; digital twin models from the main repository [10] provide simulation-based validation before clinical deployment
Operational Reliability	Automatic detection and exclusion of non-responsive sites from aggregation rounds; checkpoint-based recovery enabling restart from last successful aggregation; redundant aggregation servers with automatic failover; health monitoring of all federated learning components with 30-second heartbeat intervals	System recovery from single-site failure within 60 seconds; recovery from aggregation server failure within 5 minutes; 99.9 percent successful round completion rate over 30-day periods; zero data loss during any failure scenario	MCP Safety Server monitors federated learning component health; MCP Stream Router provides reliable message delivery between federated learning components; MCP Dashboard Generator displays real-time federated learning operational status

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Pillar	Implementation Mechanisms	Validation Criteria	Platform Integration
Scalability	Asynchronous federated learning allowing sites to participate in rounds at different times within a configurable window; gradient compression reducing communication bandwidth by up to 90 percent; hierarchical aggregation where regional hubs pre-aggregate before national aggregation; dynamic batch sizing that adapts to available bandwidth at each site	Linear scaling of communication overhead with number of sites up to 100 sites; aggregation round completion within 1 hour for up to 50 sites; bandwidth consumption below 100 MB per site per round after compression; no degradation in model quality with hierarchical versus flat aggregation	MCP regional hub servers serve as hierarchical aggregation points; MCP Capability Negotiator manages bandwidth allocation for federated learning traffic; MCP Performance Scorer evaluates aggregation efficiency as part of system performance metrics

The Privacy and Security pillar’s differential privacy implementation uses a calibrated noise mechanism where the epsilon parameter controls the privacy-utility tradeoff. Lower epsilon values (closer to 0.1) provide stronger privacy guarantees but introduce more noise into model updates, potentially slowing convergence. Higher epsilon values (closer to 1.0) preserve more utility but offer weaker individual privacy guarantees. The framework allows site-level configuration of epsilon values within bounds set by the national governance policy, enabling sites operating under stricter state privacy laws (such as California’s CMIA and CCPA) to select lower epsilon values while sites in states with less restrictive requirements may select higher values to accelerate model improvement. Secure aggregation through Shamir secret sharing ensures that the aggregation server never observes individual site model updates in plaintext, providing cryptographic protection against a compromised aggregation server.

The Regulatory Compliance pillar implements automated compliance mapping that links each federated learning operation to specific provisions of the three adapted regulatory standards. When a training round begins, the system automatically verifies that all participating sites have current IRB approval for data use in model training, that all patients whose data will be used have active informed consent that explicitly covers federated learning participation, and that the training parameters conform to the trial protocol specifications registered with the MCP Protocol Manager. This automated verification prevents inadvertent regulatory violations that could occur if federated learning operations proceed without current authorizations.

The Clinical Validity pillar addresses the concern that federated model updates could introduce subtle biases or performance degradation that would not be detected through automated metrics alone. The clinical review board evaluation requirement ensures that human clinical

experts assess model recommendations before deployment, providing a check against algorithmic bias and ensuring that model outputs align with current clinical best practices. The bias detection algorithms compare model performance across demographic subgroups defined by age, sex, race, ethnicity, and cancer type, flagging any statistically significant performance disparities for clinical review board investigation. This approach ensures that federated learning benefits are distributed equitably across patient populations, supporting the health equity goals of the nationwide deployment strategy described in Section 14.

The Operational Reliability pillar ensures that the federated learning system degrades gracefully under adverse conditions. The automatic site exclusion mechanism removes non-responsive sites from the current aggregation round without requiring manual intervention, preventing a single site’s outage from blocking model improvement across the entire network. Checkpoint-based recovery stores the state of each aggregation round at configurable intervals, enabling restart from the last successful checkpoint rather than requiring a full round restart. The redundant aggregation servers operate in an active-standby configuration where the standby server continuously receives state updates from the active server, enabling failover within the 5-minute recovery target.

The Scalability pillar addresses the practical challenge of coordinating federated learning across dozens or hundreds of trial sites with varying computational resources and network connectivity. Asynchronous participation allows sites to complete local training and submit updates within a configurable time window (default: 4 hours) rather than requiring simultaneous participation. Gradient compression uses top-k sparsification and quantization to reduce the size of transmitted model updates by up to 90 percent, making federated learning feasible even over bandwidth-constrained links. Hierarchical aggregation leverages the existing MCP regional hub infrastructure, with regional hubs performing local aggregation before forwarding to the national aggregation server, reducing the national server’s computational load and network bandwidth requirements.

12.13 Extended Triple AI Peer Review Process

The triple AI peer review pipeline described in Section 12.6 warrants detailed examination of each reviewer’s specific role, review methodology, and contribution to the overall quality assurance process. The three-reviewer architecture was designed to exploit the complementary strengths of different AI systems, creating a review process more thorough than any single reviewer could provide.

12.13.1 Primary Reviewer: Claude Opus 4.6

Claude Opus 4.6 [14] serves as the primary reviewer with responsibility for three review dimensions: code quality, safety implications, and regulatory compliance. In the code quality dimension, Claude Opus 4.6 examines the implementation of each federated learning update for adherence to coding standards, proper error handling, resource management (memory leaks, connection cleanup, file handle closure), and documentation completeness. The reviewer generates structured findings with severity levels (critical, major, minor, informational) and specific remediation recommendations.

In the safety dimension, Claude Opus 4.6 evaluates whether model updates could introduce unsafe behavior in Physical AI systems. This includes checking that model output ranges remain within clinically safe bounds, that error conditions produce fail-safe behaviors rather

than unpredictable outputs, and that model confidence scores are properly calibrated to prevent overconfident recommendations in edge cases. The safety review draws on the safety requirements documented in the adapted 21 CFR Part 50 [2] Subpart C and the Physical AI safety matrices from the site documentation [5].

In the regulatory compliance dimension, Claude Opus 4.6 verifies that model updates maintain compliance with all three adapted regulatory standards. This includes checking that audit trail entries are generated for every model change, that patient data handling conforms to HIPAA [17] requirements, and that the model update process follows the clinical trial protocol specifications. The primary reviewer produces a comprehensive report that becomes the baseline against which secondary and tertiary reviews add their findings.

12.13.2 Secondary Reviewer: GPT-5.4

GPT-5.4 [15] serves as the secondary reviewer with responsibility for mathematical correctness, statistical validity, and performance benchmarks. In the mathematical correctness dimension, GPT-5.4 verifies that aggregation algorithms produce mathematically correct results, that gradient computations are numerically stable, that convergence guarantees hold under the specified conditions, and that differential privacy noise injection uses correctly calibrated distributions. This review dimension requires deep mathematical analysis that complements the implementation-focused primary review.

In the statistical validity dimension, GPT-5.4 evaluates whether model performance claims are supported by statistically rigorous evidence. This includes verifying that performance comparisons use appropriate statistical tests, that confidence intervals are correctly computed, that sample sizes are sufficient for the claimed significance levels, and that multiple comparison corrections are applied when evaluating performance across subgroups. The statistical review ensures that claims of model improvement are genuine rather than artifacts of statistical noise or selection bias.

In the performance benchmarks dimension, GPT-5.4 compares model update performance against established baselines. Each model update must demonstrate improvement on standardized benchmark datasets before approval for deployment. The secondary reviewer tracks performance trajectories across multiple aggregation rounds, identifying whether the federated learning process is producing consistent improvement or exhibiting signs of divergence, oscillation, or convergence to suboptimal solutions.

12.13.3 Tertiary Reviewer: Gemini

Gemini [16] serves as the tertiary reviewer with responsibility for cross-platform compatibility, integration testing, and deployment readiness. In the cross-platform compatibility dimension, Gemini verifies that model updates function correctly across all Physical AI system types deployed in the trial network. A model update trained primarily on data from da Vinci surgical robots must not degrade performance when applied to Franka Panda laboratory automation systems or vice versa. This compatibility review ensures that federated learning benefits are universal across the Physical AI ecosystem.

In the integration testing dimension, Gemini executes automated test suites that validate model updates against the MCP server infrastructure. This includes verifying that updated models correctly interface with the Safety Server's anomaly detection, that model outputs are properly formatted for the Data Flow Server's routing, and that model metadata is correctly

logged through the Governance Server’s audit trail. Integration testing catches interface issues that would not be detected through isolated model evaluation.

In the deployment readiness dimension, Gemini evaluates whether the operational prerequisites for model deployment are satisfied. This includes verifying that deployment scripts are correct, that rollback procedures are tested, that monitoring configurations will capture relevant metrics from the updated model, and that all participating sites have acknowledged readiness for the update. The deployment readiness review prevents premature deployment of model updates that have passed quality and performance reviews but lack the operational infrastructure for safe rollout.

12.13.4 Review Consensus and Conflict Resolution

The three reviewers operate independently, producing separate reports that are then reconciled through a structured consensus process. When all three reviewers approve an update without critical findings, the update proceeds to deployment. When any reviewer identifies a critical finding, deployment is blocked until the finding is resolved and the blocking reviewer confirms resolution. When reviewers produce conflicting assessments, a conflict resolution protocol examines the specific points of disagreement and applies domain-specific precedence rules: safety findings from Claude Opus 4.6 take precedence in safety-related conflicts, mathematical findings from GPT-5.4 take precedence in correctness-related conflicts, and compatibility findings from Gemini take precedence in integration-related conflicts.

The 31-for-31 resolution record during framework development demonstrates the effectiveness of this multi-reviewer approach. Each recommendation was traceable to a specific reviewer, addressed through documented changes, and verified by the originating reviewer before closure. This audit trail of review findings and resolutions provides evidence of thorough quality assurance that supports regulatory confidence in the federated learning framework’s outputs.

13 Financial and Economic Impact Analysis

Physical AI oncology trials offer substantial, documented financial benefits that make a compelling case for nationwide adoption. The analysis draws from cost data in A Cancer Patient’s Journey, Physical AI [6], the 24-hour simulation in Clinical Trial Site Complete Documentation [5], and external economic data from Tufts CSDD [18] on drug development delays. The financial case extends beyond cost reduction to encompass accelerating the time to drug approval, which directly translates to earlier patient access to life-saving treatments.

13.1 Cost of Drug Development Delays

Tufts CSDD [18] data establishes that each day of delay in drug development carries significant economic consequences. For oncology drugs, the stakes are particularly high because delayed approvals mean delayed patient access to potentially life-saving treatments. The cumulative cost of delays across the drug development pipeline, from preclinical through Phase 3 and regulatory review, represents a primary target for Physical AI cost reduction.

Physical AI addresses delays at multiple points in the pipeline. Phase 0 simulation validation enables parallel testing of Physical AI systems while traditional preclinical work proceeds, reducing sequential delays. Automated screening and enrollment accelerate patient recruitment, historically one of the most time-consuming phases of clinical trials. Robotic treatment delivery enables 24-hour operations, multiplying throughput per site. Automated documentation and real-time regulatory reporting reduce the lag between data generation and submission. NCI data on trial costs [33] and modernization efforts [26] provide additional context for the current cost landscape.

13.2 Physical AI Cost Advantages: Per-Patient Analysis

The per-patient cost comparison between traditional oncology trials and Physical AI trials reveals advantages across multiple cost categories. Staffing costs decrease because Physical AI enables treating more patients with more robots and fewer workers. Equipment utilization costs decrease because robots operate continuously rather than during limited shift hours. Facility operations costs decrease per patient because higher throughput amortizes fixed costs across more patients. Data management costs decrease through automated capture, validation, and reporting. Regulatory compliance costs decrease through standardized processes and automated documentation.

Table 32: Per-Patient Cost Comparison: Traditional vs. Physical AI Trials

Cost Category	Traditional Trial	Physical AI Trial
Clinical staff per patient	Multiple staff per procedure	Reduced staff with robot support
Equipment utilization	8-12 hour daily operation	24-hour continuous operation
Data management	Manual entry, QC reviews	Automated capture and validation
Regulatory documentation	Manual report generation	Real-time automated reporting
Adverse event monitoring	Periodic human review	Continuous AI-assisted monitoring
Patient scheduling	Manual coordination	AI-optimized scheduling
Quality control	Periodic audits	Continuous automated verification

While initial capital investment for robotic systems is higher than traditional equipment, the operational cost per patient is lower due to reduced staffing, faster procedures, automated documentation, continuous monitoring without human fatigue, and reduced error-related costs. The break-even analysis depends on patient volume, with higher-volume sites achieving faster return on investment.

13.3 Physical AI Speed Advantages: Timeline Compression

Physical AI compresses trial timelines at multiple stages, delivering cumulative time savings that translate to earlier drug approval.

13.3.1 Enrollment Speed

Physical AI systems accelerate patient enrollment through automated screening using AI algorithms that review medical records against eligibility criteria in minutes rather than days, automated eligibility verification that cross-references patient data across multiple databases simultaneously, and streamlined consent processes using AI-enhanced materials and robot demonstrations that help patients understand trial requirements faster. Patient Instructions: Physical AI Trials [7] reduces enrollment friction by providing clear, accessible information that addresses patient concerns proactively.

13.3.2 Treatment Throughput

Physical AI systems increase treatment throughput through 24-hour operations as demonstrated in the 24-hour simulation [5] where 168 patients were treated continuously, reduced per-procedure time through precision robotics that eliminate setup delays and manual positioning, and elimination of scheduling bottlenecks caused by staff availability constraints. The 99.7 percent uptime achieved in the simulation demonstrates that Physical AI systems can maintain near-continuous operations.

13.3.3 Data Processing and Reporting

Physical AI systems accelerate data processing through automated documentation that captures procedure data in real time, data validation that occurs concurrently with data collection through MCP server quality checks, and integrated regulatory reporting through MCP servers [8] that generate submission-ready reports from operational data without manual transcription.

13.4 Scale Economics: From Single Patient to 168 Patients

The financial evidence spans two scales. The single-patient journey [6] provides detailed per-stage cost data for one patient through the complete ten-stage pipeline. The 24-hour simulation [5] provides operational cost data for 168 patients treated simultaneously. The comparison demonstrates favorable scale economics: per-patient costs decrease as patient volume increases because fixed costs (infrastructure, robot procurement, software licensing) are amortized across more patients while variable costs per patient decrease through operational efficiency.

The 168-patient simulation provides a more realistic operational cost picture than the single-patient case because it accounts for concurrent resource utilization, robot scheduling optimization, and infrastructure sharing. At the 168-patient daily throughput level, Physical

AI trial costs per patient are projected to be substantially below traditional per-patient costs, creating a compelling economic argument for nationwide deployment where multiple sites collectively serve thousands of patients.

13.5 Infrastructure Investment and Return

The infrastructure investment for a Physical AI trial site includes building modifications for robot operations (reinforced flooring, power systems, network infrastructure), robot procurement and maintenance across multiple categories, MCP server deployment [8] including hardware, software, and network connectivity, federated learning infrastructure [9] including computing resources and secure communication systems, staff training for Physical AI operations, and regulatory compliance activities including Phase 0 simulation validation.

The return on investment depends on patient throughput, trial duration, and the number of concurrent trials a site supports. Sites operating multiple concurrent Physical AI trials can share infrastructure costs, improving the financial case. The open-source availability of the platform [10, 11, 12] eliminates software licensing costs, with investment concentrated in hardware, infrastructure, and training.

13.6 Patient Financial Benefits

Patient benefits from Physical AI trials include convenience through reduced number of required visits as continuous monitoring reduces follow-up frequency, lower treatment costs passed through from operational savings, reduced travel burden from faster procedures requiring fewer and shorter clinic visits, and reduced lost wages from shorter treatment times. NCI information on who pays for clinical trials [33] provides baseline comparison for understanding how Physical AI reduces direct patient costs.

13.7 Pharmaceutical Industry Impact

Physical AI impacts pharmaceutical industry economics through reduced per-trial costs enabling companies to conduct more trials within existing budgets, faster time to market for new oncology drugs through accelerated enrollment and treatment delivery, improved data quality reducing regulatory review cycles and rejection rates, and competitive advantages for early adopters who establish Physical AI trial capabilities. FDA adaptive designs guidance [27] provides context for how trial design innovations affect industry economics, and Physical AI represents the next evolution in trial design optimization.

13.8 National Economic Impact

The national economic impact of widespread Physical AI oncology trial adoption includes aggregate drug development cost savings across the pharmaceutical industry, earlier availability of cancer treatments to the broader population through accelerated approval timelines, reduced healthcare system burden through more effective treatments reaching patients sooner, job creation in Physical AI manufacturing, maintenance, and operations (new roles that partially offset reduced traditional staffing), and technology export potential as the U.S. establishes leadership in Physical AI clinical trial infrastructure.

13.9 Comparison with Existing FDA Economic Analyses

This financial analysis is more comprehensive than existing FDA economic assessments because it covers the entire trial lifecycle rather than individual components, includes patient-side economics not typically captured in sponsor-focused analyses, incorporates infrastructure investment amortization over multi-year operational periods, and projects national-scale impact beyond individual trial economics. FDA guidance documents [21] and NCI resources [20] provide valuable but narrower economic perspectives that this analysis extends to the full Physical AI context.

13.10 Financial Projections by Implementation Phase

Table 33 presents estimated financial parameters for each implementation phase, providing a framework for investment planning.

Table 33: Financial Projections by Implementation Phase

Parameter	Phase 1 (California)	Phase 2 (Multi-site)	Phase 3 (Nation-wide)
Number of sites	1	5-10	50 or more
Robots per site	20-30	20-30	15-30
Annual patients per site	Up to 61,000	Up to 61,000	Variable
Infrastructure investment	Site construction, robot procurement, MCP servers	Incremental per site, shared MCP hubs	Regional hub infrastructure, standardized deployment
Staffing model	Mixed traditional and Physical AI	Evolving toward Physical AI-primary	Physical AI-primary with specialized human roles
Timeline to break-even	18-24 months projected	12-18 months (economies of learning)	6-12 months (standardized deployment)
Open-source savings	Full platform available	Shared improvements across sites	Community-maintained platform

The projections assume that each site operates at throughput levels demonstrated in the 24-hour simulation [5], with 168 patients per 24-hour period representing the demonstrated operational capability. Actual throughput will depend on site size, cancer type mix, and local staffing conditions. Phase 1 investment is highest per site due to first-mover costs including site construction customization, initial staff training, and regulatory pathway development. Phase 2 and Phase 3 benefit from economies of learning as standardized processes, shared training materials, and community-maintained open-source code [10, 11, 12] reduce per-site costs.

13.11 Return on Investment Analysis

Table 34: Return on Investment Framework for Physical AI Trial Sites

ROI Component	Description
Direct cost savings	Reduced per-patient costs through automation, 24-hour operations, and fewer staff
Time-value savings	Earlier drug approval per Tufts CSDD data on value of reduced delays
Quality improvement	Reduced adverse events, better data quality, fewer regulatory deficiencies
Multi-trial leverage	Same infrastructure supporting multiple concurrent trials
Training efficiency	AI-assisted training reducing time-to-competency for new staff
Data asset value	Federated learning models improving with each trial, creating compounding value

The ROI analysis demonstrates that Physical AI trial site investment is financially justified across multiple return dimensions. The most significant return comes from time-value savings: if Physical AI enables even modest acceleration of drug approval timelines, the value per day of acceleration as quantified by Tufts CSDD [18] exceeds the per-day amortized cost of Physical AI infrastructure. The compounding value of federated learning models represents a unique ROI dimension: each trial conducted at a Physical AI site contributes to model improvement that benefits all future trials across the network, creating increasing returns to scale that traditional trial infrastructure cannot match.

13.12 Detailed Cost Projections by Phase

Table 35 expands the phase-level projections with granular cost breakdowns across all major expenditure categories. These projections encompass capital expenditures, recurring operational costs, and projected savings relative to traditional trial operations. All figures are expressed in 2026 U.S. dollars and represent estimated ranges based on the simulation evidence [6, 5], published drug development cost data [18], and NCI trial cost information [33].

Table 35: Detailed Cost Projections by Implementation Phase (Estimated Ranges in Millions USD)

Cost Category	Phase 1: Cali- fornia (Year 1- 2)	Phase 2: Multi- Site (Year 2-4)	Phase 3: Na- tionwide (Year 4-7)
<i>Capital Expenditures</i>			
Facility construction and modification	\$8M - \$12M (1-2 sites)	\$35M - \$55M (5-10 sites)	\$180M - \$300M (50-100 sites)
Robot procurement (all categories)	\$5M - \$8M (25-35 robots per site)	\$25M - \$45M (scaled fleet)	\$150M - \$250M (national fleet)

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Cost Category	Phase 1: California (Year 1-2)	Phase 2: Multi-Site (Year 2-4)	Phase 3: Nationwide (Year 4-7)
MCP server infrastructure	\$1M - \$2M (local spoke servers)	\$5M - \$8M (regional hubs added)	\$20M - \$35M (national topology)
Federated learning infrastructure	\$0.5M - \$1M (single node)	\$3M - \$5M (regional network)	\$12M - \$20M (national network)
Network and cybersecurity	\$1M - \$1.5M (site network)	\$4M - \$7M (inter-site links)	\$15M - \$25M (national backbone)
Total Capital	\$15.5M - \$24.5M	\$72M - \$120M	\$377M - \$630M
<i>Annual Recurring Costs</i>			
Clinical and technical staffing	\$3M - \$5M per year	\$12M - \$22M per year	\$60M - \$100M per year
Robot maintenance and calibration	\$0.8M - \$1.2M per year	\$4M - \$7M per year	\$20M - \$35M per year
Software licensing and support	\$0.2M - \$0.4M per year	\$1M - \$2M per year	\$5M - \$8M per year
Regulatory compliance operations	\$0.5M - \$0.8M per year	\$2M - \$4M per year	\$10M - \$18M per year
Training and workforce development	\$0.3M - \$0.5M per year	\$1.5M - \$3M per year	\$8M - \$12M per year
Facility operations and utilities	\$1M - \$1.5M per year	\$5M - \$8M per year	\$25M - \$40M per year
Total Annual Recurring	\$5.8M - \$9.4M/yr	\$25.5M - \$46M/yr	\$128M - \$213M/yr
<i>Projected Annual Savings vs. Traditional Trials</i>			
Staffing efficiency gains	\$1M - \$2M per year	\$6M - \$12M per year	\$35M - \$60M per year
Throughput-driven cost amortization	\$2M - \$3M per year	\$10M - \$18M per year	\$55M - \$90M per year
Automated documentation savings	\$0.5M - \$1M per year	\$3M - \$5M per year	\$15M - \$25M per year
Reduced error and rework costs	\$0.3M - \$0.6M per year	\$2M - \$4M per year	\$10M - \$18M per year
Accelerated timeline value	\$5M - \$10M per year	\$25M - \$50M per year	\$150M - \$300M per year
Total Annual Savings	\$8.8M - \$16.6M/yr	\$46M - \$89M/yr	\$265M - \$493M/yr

The accelerated timeline value represents the largest projected savings category across all phases. This value derives from Tufts CSDD [18] data on the cost of drug development delays, applied to the timeline compression that Physical AI enables through 24-hour operations, automated enrollment, and real-time regulatory reporting. For each day of acceleration in drug

approval, the sponsoring pharmaceutical company avoids one day of lost revenue from the approved drug, and patients gain one day of earlier access to the treatment. At the nationwide scale of Phase 3, even modest per-trial timeline reductions generate substantial aggregate value across the dozens of concurrent Physical AI oncology trials operating in the network.

The capital expenditure estimates for Phase 3 reflect the significant infrastructure investment required for nationwide deployment, with facility construction and robot procurement comprising the majority of costs. However, these costs must be evaluated in context: the total Phase 3 capital expenditure of \$377M to \$630M represents a fraction of the annual U.S. spending on oncology clinical trials, and the projected annual savings of \$265M to \$493M suggest that the infrastructure investment could be recovered within two to three years of full-scale operation.

13.13 Detailed Return on Investment Analysis

Table 36 provides quantified ROI metrics for each implementation phase, calculating the net financial position over the phase duration and the projected payback period for capital investment.

Table 36: Quantified Return on Investment Analysis by Implementation Phase

ROI Metric	Phase 1: Cali- fornia	Phase 2: Multi- Site	Phase 3: Na- tionwide
Phase duration	2 years	3 years (Year 2-4)	4 years (Year 4-7)
Total capital investment (midpoint)	\$20M	\$96M	\$504M
Total recurring costs over phase	\$15.2M (2 years)	\$107M (3 years)	\$682M (4 years)
Total savings over phase	\$25.4M (2 years)	\$203M (3 years)	\$1,516M (4 years)
Net financial position at phase end	-\$9.8M (invest- ment phase)	+\$0.3M (break- even)	+\$330M (positive return)
Cumulative net position	-\$9.8M	-\$9.5M	+\$320.5M
Payback period from phase start	Not achieved in Phase 1	Achieved at end of Phase 2	Fully recovered in Year 5
Annualized ROI (at phase end)	-24.5% (pre- return)	+0.1% (break- even)	+16.4% (positive)
Patient throughput (annual)	500 - 1,000 pa- tients	3,000 - 8,000 pa- tients	20,000 - 60,000 pa- tients
Cost per patient (all-in)	\$17,600 - \$20,000	\$8,400 - \$12,800	\$5,700 - \$9,500
Traditional cost per patient	\$30,000 - \$50,000	\$30,000 - \$50,000	\$30,000 - \$50,000
Per-patient savings	\$10,000 - \$30,000	\$17,200 - \$41,600	\$20,500 - \$44,300

The ROI analysis demonstrates a characteristic investment-return curve for large-scale infrastructure deployment. Phase 1 operates at a net loss because the capital investment and recurring costs for a single-site or two-site deployment are not offset by the savings generated from a relatively small patient population. This is expected and acceptable because Phase 1 serves

primarily as a validation phase, generating operational evidence that de-risks subsequent phases. The -24.5 percent annualized ROI in Phase 1 reflects the front-loaded nature of infrastructure investment.

Phase 2 achieves approximate break-even as multi-site operations generate sufficient throughput to offset both Phase 2 costs and the remaining Phase 1 deficit. The critical factor enabling break-even in Phase 2 is the accelerated timeline value, which grows disproportionately with multi-site deployment because multiple concurrent trials at different sites generate independent timeline acceleration benefits. Each additional site that achieves operational status contributes incremental savings without proportional increases in shared infrastructure costs (regional hub servers, national coordination layer, and federated learning infrastructure serve multiple sites simultaneously).

Phase 3 achieves strongly positive returns as nationwide scale amplifies every savings category. The cost per patient decreases from \$17,600 to \$20,000 in Phase 1 to \$5,700 to \$9,500 in Phase 3, reflecting the powerful scale economics of shared infrastructure. Traditional oncology trial costs of \$30,000 to \$50,000 per patient, as informed by NCI data [33] and industry analyses, provide the comparison baseline that makes Physical AI per-patient savings increasingly compelling at scale.

The cumulative net position turns positive during Phase 3, with full payback of all prior investment achieved approximately in Year 5 of the overall program. From Year 5 forward, the Physical AI trial infrastructure generates ongoing positive returns that can be reinvested in technology upgrades, geographic expansion, and new therapeutic applications.

13.14 Quantified Patient Financial Benefits

The patient financial benefits described qualitatively in Section 13.6 can be quantified through specific scenarios that illustrate the magnitude of savings for individual patients participating in Physical AI oncology trials.

Table 37: Quantified Patient Financial Benefits: Physical AI vs. Traditional Trials

Benefit Category	Traditional Trial Patient	Physical AI Trial Patient
Clinic visits per treatment cycle	12-20 visits over 6 months	6-10 visits over 6 months (continuous remote monitoring reduces visit frequency)
Average wait time per visit	45-90 minutes	10-20 minutes (AI-optimized scheduling)
Total clinic time per cycle	60-100 hours	20-35 hours

Continued

Benefit Category	Traditional Trial Patient	Physical AI Trial Patient
Travel cost per visit (avg.)	\$25-\$75 (parking, fuel, transit)	\$25-\$75 per visit but fewer visits
Lost wages per visit (avg.)	\$150-\$300 (half to full day)	\$75-\$150 (shorter visits)
Total lost wages per cycle	\$1,800-\$6,000	\$450-\$1,500
Caregiver lost wages per cycle	\$900-\$3,000 (companion for each visit)	\$225-\$750 (fewer visits, shorter duration)
Childcare costs per cycle	\$600-\$2,000 (for visits requiring child-care)	\$150-\$500
Out-of-pocket medical costs	\$2,000-\$5,000 (co-pays, prescriptions, supplies)	\$1,500-\$3,500 (reduced complications, fewer follow-ups)
Total estimated patient savings per 6-month treatment cycle	-	-

The visit frequency reduction deserves detailed explanation. Traditional oncology trials require frequent in-person visits for treatment administration, monitoring, laboratory work, and safety assessments. Physical AI trials reduce visit frequency through three mechanisms. First, continuous remote monitoring via Physical AI sensor systems and companion robot health tracking (described in A Cancer Patient’s Journey, Physical AI [6]) provides real-time patient status data that eliminates the need for many routine check-in visits. Second, robot-assisted treatment delivery achieves more precise dosing and positioning, reducing the need for dose adjustment visits that are common in traditional trials. Third, automated laboratory processing through Physical AI laboratory robots enables faster result availability, allowing clinical decisions to be made without requiring the patient to return for a separate results visit.

The wait time reduction reflects the AI-optimized scheduling capabilities described in the MCP Analytics Server’s operational dashboards (Section 11). Traditional trial clinics experience significant wait times due to scheduling variability, procedure overruns, and resource conflicts. Physical AI scheduling algorithms predict procedure durations with higher accuracy based on patient-specific factors and robot performance data, reducing overbooking and minimizing idle time between patients. The 24-hour simulation [5] demonstrated continuous patient processing with minimal inter-patient delays, suggesting that wait times in Physical AI operations can be substantially shorter than in traditional settings.

The lost wages savings represent a particularly important benefit for patients in lower-income brackets who face difficult choices between trial participation and maintaining employment. By reducing both the number of visits and the duration of each visit, Physical AI trials lower the employment barrier to trial participation. This has health equity implications: traditional trials disproportionately exclude patients who cannot afford to take time off work for frequent

clinic visits. Physical AI's reduced time burden could broaden the demographic diversity of trial participants, improving the generalizability of trial results.

Caregiver financial burden is an often-overlooked component of trial costs. Many oncology patients require a companion for clinic visits due to treatment side effects, transportation needs, or emotional support requirements. Each visit that is eliminated also eliminates the caregiver's time and financial burden. The caregiver savings estimated in Table 37 represent a conservative estimate because they do not account for the emotional and psychological costs of repeated caregiving disruptions.

The out-of-pocket medical cost reduction derives from Physical AI's precision treatment delivery and continuous monitoring capabilities. Precision robotics reduce treatment-related complications that generate additional medical costs (emergency visits, hospitalizations, additional medications). Continuous monitoring enables earlier detection of adverse events, allowing intervention before conditions escalate to more costly treatment stages. The adapted ICH E6(R3) [1] adverse event reporting requirements ensure that any complications are promptly documented and addressed, further reducing the risk of escalating medical costs.

The total estimated patient savings of \$3,125 to \$10,500 per six-month treatment cycle represent a meaningful financial benefit, particularly for patients managing cancer treatment alongside other financial obligations. When multiplied across the thousands of patients projected to participate in Physical AI trials at Phase 3 nationwide scale, the aggregate patient savings represent tens of millions of dollars annually returned to patients and their families. NCI information on who pays for clinical trials [33] documents the current financial burden on patients; Physical AI represents a significant mechanism for reducing that burden while simultaneously improving treatment quality and trial efficiency.

14 National Implementation Strategy

The National Platform provides a clear, actionable strategy for implementing Physical AI oncology trials across the United States. The strategy builds on the validated evidence from both simulations, the comprehensive regulatory framework, the quantitative standards (PSL and USL), and the national infrastructure (MCP servers and federated learning) described in preceding sections. The evidence base provides compelling reasons for nationwide implementation, and this section outlines the practical path forward.

14.1 Implementation Phases

14.1.1 Phase 1: First Site Activation (California)

The first Physical AI oncology trial site activates in California, leveraging the complete documentation package from Clinical Trial Site Complete Documentation [5]. Activation follows the PSL framework with all three dimensions (Legislative Authorization, Regulatory Compliance, Operational Readiness) verified. State-specific legislation (SB 1042 for authorization, AB 2847 for patient rights, SB 892 for data protection) provides the legal foundation.

Phase 1 activities include: site construction and modification per building and premises codes, robot procurement with USL score verification (prioritizing da Vinci dVRK at USL 7.1 for surgical procedures and Franka Panda at USL 7.4 for laboratory automation), staff hiring and Physical AI-specific training, MCP server installation and configuration, Phase 0 simulation validation replicating and extending the 24-hour simulation, IND submission with Physical AI System Description, IRB review with Physical AI-specific evaluation, and initial patient enrollment.

Evidence from both simulations [6, 5] supports the operational viability of Phase 1. The 24-hour simulation demonstrated 99.7 percent uptime with 168 patients, and the single-patient journey demonstrated complete ten-stage pipeline execution. Phase 1 targets a smaller initial patient cohort to validate real-world operations against simulation predictions.

14.1.2 Phase 2: Multi-Site Expansion

Phase 2 expands from single-site to multi-site operations, activating additional Physical AI trial sites using the national MCP server infrastructure [8] and federated learning framework [9]. Site selection criteria are based on PSL scoring, ensuring that new sites meet the same legislative, regulatory, and operational standards as the California prototype. Robot deployment decisions at each site are based on USL evaluation, ensuring that only robots meeting minimum readiness thresholds are deployed.

Multi-site coordination is managed through the five-server MCP architecture with regional hub servers aggregating data from sites within geographic clusters. Federated learning enables model improvement across sites without data centralization, with the triple AI peer review pipeline validating all model updates before deployment. Phase 2 targets expansion to five to ten sites across multiple states, requiring adaptation of the California-specific legislation (SB 1042, AB 2847, SB 892) to each state's regulatory environment.

14.1.3 Phase 3: Nationwide Deployment

Phase 3 achieves full nationwide coverage with Physical AI trial sites distributed across all major geographic regions. Geographic distribution ensures equitable access for patients in different parts of the country. The national MCP deployment topology with hub-and-spoke architecture supports hundreds of connected trial sites. Governance transitions from centralized management to a distributed model with regional authorities and national coordination.

14.2 Regulatory Pathway

The regulatory pathway leverages the three adapted standards. The Adaption: ICH Harmonised Guideline [1] establishes GCP requirements that investigators and sponsors follow at every phase. The Physical AI Adaption: 21 CFR Part 50 [2] establishes human subjects protections that IRBs enforce through Physical AI-specific review. The Physical AI Adaption: 21 CFR Part 312 [3] establishes IND requirements including Phase 0 simulation validation, Physical AI adverse event reporting, and the new Subpart J provisions.

These adapted standards build on existing, well-established frameworks and add credibility to Physical AI implementations. They demonstrate that Physical AI trials can operate within the existing regulatory infrastructure with targeted adaptations rather than requiring entirely new legislation at the federal level.

14.3 Standards Compliance Framework

Implementation sites must achieve minimum PSL scores across all three dimensions before activation. PSL Dimension 1 requires appropriate state legislative authorization (California sites use SB 1042; other states require analogous legislation). PSL Dimension 2 requires compliance with applicable state and federal regulations. PSL Dimension 3 requires completion of all operational readiness activities. Robots must achieve minimum USL scores: the platform recommends Advanced band (USL 7.0 or higher) for surgical robots, Intermediate band (USL 5.0 or higher) for patient-contact support robots, and Foundational band (USL 3.0 or higher) for logistics-only robots.

14.4 Infrastructure Deployment

Technical infrastructure deployment follows a standardized process at each site: MCP server installation with five-server configuration, network connectivity meeting bandwidth and latency requirements for real-time Physical AI data exchange, federated learning node setup with secure communication channels, cybersecurity hardening per the requirements in the adapted 21 CFR Part 312 [3] Subpart J, and integration testing with existing hospital IT systems through FHIR [30] and DICOM [31] adapters. The three code repositories [10, 11, 12] provide the open-source foundation for all infrastructure components.

14.5 Workforce Transition

Physical AI enables treating more patients with more robots and fewer traditional clinical workers. The workforce transition creates new roles: Physical AI System Operators who monitor and control robots during procedures, Robot Maintenance Technicians who service and calibrate Physical AI systems, AI Safety Monitors who oversee automated decision-making processes,

and Physical AI Trial Coordinators who manage the integration of robotic systems with clinical trial operations.

Existing roles evolve rather than disappear: clinical trial coordinators develop Physical AI oversight competencies, research nurses add robot monitoring to their responsibilities, and data managers expand their scope to include Physical AI-generated data streams. Training requirements follow the adapted ICH E6(R3) [1] training standards, with Physical AI-specific certifications ensuring that all personnel are qualified for their expanded responsibilities.

14.6 Patient Engagement Strategy

Patient engagement and education during implementation include public communication about Physical AI trials through institutional websites, media outreach, and community events. Community outreach addresses concerns about robotic healthcare through transparent information and demonstration opportunities. Patient advocacy group partnerships ensure that patient perspectives shape implementation decisions. Patient Instructions: Physical AI Trials [7] serves as the primary patient education tool, with materials adapted for local contexts.

Patient trust is essential for successful implementation. The shift of control towards patients through comprehensive rights (AB 2847), transparent information (patient instructions), and ongoing consent (adapted 21 CFR Part 50 [2]) is designed to build and maintain trust. NCI clinical trial safety information [34] and clinical trial information [20] provide evidence-based messaging frameworks that can be adapted for Physical AI communication.

14.7 Quality Assurance and Continuous Improvement

The quality assurance framework includes regular PSL re-evaluation (quarterly review of all three dimensions), periodic USL re-scoring of deployed robots (semi-annual assessment against updated scoring criteria), federated learning model performance monitoring (continuous tracking of model accuracy and safety metrics), MCP server uptime tracking (real-time availability monitoring with 99.5 percent minimum threshold), adverse event trend analysis (monthly review of Physical AI-specific events), and regulatory compliance auditing (annual comprehensive audit against all three adapted standards).

14.8 Risk Mitigation

Key implementation risks and their mitigation strategies include: technology failure risks mitigated by emergency preparedness plans from the site documentation [5] and redundant safety systems; regulatory risks mitigated by the comprehensive adapted standards that provide clear compliance pathways; patient safety risks mitigated by the multi-layer safety architecture spanning Physical AI safety matrices, MCP safety servers, and human oversight requirements; data privacy risks mitigated by federated learning [9] and HIPAA [17] compliance infrastructure; and workforce transition risks mitigated by training programs, phased deployment, and role evolution rather than elimination.

14.9 Success Metrics

Implementation success is evaluated through: patient enrollment rates compared to traditional trial benchmarks, treatment outcomes compared to historical non-Physical AI trials, cost per

patient compared to industry averages, time to drug approval compared to historical timelines, adverse event rates compared to traditional trials and the simulation benchmarks from A Cancer Patient’s Journey, Physical AI [6], PSL and USL score trends over time, MCP server uptime and reliability, federated learning model improvement rates, patient satisfaction scores, and regulatory compliance audit results. These metrics provide a comprehensive framework for assessing whether Physical AI oncology trials deliver the benefits projected by the simulation evidence and economic analysis.

14.10 Implementation Timeline

Table 38: Implementation Timeline: Key Milestones by Phase

Phase	Period	Milestones
Phase 1	Year 1, Q1-Q2	Site construction, robot procurement, staff recruitment, training program launch
Phase 1	Year 1, Q3	MCP server installation, Phase 0 simulation validation, IND submission
Phase 1	Year 1, Q4	IRB approval, site activation, first patient enrollment
Phase 1	Year 2	Operational optimization, 12-month outcomes analysis, Phase 2 planning
Phase 2	Year 2-3	Site selection (5-10 sites), state legislative adaptation, regional MCP hub deployment
Phase 2	Year 3	Multi-site federated learning activation, cross-site model improvement
Phase 2	Year 3-4	Multi-site outcome comparison, standardized deployment package refinement
Phase 3	Year 4-5	Nationwide expansion to 50 or more sites, national MCP coordination
Phase 3	Year 5-6	Full nationwide coverage, international pilot exploration
Phase 3	Year 6 onward	Continuous improvement, next-generation robot integration, outcome-driven refinement

14.11 Workforce Transition Detail

The workforce transition creates new roles while evolving existing ones. Table 39 presents the complete role mapping.

Table 39: Workforce Transition: New Roles and Evolved Roles

Type	Role	Responsibilities
New	Physical AI System Operator	Robot monitoring, control, and intervention during procedures
New	Robot Maintenance Technician	Hardware service, calibration, software updates, parts replacement
New	AI Safety Monitor	Oversight of AI decision-making, anomaly detection review, safety signal evaluation
New	Physical AI Trial Coordinator	Integration of robotic workflows with clinical operations
New	MCP Systems Administrator	MCP server management, network operations, data flow monitoring
New	Federated Learning Engineer	Model training coordination, privacy compliance, performance optimization
Evolved	Clinical Research Coordinator	Adds Physical AI documentation, robot scheduling, patient-robot interaction oversight
Evolved	Research Nurse	Adds robot telemetry monitoring, Physical AI adverse event documentation
Evolved	Data Manager	Adds Physical AI data streams, automated quality review, MCP data validation
Evolved	Regulatory Affairs Specialist	Adds Physical AI IND content, Subpart J compliance, cybersecurity reporting
Evolved	Quality Assurance Manager	Adds PSL scoring, USL monitoring, Physical AI system audit
Evolved	Site Director	Adds Physical AI governance, robot fleet management, PSL compliance oversight

The training pipeline for new roles follows a structured curriculum. Physical AI System Operators complete 40 hours of classroom instruction covering robot categories, safety systems, and emergency procedures, followed by 80 hours of supervised practical training on each robot type they will operate. Robot Maintenance Technicians complete manufacturer-specific certification programs supplemented by 40 hours of Physical AI-specific training covering clinical environment requirements, data security for robot maintenance activities, and documentation standards per the adapted ICH E6(R3) [1].

For evolved roles, training focuses on the incremental competencies needed for Physical AI oversight. Clinical Research Coordinators complete a 24-hour Physical AI supplement covering robot identification, basic troubleshooting, patient-robot interaction protocols, and Physical AI adverse event recognition. Research Nurses complete a 16-hour supplement covering telemetry interpretation, safety alert response, and Physical AI-specific documentation. Data Managers complete a 32-hour supplement covering Physical AI data schemas, MCP data validation, and automated quality review workflows.

The workforce transition plan includes retention incentives for existing staff who develop Physical AI competencies, ensuring that institutional knowledge is preserved while new capabilities are added. Salary adjustments reflect the expanded scope of evolved roles. New role compensation is benchmarked against comparable technology-healthcare hybrid positions in the labor market.

15 Discussion

This section synthesizes how the National Platform for Physical AI Oncology Trials represents a paradigm shift in clinical trial regulation and practice, providing a unified framework that is more comprehensive than any existing FDA approach. The discussion draws from all preceding sections and all source documents to make the case that Physical AI oncology trials should be adopted nationwide.

15.1 Paradigm Shift in Clinical Trial Regulation

The National Platform differs fundamentally from existing FDA approaches in scope and integration. While FDA issues individual guidance documents addressing specific topics (AI in drug development, adaptive trial designs, decentralized trial elements, PCCP for AI devices), this platform provides an integrated, end-to-end framework that simultaneously addresses regulatory adaptation, site establishment, patient care, national infrastructure, and economic impact.

The current regulatory framework, as documented in the U.S. government branch operations analysis (Section 2) and the dual-jurisdiction regulatory landscape (Section 3), was not designed for Physical AI. It effectively regulates traditional clinical trials through a layered framework of statutes, regulations, and guidances. But Physical AI introduces challenges that cross the boundaries of existing regulatory categories: robots that simultaneously function as medical devices, electronic data systems, and research tools; AI algorithms that make treatment decisions; and data streams that combine clinical, operational, and robotic information.

The three adapted standards [1, 2, 3] demonstrate that existing frameworks can be adapted rather than replaced. ICH E6(R3) [13], 21 CFR Part 50 [24], and 21 CFR Part 312 [23] have served the pharmaceutical and regulatory industries for decades. By building Physical AI provisions onto these established foundations, the adapted standards leverage decades of established practice, institutional familiarity, and legal precedent.

15.2 Credibility Through Existing Standards

The three adapted regulatory drafts add credibility to Physical AI implementations because they build on recognized standards. ICH E6(R3) is the internationally accepted GCP standard used across dozens of countries. Any investigator, sponsor, or regulator familiar with ICH E6(R3) can understand the Physical AI adaptations because they follow the same structure, use the same principles, and extend rather than replace existing requirements. The 30 Physical AI-specific definitions in the adapted ICH standard create standardized terminology without discarding established GCP vocabulary.

21 CFR Part 50 is the foundational U.S. human subjects protection regulation. The adaptation preserves all existing protections and adds Physical AI-specific elements, meaning that no current patient protection is weakened by the adaptation. The new Subpart C for Physical AI safety requirements represents a structural addition that enhances rather than replaces existing protections.

21 CFR Part 312 is the core IND regulation governing most U.S. interventional drug clinical trials. The adaptation spanning ten subparts (including the new Subpart J) demonstrates that the existing subpart structure can accommodate Physical AI with minimal structural disruption. The Phase 0 simulation validation stage is the most significant innovation, establishing a new

mandatory pre-enrollment phase that has no precedent in the original regulation but fills a genuine need for Physical AI safety verification.

15.3 PSL and USL: Quantitative Standards for Robot Usage

PSL (three dimensions) and USL (four dimensions) complement each other for responsible robot usage. PSL ensures site-level compliance across legislative authorization, regulatory compliance, and operational readiness. USL evaluates individual robot readiness across simulation framework switching, AI integration, cross-robot sharing, and clinical trial collaboration. Together they create a dual-layer quantitative evaluation that no existing framework provides.

The USL results (Table 17) reveal important patterns. Open-source ecosystem maturity is the strongest predictor of readiness. Clinical trial collaboration (Dimension D) is the weakest dimension field-wide. Hardware excellence does not predict readiness. These findings have practical implications for implementation: investment should prioritize software ecosystem development and regulatory infrastructure rather than hardware capability alone.

15.4 Evidence from Simulations

The evidence for Physical AI oncology trial transition rests on two pivotal completed examples. The 24-hour simulation [5] demonstrated that 168 patients could be treated with 29 robots across 15 cancer types at 99.7 percent uptime with zero patient harm events and 7 adverse events documented per ICH E6(R3). A Cancer Patient's Journey, Physical AI [6] demonstrated that a single patient could be managed through all ten stages of a regulated trial from prescreening through closeout.

Both simulations demonstrate that state-of-the-art AI has the ability to understand the Physical AI application, proficiently work with the applications, and scale beyond human and prior technology capabilities. The evidence spans multiple dimensions: safety (zero patient harm, documented adverse events per established standards), efficiency (24-hour continuous operations), scalability (168 patients in one day), regulatory compliance (mapped to all three adapted standards), and cost-effectiveness (documented per-patient savings).

15.5 Patient-Centered Design

Control has shifted towards the patient's side, with more patient rights than any existing clinical trial framework provides. The adapted 21 CFR Part 50 [2] requires comprehensive informed consent with Physical AI-specific elements and ongoing consent as technology evolves. AB 2847 from the site documentation package [5] establishes legally enforceable rights including the right to refuse robotic procedures, the right to human alternatives, and the right to real-time information about robot operations. Patient Instructions: Physical AI Trials [7] translates these rights into accessible language for each of the ten robot types.

Patient benefits extend beyond rights to include practical advantages: convenience through reduced wait times and automated scheduling, lower cost of treatments through reduced staffing overhead, higher quality of treatment through precision robotics and continuous monitoring, emotional support through companion robots, faster recovery through rehabilitation exoskeletons, and comprehensive data-driven care. NCI information on trial safety [34] and trial costs [33] provides context for understanding the magnitude of these improvements.

15.6 National Infrastructure Readiness

The five-server MCP architecture [8] and the federated learning framework [9] provide the technical foundation for nationwide deployment. The MCP infrastructure solves the fragmentation problem through standardized communication protocols, while federated learning enables multi-site model improvement without compromising patient privacy.

Integration with existing healthcare systems through FHIR [30], DICOM [31], and HIPAA [17] compliance ensures that Physical AI infrastructure does not require replacing existing hospital IT systems but rather connects to them through standardized adapters. The open-source availability of all infrastructure components [10, 11, 12] ensures that any institution can evaluate and deploy the technology.

15.7 Governance Across Government Branches

The governance framework adapted from U.S. government branch operations (Section 2) provides institutional legitimacy and accountability. Legislative, executive, and judicial branch mechanisms all apply to Physical AI regulation, ensuring that no single branch has unchecked authority over Physical AI trial governance. Congressional oversight through budget processes and the Congressional Review Act, executive implementation through FDA and HHS, and judicial review through administrative law all operate as designed, with Physical AI adding a new subject matter to established governance processes.

15.8 Comparison with International Approaches

This National Platform positions the United States as a global leader in Physical AI oncology trial regulation. While other countries may be exploring AI in clinical trials, no other jurisdiction has produced a comprehensive framework spanning regulatory adaptation, site establishment, validated simulations, quantitative standards, and national infrastructure. The ICH E6(R3) [13] basis of the adapted GCP standard provides a bridge for international harmonization, as countries that recognize ICH E6(R3) could adapt the Physical AI extensions for their own regulatory environments.

15.9 Addressing Concerns and Criticisms

Potential concerns about Physical AI in oncology trials are addressed by specific platform components. Patient safety concerns are addressed by the multi-layer safety architecture (Physical AI safety matrices, MCP safety servers, human oversight requirements, emergency preparedness plans). Job displacement concerns are addressed by the workforce transition strategy with new role creation and existing role evolution. Technology reliability concerns are addressed by the 24-hour simulation demonstrating 99.7 percent uptime. Regulatory adequacy concerns are addressed by the three adapted standards. Cost barrier concerns are addressed by the financial analysis demonstrating favorable economics at scale. Health equity concerns are addressed by the nationwide deployment strategy ensuring geographic access. Cybersecurity concerns are addressed by Subpart J of the adapted 21 CFR Part 312 [3].

15.10 Limitations of the National Platform

The platform has acknowledged limitations. The simulations are not real clinical trials and may not capture all real-world challenges. The regulatory adaptations have not been formally adopted by FDA or ICH. The economic projections are based on simulation data rather than actual operational costs. The technology requirements may create access barriers for under-resourced institutions. The USL scores represent a February 2026 snapshot of a rapidly evolving field. The single-evaluator methodology for USL scoring introduces potential bias.

For each limitation, the platform provides a foundation that can be refined through real-world implementation. The adapted standards offer regulatory proposals that can be formally reviewed. The simulations provide benchmarks against which real operations can be compared. The open-source implementation enables community improvement and validation.

Table 40: National Platform vs. Existing FDA Approach: Comprehensive Comparison

Dimension	Existing FDA Approach	National Platform
Regulatory scope	Individual guidance documents for specific AI topics	Three simultaneously adapted core standards (ICH, CFR 50, CFR 312)
Standards framework	No quantitative robot or site readiness scoring	PSL (3 dimensions) and USL (4 dimensions) with 9 robot evaluations
Simulation evidence	No required pre-enrollment simulation	Phase 0 validation with two completed simulations (1 patient, 168 patients)
Site establishment	General site requirements across various guidances	Complete 11-document package with legislation, codes, and SOPs
Patient protections	Standard informed consent per existing CFR 50	Enhanced consent with 17 Physical AI definitions, three-tier system, ongoing consent
Infrastructure	No national Physical AI coordination infrastructure	Five MCP servers with 23 tools plus federated learning framework
Economic analysis	Individual cost-effectiveness studies per guidance	End-to-end financial analysis from per-patient to nationwide scale
Implementation	No phased Physical AI deployment guidance	Three-phase strategy from California to nationwide deployment
Open source	N/A	Three repositories under MIT license

15.11 Addressing Stakeholder Concerns

Different stakeholders have distinct concerns about Physical AI adoption. The following analysis addresses each stakeholder group’s primary concerns with specific platform components that provide mitigation.

Table 41: Stakeholder Concerns and Platform Responses

Stakeholder	Primary Concern	Platform Response
Patients	Safety of robot interactions	Three-tier safety classification, Physical AI safety matrix, right to refuse, ongoing consent
Oncologists	Loss of clinical autonomy	Human oversight requirement in all adapted standards, physician override at any time
Nurses	Job displacement	Workforce transition plan with role evolution, new Physical AI coordinator roles
Sponsors	Regulatory uncertainty	Three adapted standards provide clear compliance pathways
IRBs	Ability to evaluate Physical AI risks	Physical AI-specific IRB review criteria, NCI CIRB model
FDA	Adequate oversight authority	Adapted standards build on existing FDA authority, Subpart J for Physical AI
Insurers	Treatment cost and liability	Financial analysis showing cost reduction, standardized safety documentation
Hospital IT	System integration complexity	MCP servers with FHIR, DICOM, HL7 adapters, standard protocols

15.12 Future Research Directions

Several research directions emerge from the National Platform that warrant investigation. First, real-world validation of the adapted standards through actual Physical AI oncology trials will provide empirical data on the practical adequacy of the regulatory framework. Second, expansion of USL scoring to additional robot categories (autonomous mobile robots, surgical navigation systems, telepresence robots) will broaden the readiness assessment coverage. Third, international adaptation of the platform for regulatory environments beyond the United States, leveraging the ICH E6(R3) basis for global harmonization, will enable worldwide Physical AI trial deployment. Fourth, long-term outcome studies comparing Physical AI trial results with traditional trial benchmarks will provide definitive evidence for regulatory and clinical decision-making. Fifth, advances in federated learning techniques, including homomorphic encryption and personalized federation, will strengthen the privacy-preserving capabilities of the multi-site coordination framework.

16 Conclusion

16.1 Summary of the National Platform

The National Platform for Physical AI Oncology Trials provides the first comprehensive, end-to-end framework for integrating Physical AI (robotics and AI systems) into every phase of the oncology clinical trial lifecycle. The platform comprises three adapted regulatory standards that build on internationally recognized frameworks, the Adaption: ICH Harmonised Guideline [1] for good clinical practice, the Physical AI Adaption: 21 CFR Part 50 [2] for human subjects protection, and the Physical AI Adaption: 21 CFR Part 312 [3] for investigational new drug applications. These adapted standards establish that Physical AI trials can operate within existing regulatory infrastructure through targeted adaptations.

The platform defines quantitative standards through PSL (three dimensions for site compliance) and USL (four dimensions for robot readiness) from the Physical AI Unification Standard Level [4] and Clinical Trial Site Complete Documentation [5]. Patient-centered design is embodied in A Cancer Patient’s Journey, Physical AI [6] and Patient Instructions: Physical AI Trials [7]. National infrastructure is provided through Physical AI National MCP Servers [8] and Physical AI Federated Learning [9]. The entire platform is implemented across three open-source repositories [10, 11, 12] under the MIT license.

16.2 Key Findings

1. **Regulatory compatibility.** Physical AI oncology trials can operate within existing regulatory frameworks through targeted adaptations of ICH E6(R3), 21 CFR Part 50, and 21 CFR Part 312, adding credibility through association with established standards recognized internationally.
2. **Simulation evidence.** The 24-hour simulation (168 patients, 29 robots, 99.7 percent uptime, zero patient harm events) and single-patient journey (10 stages, prescreening through closeout) provide compelling evidence that Physical AI can safely manage oncology trial operations at scale.
3. **Quantitative standards.** PSL (3 dimensions) and USL (4 dimensions) provide complementary quantitative frameworks for evaluating site readiness and robot readiness respectively, enabling standardized nationwide assessment with reproducible scoring.
4. **Patient empowerment.** Physical AI shifts control towards patients through comprehensive rights, transparent information, ongoing consent, and multiple benefits including lower wait times, reduced treatment costs, and higher quality of care through precision robotics.
5. **National infrastructure.** Five MCP servers and federated learning enable multi-site coordination while preserving patient privacy under HIPAA and state privacy laws, with standardized communication protocols and privacy-preserving model improvement.
6. **Economic viability.** Financial analysis demonstrates per-patient cost savings and timeline compression, with benefits compounding at scale from single sites to nationwide deployment, supported by Tufts CSDD data on the value of reducing drug development delays.
7. **Governance legitimacy.** The governance framework adapted from U.S. government branch operations provides institutional legitimacy through legislative, executive, and judicial oversight mechanisms that apply to Physical AI regulation.

16.3 Implications for the Pharmaceutical Industry

Pharmaceutical companies should prepare for Physical AI oncology trials by evaluating their trial portfolios for Physical AI suitability, identifying trials where robotic assistance could improve enrollment speed, treatment precision, or data quality. Companies should invest in Physical AI infrastructure including robotic systems, MCP server deployments, and staff training. Early engagement with regulatory authorities on Physical AI adoption positions companies to influence the regulatory framework rather than react to it. Partnership with the open-source community through the three repositories [10, 11, 12] provides access to the complete platform without proprietary licensing barriers.

16.4 Implications for Regulatory Authorities

FDA and other regulatory authorities should engage with the National Platform by reviewing the three adapted standards for potential formal adoption or as the basis for official guidance development. Establishing Physical AI review teams within CDER and CDRH ensures that the necessary technical expertise is available for evaluating Physical AI trial applications. Developing Physical AI-specific guidance informed by this platform's comprehensive framework provides industry with the regulatory clarity needed for investment decisions. Creating regulatory pathways for Physical AI trial site approval streamlines the activation process.

FDA clinical trials guidance [21], FDA oncology guidance [22], and NCI modernization efforts [26] all recognize the need for innovation in clinical trial conduct. The National Platform provides a comprehensive blueprint for how that innovation can be implemented responsibly and at scale.

16.5 Call to Action

The evidence from validated simulations, the comprehensive regulatory framework, the quantitative standards, the patient-centered design, and the national infrastructure all support the conclusion that Physical AI oncology trials should be implemented nationwide. The 24-hour simulation demonstrated that 168 patients can be treated with 29 robots at 99.7 percent uptime. The ten-stage patient journey demonstrated complete trial lifecycle management. Three adapted regulatory standards demonstrate compatibility with existing frameworks. PSL and USL provide objective, reproducible readiness assessment. Five MCP servers and federated learning provide national-scale infrastructure.

The question is no longer whether Physical AI should be applied to oncology clinical trials. The evidence presented in this National Platform demonstrates that Physical AI can deliver safer, faster, more efficient, and more patient-centered trials. The question is how quickly it can be deployed to begin accelerating oncology drug development, reducing costs, and expanding patient access to life-saving treatments.

This platform provides the pharmaceutical and regulatory industries with everything needed to answer that question: the regulatory framework, the standards, the evidence, the infrastructure, and the implementation strategy. The National Platform for Physical AI Oncology Trials stands as a comprehensive resource for the entire industry, offering an adapted regulatory foundation and operational guidelines to enable oncology drugs to be approved significantly faster with state-of-the-art AI and robotics. The open-source availability of all components ensures that no institutional barrier prevents any organization from evaluating, adopting, and contributing to this transformative vision for oncology clinical trials.

16.6 Future Directions

Several future directions will strengthen and extend the National Platform.

1. **Real-world validation.** Conducting the first actual Physical AI oncology trial at the California site using the complete documentation package, adapted standards, and PSL/USL scoring frameworks will provide empirical validation of the platform's practical adequacy.
2. **USL expansion.** Extending USL scoring to additional robot categories including autonomous mobile robots, surgical navigation systems, telepresence robots, and microsurgical robots will broaden the readiness assessment coverage beyond the current nine robots.
3. **International harmonization.** Adapting the platform for regulatory environments beyond the United States, including EU MDR, Japan PMDA, and China NMPA frameworks, leveraging the ICH E6(R3) basis for global consistency.
4. **Advanced federated learning.** Incorporating homomorphic encryption, split learning, and personalized federation techniques to strengthen privacy protections while improving model performance.
5. **Multi-evaluator USL validation.** Conducting inter-rater reliability studies with multiple independent evaluators scoring the same robots to validate and calibrate the USL scoring methodology.
6. **Long-term outcome tracking.** Establishing registries to track long-term oncology outcomes (recurrence, survival) for Physical AI trial participants compared to traditional trial participants.
7. **Pediatric Physical AI specialization.** Developing pediatric-specific Physical AI protocols, robot configurations, and patient materials that address the unique needs of children with cancer.
8. **Rare cancer applications.** Exploring how Physical AI can improve trial access for rare cancer patients who currently face limited enrollment opportunities due to geographic and institutional barriers.

16.7 Recommended Actions by Stakeholder Group

Table 42: Recommended Next Actions by Stakeholder Group

Stakeholder	Recommended Actions
Pharmaceutical sponsors	Evaluate trial portfolios for Physical AI suitability; invest in Physical AI infrastructure and staff training; engage with FDA on Physical AI IND submissions
FDA/regulatory agencies	Review adapted standards for potential formal guidance; establish Physical AI review teams; develop Physical AI-specific regulatory pathways
Academic medical centers	Establish Physical AI research programs; train investigators in Physical AI oversight; partner with open-source community through the three repositories
Cancer patients	Learn about Physical AI trial options; engage with patient advocacy organizations; exercise rights under AB 2847 patient rights framework
NCI/NIH	Support Physical AI trial funding through cooperative group networks; develop centralized IRB capacity for Physical AI review; fund long-term outcome studies
Robot manufacturers	Increase open-source ecosystem participation to improve USL scores; develop clinical-grade systems meeting Tier 2 and Tier 3 requirements; pursue FDA clearance for oncology applications
Hospital administrators	Assess facilities for Physical AI readiness per PSL framework; plan infrastructure investments per building and premises codes; develop workforce transition plans
State legislatures	Consider legislation analogous to SB 1042, AB 2847, and SB 892 for their states; coordinate with federal regulatory framework

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Appendices

A Complete Source File Directory

This appendix lists every source file from `national-platform/` that was used to build this paper, organized by directory. For each file, the filename, line count, and a description of its content are provided.

national-platform/21cfr312_adapt/

- `01_preamble_scope_definitions.tex` (427 lines) - Subpart A: General Provisions including 22 Physical AI definitions
- `02_ind_content_phases.tex` (274 lines) - Subpart B: IND Application including Phase 0 simulation validation
- `03_protocol_amendments_reporting.tex` (328 lines) - Information amendments, safety reporting, Subpart C start
- `04_annual_reports_withdrawal.tex` (463 lines) - Inactive status, meetings, Subpart D: Sponsor/Investigator responsibilities
- `05_clinical_holds_appendices_closing.tex` (783 lines) - Subparts E through J including new Physical AI Systems subpart

national-platform/21cfr50_adapt/

- `01_preamble_scope_definitions_consent.tex` (438 lines) - Subparts A and B: Scope, definitions, informed consent
- `02_irb_review_pediatric.tex` (236 lines) - Subparts C and D: Physical AI safety requirements, pediatric protections
- `03_additional_safeguards_closing.tex` (73 lines) - Glossary and document closing

national-platform/ich_e6r3_adapt/

- `01_preamble_principles_investigator.tex` (337 lines) - Sections 1-2: Principles and investigator responsibilities
- `02_sponsor_responsibilities.tex` (346 lines) - Section 3: Sponsor responsibilities in Physical AI trials
- `03_data_governance.tex` (228 lines) - Section 4: Data governance for Physical AI trials
- `04_appendices_glossary.tex` (389 lines) - Appendices A-C and glossary

national-platform/new_trial_psl/

- `01_preamble_and_sb1042.tex` (468 lines) - SB 1042: Trial Authorization and Site Establishment Act
- `02_ab2847_patient_rights.tex` (326 lines) - AB 2847: Patient Rights and Robotic Safety Act
- `03_sb892_data_protection.tex` (307 lines) - SB 892: Data Protection and Transparency Act
- `04_sf_municipal_code.tex` (262 lines) - San Francisco Municipal Code Update
- `05_title22_regulations.tex` (322 lines) - Title 22 Regulations
- `06_fda_compliance_guide.tex` (317 lines) - FDA National Compliance Guide
- `07_building_code.tex` (287 lines) - Building Code Standards
- `08_premises_code.tex` (276 lines) - Premises Code
- `09_parking_transportation.tex` (250 lines) - Parking and Transportation Standards
- `10_activation_sops.tex` (272 lines) - Activation and Standard Operating Procedures
- `11_emergency_preparedness.tex` (290 lines) - Emergency Preparedness Plan

national-platform/patient_journey/

- 01_preamble_introduction_stages1to4.tex (296 lines) - Introduction and pre-trial stages
- 02_stages5to8.tex (358 lines) - Treatment stages and outcomes
- 03_stages9to10_discussion_closing.tex (222 lines) - Closeout, discussion, and conclusions

national-platform/patient_robot/

- 01_preamble_robots1to5.tex (197 lines) - Patient instructions for robot types 1-5
- 02_robots6to10_closing.tex (173 lines) - Patient instructions for robot types 6-10

national-platform/national_mcp/

- chunk1_preamble_and_introduction.tex (181 lines) - Introduction and fragmentation analysis
- chunk2_methods_and_results_part1.tex (284 lines) - Methods and five-server architecture
- chunk3_results_part2.tex (224 lines) - Safety modules and integration
- chunk4_discussion_and_conclusion.tex (322 lines) - National deployment and governance

national-platform/federated_learning/

- main_chunk1_preamble_intro_methods.tex (215 lines) - Introduction and methods
- main_chunk2_results_architecture_pillars_examples.tex (318 lines) - Architecture and five pillars
- main_chunk3_results_peerreview_trust_analytics.tex (243 lines) - Peer review and analytics
- main_chunk4_discussion_limitations_conclusion.tex (154 lines) - Discussion and conclusions

national-platform/usl_standard/

- 01_preamble_introduction_framework.tex (249 lines) - Introduction and USL methodology
- 02_results_discussion_closing.tex (227 lines) - Results and discussion

national-platform/research_a/

- 01_research_a_part1.txt (113 lines) - U.S. government framework and AI clinical trial regulation
- 02_research_a_part2.txt (166 lines) - Reference URLs and citations

national-platform/research_b/

- 01_research_b_part1.txt (95 lines) - California and federal regulatory analysis
- 02_research_b_part2.txt (109 lines) - Reference URLs and citations

B Glossary of Physical AI Terms

This glossary compiles unified Physical AI terms used throughout the paper, drawn from the glossaries in all three adapted regulatory standards. Terms are listed alphabetically with definitions and primary source references.

Term	Definition
Agentic AI	AI systems capable of autonomous multi-step task execution, coordinating complex clinical workflows without continuous human direction.
Autonomous Mode	Operation of a Physical AI system where AI algorithms direct robotic actions without real-time human input, subject to safety boundaries and override capability.
Calibration	The process of adjusting Physical AI system parameters (position sensors, force sensors, actuators) to meet specified accuracy and precision standards.
Cobot	Collaborative robot designed to work safely alongside humans in shared workspaces, with force-limiting and collision-detection safety features.
Cybersecurity Incident	Any unauthorized access, disruption, or modification of Physical AI systems that may affect patient safety, data integrity, or trial operations.
Digital Twin	A computational model replicating a physical patient, tumor, or system for simulation, prediction, and treatment planning purposes.
Emergency Stop (E-Stop)	A safety mechanism that immediately halts all Physical AI system motion and operations, available through hardware buttons, software commands, and wireless triggers.
Federated Learning	A machine learning approach where models are trained across decentralized data sources without exchanging raw patient data, preserving privacy while enabling model improvement.
Humanoid Robot	A robot with human-like form factor designed for general-purpose tasks including patient guidance, logistics support, and facility operations.
MCP Server	A Model Context Protocol server providing standardized AI-to-tool communication for Physical AI system coordination across clinical trial operations.
Needle-Placement Robot	A robotic system that guides needles for biopsies and targeted drug delivery with precision exceeding manual capability.
Phase 0 Simulation	A mandatory pre-enrollment phase where Physical AI systems demonstrate safety and preliminary efficacy through validated simulations before human subjects contact.
Physical AI Adverse Event	Any untoward occurrence during a clinical investigation that is attributable to Physical AI system operation, including robot malfunctions, AI decision errors, and sensor failures.

Term	Definition
Physical AI Operator	A qualified individual responsible for monitoring and controlling Physical AI systems during clinical procedures, with training per adapted ICH E6(R3) requirements.
Physical AI Safety Matrix	A pre-procedure verification checklist ensuring all Physical AI safety systems (mechanical, sensor, E-stop, communication, AI model) are operational before patient interaction.
Physical AI System	An integrated robotic and AI platform designed for direct or indirect patient interaction in clinical trial settings, encompassing hardware, software, and AI components.
PSL	Physical AI Standard Level, a three-dimension framework evaluating trial site compliance across legislative authorization, regulatory compliance, and operational readiness.
Robotic Procedure	Any clinical intervention performed wholly or partly by a Physical AI system, including surgical, diagnostic, therapeutic, and support activities.
Social Companion Robot	A robot designed to provide emotional support, information delivery, and well-being monitoring for patients during clinical trial participation.
Supervised Mode	Operation of a Physical AI system where human operators maintain continuous real-time control over all robotic actions.
Surgical Robot	A robotic system designed for surgical procedures, operated under direct surgeon control (teleoperated) or with supervised autonomous capabilities.
USL	Unification Standard Level, a four-dimension 1.0-10.0 scoring framework evaluating robot readiness for multi-site clinical trial deployment.
USL Band	Classification tier based on composite USL score: Exemplary (9.0-10.0), Advanced (7.0-8.9), Intermediate (5.0-6.9), Foundational (3.0-4.9), Initial (1.0-2.9).

C Regulatory Cross-Reference Matrix

This matrix shows how each section of this paper maps to source documents, source files, applicable regulatory standards, and key references.

Section	Source Docs	Source Files	Standards	Key References
§ 1 Intro	All source docs	All directories	All three	[18, 21, 20]
§ 2 Gov Framework	Research A	research_a/ (2 files)	CFR 312, 50, ICH	[23, 24, 13, 17]
§ 3 Regulatory	Research B	research_b/ (2 files)	All three	[1, 2, 3]
§ 4 ICH E6(R3)	ICH Adaptation	ich_e6r3_adapt/ (4 files)	ICH E6(R3)	[1, 13, 4]
§ 5 CFR Part 50	CFR 50 Adapt	21cfr50_adapt/ (3 files)	21 CFR 50	[2, 24]
§ 6 CFR Part 312	CFR 312 Adapt	21cfr312_adapt/ (5 files)	21 CFR 312	[3, 23]
§ 7 PSL/USL	USL Standard, Site Docs	usl_standard/ (2), new_trial_psl/ (11)	All three	[4, 5]
§ 8 Site Est.	Site Docs	new_trial_psl/ (11 files)	All three	[5, 1, 2]
§ 9 Journey	Patient Journey	patient_journey/ (3 files)	All three	[6, 4]
§ 10 Patient Inst	Patient Inst	patient_robot/ (2 files)	CFR 50, ICH	[7, 2]
§ 11 MCP	National MCP	national_mcp/ (4 files)	All three	[8, 28, 11]
§ 12 FL	FL Paper	federated_learning/ (4 files)	HIPAA, ICH	[9, 12, 32]
§ 13 Financial	Journey, Site	patient_journey/, new_trial_psl/	N/A	[18, 33]
§ 14 Strategy	All sources	All directories	All three	[10, 11, 12]
§ 15 Discussion	All sources	All directories	All three	All references
§ 16 Conclusion	All sources	All directories	All three	All references

D PSL and USL Scoring Reference

PSL Scoring Reference

Table 45: PSL Three-Dimension Scoring Criteria

Dim	Name	Key Criteria	Status
1	Legislative	State authorization act, patient rights bill, data protection act	Pass/Fail
2	Regulatory	Title 22 compliance, FDA guide, building/premises codes	Pass/Fail
3	Operational	Activation SOPs, emergency plan, staff training, equipment verified	Pass/Fail

A site must achieve Passing status on all three dimensions before Physical AI trial operations may begin.

USL Scoring Reference

Table 46: USL Four-Dimension Scoring Summary

Dim	Name	Weight	Scoring Factors
A	Simulation Switching	25%	Framework count, GPU support, transfer testing, model formats
B	AI Integration	25%	LLM planning, VLA support, diffusion policy, MCP, agentic AI
C	Cross-Robot Sharing	25%	ONNX export, ROS 2, checkpoints, inter-org transfer
D	Clinical Trial Collab	25%	FDA clearance, federated learning, audit trail, multi-site

USL Band Classification: Exemplary (9.0-10.0), Advanced (7.0-8.9), Intermediate (5.0-6.9), Foundational (3.0-4.9), Initial (1.0-2.9). Final score computed as weighted average of four dimension scores, clamped to range 1.0-10.0.

E Simulation Evidence Summary

Table 47: Single-Patient Journey Evidence Summary

Parameter	Value
Patient diagnosis	Stage IIIB non-small cell lung cancer (NSCLC)
Trial duration	1,120 days (prescreening through closeout)
Pipeline stages	10 stages, fully autonomous management
Robot types involved	Surgical, cobots, imaging, companion, rehab exoskeleton
Regulatory coverage	All three adapted standards at every stage
ASCII diagrams	30 three-perspective diagrams
Cost analysis	Per-stage cost savings documented
Source	[6]

Table 48: 24-Hour Multi-Patient Simulation Evidence Summary

Parameter	Value
Patients treated	168 patients
Cancer types	15 different cancer types
Robots deployed	29 robots across multiple categories
System uptime	99.7 percent continuous operation
Patient harm events	Zero patient harm events
Adverse events	7 events documented per ICH E6(R3)
Operating duration	24 continuous hours
Documentation	72 ASCII diagrams, 11 regulatory documents
PSL validation	All three dimensions assessed
Source	[5]